



Supplementary Materials for

Quantum metric nonlinear Hall effect in a topological antiferromagnetic heterostructure

Anyuan Gao et al.

Corresponding author: Su-Yang Xu, suyangxu@fas.harvard.edu

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The PDF file includes:

Materials and Methods
Supplementary Text
Figs. S1 to S50
Tables S1 to S3
References

Materials and Methods

Bulk crystal growth

Our MnBi_2Te_4 bulk crystals were grown by two methods: the Bi_2Te_3 flux method [53] and solid-state reaction method with extra Mn and I_2 . In the Bi_2Te_3 flux method, elemental Mn, Bi and Te were mixed at a molar ratio of 15 : 170 : 270, loaded in a crucible, and sealed in a quartz tube under one-third atmospheric pressure of Ar. The ampule was first heated to 900°C for 5 hours. It was then moved to another furnace where it slowly cooled from 597°C to 587°C and stayed at 587°C for one day. Finally, MnBi_2Te_4 were obtained by centrifuging the ampule to separate the crystals from Bi_2Te_3 flux. In the solid-state reaction method, elemental form of Mn, Bi, Te and I_2 were first mixed at a stoichiometric ratio of 1.5 : 2 : 4 : 0.5 and sealed in a quartz ampoule under vacuum. The sample was heated to 900°C in 24 hours in a box furnace and stayed at the temperature for over 5 hours to ensure a good mixture. The ampoule was then air quenched and moved to another furnace preheated at 597°C, where it slowly cooled to 587°C in 72 hours and stayed at the final temperature for two weeks. The high purity bulk BP crystals were bought from Smart-elements GmbH company.

Sample fabrication

To address the sensitive chemical nature of 2D MnBi_2Te_4 flakes, all fabrication processes were completed in an argon environment without exposure to air, chemicals, or heat. Specifically, the argon-filled glovebox maintained O_2 and H_2O level below 0.01 ppm and a dew point below -96°C . The glovebox was attached to an e-beam evaporator, allowing us to make metal deposition without exposure to air. For the BP/ MnBi_2Te_4 devices, MnBi_2Te_4 was mechanically exfoliated onto a 300-nm SiO_2/Si wafer using Scotch-tape. Once identifying a proper thin MnBi_2Te_4 flake [44], a tip was used to scratch the flake to a rectangular/circular shape. After that a stencil mask technique [44] was used to make Cr/Au contacts on top of MnBi_2Te_4 without exposure to air or chemical. BP flakes were then exfoliated onto a polydimethylsiloxane (PDMS) film, and a BP flake with proper shape and thickness was then identified and transferred onto the MnBi_2Te_4 flake. Next, a 20-50 nm BN flake was transferred onto the BP/ MnBi_2Te_4 heterostructure as the top gate dielectric layer. A metal gate was evaporated onto the BP/ MnBi_2Te_4 heterostructure. For the BP/ MnBi_2Te_4 /BP devices, both the bottom layer BP and MnBi_2Te_4 were mechanically exfoliated onto 300-nm SiO_2/Si wafers. After a proper BP and MnBi_2Te_4 flakes were identified, the MnBi_2Te_4 flake was transferred onto the BP flake using the cryogenic pickup method developed in Ref. [54], where a thin piece of PDMS was cooled to -110°C by liquid nitrogen to achieve the pickup. The rest of the procedures were the same as the BP/ MnBi_2Te_4 heterostructure, which included making the contacts by shadow masks, transferring the top BP flake, and making the top gate.

BP and MnBi_2Te_4 flakes with long, straight edges were deliberately chosen (those straight edges are likely to be along the crystalline direction). When making the stack, the flakes were aligned along their straight edges. After the transport measurements were done, the samples were taken out for both Raman and SHG measurements to check the crystalline direction. The advantage of this approach is that the flakes were kept inside the glovebox throughout the fabrication process. The disadvantage is that the straight edges may turn out to be not along the crystalline direction. As a result, a few devices were made at once to make sure at least one was aligned.

Nonlinear electrical transport measurements

Electrical transport measurements were carried out in a PPMS (Quantum Design DynaCool). The base temperature is 1.65 K and maximum magnetic field is 9 T. The magnetic field was applied along the out-of-plane direction. The gate voltages were applied by Keithley 2400 source meters. Longitudinal and Hall voltages were measured simultaneously. Both first- and second-harmonic signals were collected by standard Lock-in techniques (Stanford Research Systems Model SR830) with excitation frequencies between 1-600 Hz. We have also performed the electrical sum frequency generation measurements, which will be discussed below in Supplementary Materials (SM) II.2.

Optical second harmonic generation and polarized Raman measurements

All SHG experiments were performed using a near-infrared femtosecond laser at room temperatures. The light source is an amplified Yb:KGW laser (Pharos, LightConversion) emitting 168 fs pulses at 1.2 eV with a pulse energy of 100 μ J and a default repetition rate of 100 kHz. All measurements were performed at normal incidence. The polarization of the incident laser was controlled using an achromatic half-wave plate, while a Glan-Laser (GL) polarizer prism was used as an analyzer to select the polarization of the outgoing SHG signal. Both half-wave plate and GL prism were mounted on motorized rotation stages. The SHG data and corresponding symmetry analysis will be presented in SM I.5.

The polarized Raman was performed on a Horiba LabRam HR Evolution Raman spectrometer using a 532-nm laser in a backscattering configuration at room temperature. The polarization of the incident laser beam was controlled using a rotating achromatic half-wave plate and the scattering data with all the polarized directions were received and detected by the spectrometer. An approximately 1- μ m laser beam was focused on the sample by a 100 $\times$ objective. A 1800 l/mm grating and 2 s exposure time were chosen to characterize the crystalline orientation of BP. The samples were protected by BN during the measurements of polarized Raman.

Reflection magnetic circular dichroism measurements

RMCD measurements were performed in the closed-loop magneto-optical cryostat OptiCool by Quantum Design (base temperature ~ 2 K and B field ± 7 T) using a supercontinuum laser SuperK-EXR20 by NKT photonics (wavelength 500 nm to 2500 nm). A spectrometer SpectraPro-300i by Acton Research was used to select the wavelength. The beam went through a photoelastic modulator (PEM200, Hinds instruments) operating at $\frac{\lambda}{4}$ retardation with a frequency of 50 kHz. After an optical chopper (1000 Hz) and a broadband plate beam-splitter, the beam was focused onto the sample by a 50X Mitutoyo Plan Apochromat Objective. The reflected beam went through the cryostat's top window, was collimated by the same objective, and was collected by a Si Avalanche Photodetector (APD410A, Thorlabs). The corresponding reflection APD signals were analyzed by two lock-in amplifiers at 50 kHz (the PEM frequency) and 1000 Hz (the chopper frequency), respectively. Spatial imaging were achieved using a galvo scanning mirror system.

NV center magnetometry measurements

NV wide field imaging measurement was performed in a closed-cycle optical cryostat at 5 K. Green laser used for NV initialization and readout were generated by an electrically driven 515 nm laser. The wide laser beam spot after passing the objective was about 20 μ m $\times$ 20 μ m. The NV fluorescence was imaged using a CMOS camera. Pulses to drive the green laser and to trigger the camera exposure were generated by a programmable pulse generator. Modulated microwave currents were delivered to a microwave combiner and were amplified by +50 dB before being sent to the on-chip gold wire patterned on the diamond substrate. The external magnetic field was generated by a cylindrical NdFeB permanent magnet attached to a scanning stage inside the optical cryostat.

Wireless radio frequency (RF) rectification measurements

We set up a simple experiment to harvest wireless RF signals and recorded the generated DC signals. The RF signal generator was Hittite HMC-T2220 with a frequency range of 10 MHz to 20 GHz. The samples were connected to a low temperature probe. The RF signals went into the low temperature probe through a coaxial cable and the other end of the coaxial cable was connected to an antenna. The antenna was made by a 50-mm long conducting wire with a ~ 0.2 -mm diameter. The end of the antenna was parallel to the sample and the spacing was ~ 10 mm. The electrical field direction and power of the RF signals shone on the sample were not well defined. BP/MnBi₂Te₄ samples were made on a highly doped silicon wafer covered by a 300-nm SiO₂. The DC voltage signals were first passed through a voltage amplifier SR560 and then recorded by an Agilent 34401A Digital Multimeter.

First-principles calculations

First-principles calculations were performed using the projector augmented wave (PAW) method as implemented in the VASP suite of codes [55]. The exchange-correlation part of the potential was treated within the generalized gradient approximation (GGA) scheme developed by Perdew-Burke-Ernzerhof (PBE). The spin-orbit coupling (SOC) was included following the standard implementation in the VASP package [56]. In order to treat the localized Mn 3*d* orbitals, we used an onsite $U = 5.0$ eV [36]. A $9 \times 5 \times 1$ Monkhorst-Pack k -grid was adapted for the Brillouin zone integration. The kinetic energy cutoff for the plane wave basis was set to 270 eV. The heterostructure was created by placing 2SL MnBi₂Te₄ in between monolayer BP on top and bottom, with the armchair direction of the MnBi₂Te₄ aligned along the zigzag direction of the BP. Atomic positions were relaxed until the force on each atom became less than 0.001 eV/Å. Due to the large lattice mismatch between the two materials, DFT requires a super cell. To build a 1×2 supercell, we had to artificially distort the BP unit cell by a lot ($\sim 13\%$ elongation along b). Even then, we could only do 2SL MnBi₂Te₄, because BP/6SL MnBi₂Te₄/BP with the super cell was too heavy for DFT. Given the large distortion and the fact that it is only for 2SL MnBi₂Te₄, we do not use it to fit our model parameters. We only showed this DFT result in this SM for completeness.

Theoretical modeling

We use a 40-band continuum $k \cdot p$ model to describe the low-energy electronic properties of the BP/MnBi₂Te₄/BP heterostructure. It is defined around the Γ point in the Brillouin zone (BZ) and contains terms of up $\mathcal{O}(k^3)$: $\hat{H}(\mathbf{k}) = \begin{pmatrix} \hat{h}_{\text{MBT}}(\mathbf{k}) & \hat{U}_t(\mathbf{k}) & \hat{U}_b(\mathbf{k}) \\ \hat{U}_t^\dagger(\mathbf{k}) & \hat{h}_{\text{BP},t}(\mathbf{k}) & 0 \\ \hat{U}_b^\dagger(\mathbf{k}) & 0 & \hat{h}_{\text{BP},b}(\mathbf{k}) \end{pmatrix}$ where $\mathbf{k} = (k_x, k_y)$ is an in-plane momentum. $\hat{h}_{\text{MBT}}(\mathbf{k})$, $\hat{h}_{\text{BP},t}(\mathbf{k})$ and $\hat{h}_{\text{BP},b}(\mathbf{k})$ are the Hamiltonians of MnBi₂Te₄, top BP and bottom BP, respectively. $\hat{h}_{\text{MBT}}(\mathbf{k})$, $\hat{h}_{\text{BP},t}(\mathbf{k})$ and $\hat{h}_{\text{BP},b}(\mathbf{k})$ share similar work function, which is consistent with the DFT calculation of the BP/MnBi₂Te₄/BP heterostructure (Fig. S42). The 24×24 matrix $\hat{h}_{\text{MBT}}(\mathbf{k})$ describes the low-energy band structure of 6 septuple-layer (SL) MnBi₂Te₄, where each SL i is modeled by a four-band Hamiltonian [57]: $\hat{h}_{\text{MBT},ii}(\mathbf{k}) \equiv \hat{h}_{\text{SL},i}(\mathbf{k}) = \hat{h}_{\text{N}}(\mathbf{k}) - \gamma_{af} \hat{h}_{\text{AFM},i}(\mathbf{k})$. The low-energy basis is formed by the states $\{|p_{z,\text{Bi}}^+, \uparrow\rangle, |p_{z,\text{Te}}^-, \downarrow\rangle, |p_{z,\text{Te}}^-, \uparrow\rangle, |p_{z,\text{Bi}}^+, \downarrow\rangle\}$, which are the symmetric (+) superposition of Bi p_z orbitals on the two Bi layers and the antisymmetric (−) superposition of Te p_z orbitals on the top and bottom layers in the SL. The part $\hat{h}_{\text{N}}$ accounts for the normal state, where we include a cubic $\mathcal{O}(k^3)$ warping term to obtain the correct threefold symmetric Fermi surface shape. The parameter γ_{af} controls the strength of the magnetic contribution $\hat{h}_{\text{AFM}}$ in the presence of A-type AFM magnetic order. Nearest neighbor SLs are coupled via a hopping term $\hat{h}_{\text{MBT},i,i\pm 1}(\mathbf{k}) = \hat{T}_0(\mathbf{k})$ that is derived from the k_z dispersion of bulk MnBi₂Te₄. To account for lattice strain (with strength γ_s) induced by the encapsulation with BP, we add symmetry allowed terms $\gamma_s [\hat{h}_{\text{N},s}(\mathbf{k}) - \gamma_{af} \hat{h}_{\text{AFM},s}(\mathbf{k})]$ and $\gamma_s \hat{T}_s(\mathbf{k})$. The two 8×8 blocks $\hat{h}_{\text{BP},t}$ and $\hat{h}_{\text{BP},b}$ describe the low-energy band structure of top (t) and bottom (b) BP monolayer [58]

$$\hat{h}_{\text{BP},t} = \hat{h}_{\text{BP},b} = \sum_{\sigma} \left[f_1(\mathbf{k}) \left(\hat{\phi}_{k,1\sigma}^\dagger \hat{\phi}_{k,2\sigma} + \hat{\phi}_{k,3\sigma}^\dagger \hat{\phi}_{k,4\sigma} \right) + f_2(\mathbf{k}) \left(\hat{\phi}_{k,2\sigma}^\dagger \hat{\phi}_{k,3\sigma} + \hat{\phi}_{k,1\sigma}^\dagger \hat{\phi}_{k,4\sigma} \right) + \text{h.c.} \right], \quad (\text{S1})$$

where $f_1(k_x, k_y) = 2\tilde{t}_1 e^{i\frac{ak_x}{2\sqrt{3}}} \cos\left(\frac{bk_y}{2}\right)$ and $f_2(k_x, k_y) = \tilde{t}_2 e^{-i\frac{ak_x}{\sqrt{3}}}$. Here, $\phi_{\mathbf{k}\nu,\sigma}^\dagger$ creates an electron at one of the four P p_z orbitals in the unit cell and $\tilde{t}_1$ and $\tilde{t}_2$ are set to obtain the experimentally observed BP band gap of 0.3 eV. The interaction between MnBi₂Te₄ and BP can lead to two consequences. (1) It can lead to a structural strain on MnBi₂Te₄, which we will model in the diagonal MnBi₂Te₄ Hamiltonian term $\hat{h}_{\text{MBT}}$ (see IV.6 below). (2) The bands between MnBi₂Te₄ and BP can directly hybridize. Such an electronic band coupling between MnBi₂Te₄ and BP are described by the off-diagonal 24×8 blocks $\hat{U}_t$ and $\hat{U}_b$. Both the strain and the band hybridization lead to a breaking of threefold rotation and translation symmetry. The band coupling between the Bloch states of the two layers thus needs to be derived in the real-space continuum leading to $\hat{H}_{\text{int}}^{(b)} = \sum_{\mu,\nu,\sigma} \sum_{\mathbf{k}} \left[t_{\mu\nu}^{(b)}(\mathbf{k}) \psi_{\mathbf{k}\mu,\sigma}^\dagger \phi_{b,\mathbf{k}\nu,\sigma} + \text{h.c.} \right]$. Here, $\psi_{\mathbf{k}\mu,\sigma}^\dagger$ creates an electron with momentum $\mathbf{k}$ and spin σ in one of the seven orbitals μ in the MnBi₂Te₄ unit cell and the MnBi₂Te₄-BP interface hopping amplitude is given by $t_{\mu\nu}^{(b)}(\mathbf{k}) \equiv \int d^2y t_b(\mathbf{y} + s_{\mu\nu}\hat{\mathbf{z}}) e^{-i\mathbf{k}\cdot\mathbf{y}}$. The size of the hopping is controlled by the orbital

distance with out-of-plane component $s_{\mu\nu} = (\boldsymbol{\tau}_\mu - \boldsymbol{\rho}_\nu) \cdot \hat{\mathbf{z}}$. The integration over the in-plane distance $\mathbf{y}$ is due to the lattice incommensurability. The real space hopping elements are parameterized using the Slater-Koster approach as $t_b(\mathbf{r}) = A_\sigma e^{-r/a_\sigma} \cos^2 \theta + A_\pi e^{-r/a_\pi} \sin^2 \theta$, where A_σ, A_π denote characteristic energy and a_σ, a_π denote characteristic length scales of the hopping integrals for σ and π bonding. Projection onto the relevant MnBi_2Te_4 low-energy manifold yields the coupling matrix $\hat{U}_b$. The coupling block $\hat{U}_t$ is obtained from inversion symmetry. More details of these theoretical models are shown in Secs. IV.4 – IV.9.

Supplementary Text

I. Basic characterization of BP/ MnBi_2Te_4 heterostructure

I.0. Thickness determination

Figure S1 shows the atomic force microscopy data of the main devices. Our data shows a measured thickness of $\simeq 8.5$ nm, therefore demonstrating that our sample is indeed 6SL (every SL is about 1.4 nm). We also performed RMCD of the same device (Fig. S2A). To compare and contrast, we measured RMCD of a BP/7SL MnBi_2Te_4 sample. As shown in Fig. S2B, we see a clear nonzero RMCD at $B = 0$ T, showing a finite magnetization for 7SL. Moreover, the AFM phase boundary of even-layered MnBi_2Te_4 should be around ± 2 T [42, 43]. This is seen in our RMCD data (Fig. S2A). In a following subsection (SM.I.3), we show the magnetotransport data of our heterostructure device, which also reveal the ± 2 T phase boundary.

In general, our thickness determination of MnBi_2Te_4 flakes has been repeatedly tested through a very rigorous process for many times. (1) We measure the optical contrast of the flakes under the microscope in our glovebox. We then took the flakes out of the glovebox to directly determine by atomic force microscopy. This allows us to establish a one-to-one correspondence between the optical contrast and the layer number (Fig. S3). We repeated this process many times to ensure that the correspondence is reproducible and reliable. (2) Beyond that, we have done rigorous checks by RMCD and NV center magnetometry. As shown in Fig. S4, we exfoliated 2D MnBi_2Te_4 sample and determine the layer number based on the optical contrast of the microscope image (Fig. S4A). We then measured RMCD (Fig. S4B), which matches the layer number determined by the optical contrast. (3) Moreover, similar check have been performed via NV center magnetometry (carried out by Xinyue Zhang and Brian Zhou at Boston college and Mengqi Huang and Chunhui Du at UCSD). The NV center magnetometry shown in Figs. S4C,D (taken by Xinyue Zhang and Brian Zhou) also matches the layer number determined by the optical contrast. The RMCD test was performed over 20 samples; The NV test was performed on 5 samples. Through these systematic checks, we can rigorously determine the layer number of our 2D MnBi_2Te_4 samples.

As an additional check, we have performed new NV measurements on the BP/6SL MnBi_2Te_4 /BP heterostructure. The idea is the following: our data above already demonstrate that 6SL MnBi_2Te_4 has no observable magnetization. This additional check aims to show that in the BP/6SL MnBi_2Te_4 /BP heterostructure, the 6SL MnBi_2Te_4 remains to be fully compensated with no observable magnetization. This is achieved in Fig. S5, where we fabricated a BP/6SL MnBi_2Te_4 /BP heterostructure onto a diamond substrate. Our data confirm the absence of observable magnetization in this heterostructure. We wish to highlight that in the BP/6SL MnBi_2Te_4 /BP heterostructure, the following three statements are equivalent: (1) The system has $\mathcal{PT}$ symmetry; (2) The spin-up layers and spin-down layers of 6SL MnBi_2Te_4 compensate; (3) The system has no observable net magnetization. Our NV measurements directly show (3) no observable net magnetization, which then in turn proves (1) and (2). In particular, the $\mathcal{PT}$ symmetry dictates that Berry curvature is identically zero, which is one of the ways for us to distinguish Berry curvature dipole and quantum metric dipole

I.1. The symmetry of MnBi_2Te_4 , BP/ MnBi_2Te_4 and BP/ MnBi_2Te_4 /BP

The even-layered MnBi_2Te_4 is a fully compensated antiferromagnet at low temperature. Although the lattice of MnBi_2Te_4 is centrosymmetric, the spin breaks the inversion and time reversal symmetry. Therefore, even-layered MnBi_2Te_4 breaks $\mathcal{P}$ and $\mathcal{T}$ but preserves the $\mathcal{PT}$ symmetry (Fig. S6A). After stacking BP on MnBi_2Te_4 , the inversion symmetry of the lattice is also broken. Therefore, the BP/ MnBi_2Te_4 breaks $\mathcal{P}$, $\mathcal{T}$ and also $\mathcal{PT}$ symmetry (Fig. S6B). Sandwiching MnBi_2Te_4 between two 2L BP, the lattice of the heterostructure is still centrosymmetric when the crystallographic a axes of the BP layers and the MnBi_2Te_4 layer are all

aligned. Therefore, the aligned BP/MnBi₂Te₄/BP heterostructure breaks $\mathcal{P}$ and $\mathcal{T}$ but still preserves the $\mathcal{PT}$ symmetry (Fig. S6C).

I.2. Determining the crystalline directions for MnBi₂Te₄ and BP

The crystalline directions of the MnBi₂Te₄ flakes were determined by optical SHG measurements at room temperature. At room temperature, the interior of thin MnBi₂Te₄ flake's crystal structure is centrosymmetric (group D_{3d}). Therefore, the SHG signals are only expected to originate from the surface (surface group C_{3v}). The largest SHG signals are along the mirror plane of the MnBi₂Te₄ surface according to the symmetry. Figure S7A and S7B show how our SHG data correspond to the crystalline axes of MnBi₂Te₄. These results are consistent with previous SHG results on Bi₂Se₃ (same symmetry as MnBi₂Te₄ above Neel temperature) [59].

The crystalline directions of the BP were determined by polarized Raman scattering measurements. Figure S7C and S7D show how our Raman data correspond to the crystalline axes of BP. These results are consistent with previous Raman measurements on BP [59].

I.3. Basic transport characterizations

Figure S8 shows the magneto-transport data of MnBi₂Te₄ before and after covering BP of Device-BM19. At charge neutrality, both MnBi₂Te₄ and BP/MnBi₂Te₄ heterostructure show that the Hall resistivity (R_{xy}) is nearly zero in the AFM phase and is nearly quantized in the FM phase. These data suggest that interfacing MnBi₂Te₄ with BP does not change the topological phases of even-layered MnBi₂Te₄: it is a Chern insulator in the FM phase and an Axion insulator in the AFM phase. The magneto-transport data is consistent with the DFT calculated band structure of BP/MnBi₂Te₄/BP (Fig. S42), in which the lowest conducting band and highest valance band are both derived from MnBi₂Te₄ band.

As described in SM.I.0, the AFM phase boundary for even-layered MnBi₂Te₄ should be ± 2 T. This ± 2 T transition is present in our transport data, just subtler and gate dependent. Here, we take the device studied in Figs. S8E,F) as an example to demonstrate this point. Figures S9A,B reproduce the R_{yx} and R_{xx} data in Figs. S8E,F. While the R_{yx} data does not show the ± 2 T transition, R_{xx} has a slope change at ± 2 T. Moreover, the ± 2 T feature is strongly gate-dependent. As shown in Figs. S9C-F, by changing the gate voltage, the ± 2 T becomes much more apparent. In fact, similar behavior is also seen in transport data from other groups. Ref. [40] Fig. 3a shows the 6SL transport data from an independent group. Similarly, the R_{yx} does not show the ± 2 T transition.

Figure S10 shows the gate-dependent four-point resistance of bare BP and 6SL MnBi₂Te₄. BP was found to be significantly more resistive than 6SL MnBi₂Te₄. The resistance of BP is two orders of magnitude larger than that of MnBi₂Te₄ for all of the gate voltages. The resistivity data can be explained by the electronic structures. MnBi₂Te₄ is a Dirac material while BP is a semiconductor (band gap ~ 0.3 eV for thick flake and ~ 0.7 eV for bilayer [60]). Therefore, it is reasonable to assume that the current flowing in BP layer is small.

To further confirm that, we designed and fabricated another kind of device. As shown by the schematic in Fig. S11B, we injected current through the drain electrode, then collected the current from two source electrodes (S_{MBT} and S_{BP}). Figures S11C-F show the currents collected from S_{MBT} and S_{BP} and their ratio as a function of V_{BG} and V_{TG} . These data show that the current flowing through BP layer is small. Therefore, in the BP/MnBi₂Te₄ heterostructure, the current mainly flows in MnBi₂Te₄ layer.

I.4. Angular-resolved transport

Here we use transport method to determine the resistivity anisotropy of the sample. Assuming the resistivity of the sample is ρ_a and ρ_b along a and b axis (a, b are in the in-plane crystallographic axes of the lattice, $a \perp b$). The measurements were performed in Cartesian basis and the angle between x and a is θ . The resistivity in longitudinal (ρ_{xx}) and transverse (ρ_{yx}) direction are expressed as [61]:

$$\rho_{xx} = \rho_a \cos^2(\theta) + \rho_b \sin^2(\theta) = \frac{1}{2}(\rho_a + \rho_b) + \frac{1}{2}(\rho_a - \rho_b)\cos(2\theta) \quad (\text{S2})$$

$$\rho_{yx} = (\rho_a - \rho_b)\cos(\theta)\sin(\theta) = \frac{1}{2}(\rho_a - \rho_b)\sin(2\theta) \quad (\text{S3})$$

Bare MnBi_2Te_4 has C_{3z} rotational symmetry. Therefore, $\rho_a = \rho_b$, and then $\rho_{xx} = \rho_a$, $\rho_{yx} = 0$. Hence there is no anisotropy for bare MnBi_2Te_4 , consistent with our results showing both R_{xx} and R_{yx} are independent of the measurement directions and the R_{yx} is always around zero (Fig. S12). After interfacing MnBi_2Te_4 with BP, the C_{3z} of the sample is broken. so, $\rho_a \neq \rho_b$. According to the Eqs. S2 and S3, both R_{xx} and R_{yx} depend on measurement direction and have a 180° periodicity, which is consistent with our measurement (Fig. S12). Moreover, for the angular-dependent of R_{xx} and R_{yx} , R_{xx} has maximum/minimum value when the R_{yx} crosses zero. This further confirms that MnBi_2Te_4 has anisotropy after interfacing with BP, i.e. $\rho_a \neq \rho_b$.

To further exclude the effects of nonlocal transport, we also performed two-probe measurement (Fig. S12C). The two-probe resistance showed the same angular dependence as four-probe measurements, and the overall resistance value is larger due to the additional contact resistance. Hence, two-probe measurements again confirm that interfacing MnBi_2Te_4 with BP breaks the C_{3z} rotational symmetry.

I.5. Optical second-harmonic generation

Optical SHG is an effective way for probing sample's symmetry because it is sensitive to crystal symmetry [62]. Here, we use optical SHG to demonstrate that interfacing MnBi_2Te_4 with BP can break the C_{3z} rotational symmetry of MnBi_2Te_4 . The optical SHG measurements were performed at room temperature. At room temperature, the interior of MnBi_2Te_4 is centrosymmetric which prohibits any SHG signal generation. However, the surfaces of MnBi_2Te_4 break the inversion symmetry and have the C_{3z} rotational symmetry. Therefore, for the normal incidence measurement, the SHG signals of MnBi_2Te_4 are mainly from the top surface. For BP, because both its interior and surfaces have the two-fold rotational symmetry, BP does not generate any SHG signal, which is confirmed by our measurements (Fig. 1H). Therefore, for the BP/ MnBi_2Te_4 heterostructure, the SHG signals come from the surface of MnBi_2Te_4 . Hence, optical SHG is an effective way for probing the surface symmetry of MnBi_2Te_4 .

Figure S13B shows the SHG signals of a BP/ MnBi_2Te_4 heterostructure. The SHG pattern is asymmetric which demonstrates that the C_{3z} symmetry of MnBi_2Te_4 is broken. To double check the C_{3z} symmetry breaking was induced by BP, we removed the BP layer of the heterostructure using the Scotch tape (Fig. S13C). We probed SHG from the same area of the MnBi_2Te_4 . As shown in Fig. S13D, after removing the BP, the SHG signals of MnBi_2Te_4 recovers the symmetric pattern. We repeated these measurements on three devices, all of them showed consistent results. Therefore, the optical SHG measurements demonstrate that the C_{3z} symmetry is broken after interfacing MnBi_2Te_4 with BP.

II. Addressing alternative mechanisms for the nonlinear Hall signals

II.1. Berry curvature dipole

The most important competing mechanism is the Berry curvature dipole induced nonlinear Hall effect [4]. Below, we explain why the Berry curvature dipole mechanism cannot explain our observations in BP/ MnBi_2Te_4 /BP.

- **Berry curvature dipole is zero due to $\mathcal{PT}$ symmetry:** The aligned BP/ MnBi_2Te_4 /BP device is expected to respect $\mathcal{PT}$ symmetry. $\mathcal{PT}$ enforces Berry curvature and Berry curvature dipole to vanish. Yet, we still observe large nonlinear Hall effect, which naturally cannot be explained by Berry curvature dipole. Therefore, below, let us assume that the PT symmetry is weakly broken, so that Berry curvature dipole becomes nonzero. Even then, it still cannot explain our data for the following reasons.
- **AFM order:** (1) Our temperature dependence data show that the nonlinear Hall effect vanishes above T_N . This is inconsistent with Berry curvature dipole because the D_{Berry} induced nonlinear Hall effect arises from the lattice inversion breaking, which should exist as long as the lattice (i.e., the heterostructure in our case) is stable ($>$ room temperature). (2) Our data show that the nonlinear Hall effect reverses sign upon reversing the AFM order. This cannot be explained by the Berry curvature dipole. The two AFM states (related by time-reversal $\mathcal{T}$) are expected to have the same Berry curvature dipole but opposite quantum metric dipoles. Therefore, our AFM order relationship data is inconsistent with Berry curvature dipole but consistent with quantum metric dipole.

- **Scaling data:** Our scaling data show that the nonlinear Hall effect is independent of the scattering time τ at least below ~ 15 K. This cannot be explained by the Berry curvature dipole origin. As shown above, the Berry curvature dipole induced nonlinear Hall effect is linearly proportional to τ , whereas the quantum metric dipole induced nonlinear Hall effect is independent of τ .
- **Vertical E_z field:** The vertical displacement field E_z is an external field that breaks $\mathcal{PT}$ symmetry. Our data show that the nonlinear Hall effect is already nonzero at $E_z = 0$ and it is symmetric with respect to $\pm E_z$. This cannot be explained by the Berry curvature dipole, which is expected to be antisymmetric with respect to $\pm E_z$, i.e., $D_{\text{Berry}}(+E_z) = -D_{\text{Berry}}(-E_z)$. By contrast, our observed nonlinear Hall signal is symmetric about $E_z = 0$ (see symmetry analysis in Sec. IV.2.1), as shown in Fig. S14.

We can also experimentally determine the direction of the Berry curvature dipole. Intuitively, the direction of the Berry curvature dipole D_{Berry} is determined by the sign of Berry curvature. In MnBi_2Te_4 , we can measure the sign of Berry curvature by measuring the linear Hall signals σ_{xy} [44]. We simultaneously measured σ_{xy} and $V_{yxx}^{2\omega}$ as a function of E_z field (Fig. S15). The σ_{xy} is antisymmetric as a function of E_z which indicates that D_{Berry} is also antisymmetric as a function of E_z . However, the $V_{yxx}^{2\omega}$ is symmetric as a function of E_z . Therefore, the nonlinear Hall signals observed here are not induced by Berry curvature dipole.

- **Mirror symmetry:** In Device-BMB1, by aligning the a axis of the top BP, MnBi_2Te_4 and bottom BP layers, we also preserve the mirror plane $\mathcal{M}_y$. In the AFM state, $\mathcal{M}_y$ is broken but $\mathcal{M}_y\mathcal{T}$ is a good symmetry. For Berry curvature dipole, one expects $\sigma_{yxx} \equiv 0$ and $\sigma_{xyy} \neq 0$. Hence, the D_{Berry} -induced nonlinear Hall voltage $V^{2\omega}$ should be along the x direction. For quantum metric dipole, one expects $\sigma_{yxx} \neq 0$ and $\sigma_{xyy} \equiv 0$. Hence, the D_{Metric} -induced nonlinear Hall voltage $V^{2\omega}$ should be along the y direction. In other words, $\mathcal{M}_y\mathcal{T}$ dictates the two nonlinear Hall effects to be perpendicular. Our observed nonlinear Hall voltage $V^{2\omega}$ is along y (Fig. S16), which is inconsistent with Berry curvature dipole but consistent with quantum metric dipole.
- **Scaling property:** For Berry curvature dipole, the nonlinear Hall conductivity is expected to be proportional to the scattering time, $\sigma_{yxx}^{2\omega} \propto \tau$. For quantum metric dipole, we have, $\sigma_{yxx}^{2\omega} \propto \tau^0$. Our data (Fig. 3A) is consistent with the latter.

II.2. Second-order Drude conductivity

Another important mechanism is the second-order Drude effect [22]. From the symmetry point of view, the second-order Drude effect and the quantum metric dipole induced nonlinear signals are both allowed in $\mathcal{PT}$ -symmetric AFMs (see SM Sec. IV.2.2). Therefore, our goal is to investigate which effect is more dominant in our sample.

II.2.1 Anti-symmetric vs symmetric

For the quantum metric Hall effect, its nonlinear conductivity is expected to be antisymmetric, $\sigma_{yxx}^{2\omega} = -\sigma_{xyx}^{2\omega}$. For the second-order Drude effect, its nonlinear conductivity is expected to be symmetric, $\sigma_{yxx}^{2\omega} = \sigma_{xyx}^{2\omega}$. $\sigma_{ijk}^{2\omega}$ is defined as $V_i^{2\omega} \propto \sigma_{ijk}^{2\omega} I_j^\omega I_k^\omega$. $\sigma_{xyx}^{2\omega}$ means that we flow currents along both x and y directions, which is difficult to implement experimentally.

Therefore, here we present our electrical sum frequency generation (SFG) measurements, where we inject two currents with frequencies ω_1 and ω_2 , and we detect the SFG voltage $V^{\omega_1+\omega_2}$. By separating the two currents in frequency domain, it is easier to also control their directions separately.

Data: Figure S17A shows $\sigma_{yxx}^{2\omega}$ measurement, which is achieved by passing both currents at ω_1 and ω_2 along x direction and measure $V^{\omega_1+\omega_2}$ along y direction. Fig. S17B shows $\sigma_{xyx}^{2\omega}$ measurement, which is achieved by passing a current at ω_1 along y direction while passing another current at ω_2 along x direction and measure $V^{\omega_1+\omega_2}$ along x direction. Indeed, our data show $\sigma_{yxx}^{2\omega} = -\sigma_{xyx}^{2\omega}$ (Fig. S17C), which demonstrates that the quantum metric Hall effect is dominant in our BP/ MnBi_2Te_4 samples.

Methods: The SFG voltage $V^{\omega_1+\omega_2}$ are detected by the following method: The SFG voltage can be expressed as $V^{\omega_1+\omega_2} \propto \sin(\omega_1 + \omega_2)t$. The standard lock-in detectors (SR830) cannot directly lock to the

$\omega_1 + \omega_2$ frequency. Instead, we measured $\sin \omega_1 t \sin \omega_2 t$, which directly relates to $\sin(\omega_1 + \omega_2)t$ by the angle addition theorem. We chose $\omega_1 = 547$ Hz and $\omega_2 = 1.37$ Hz, so that $\omega_1 \gg \omega_2$. To probe $\sin \omega_1 t \sin \omega_2 t$, the SFG voltage signal was fed through the first lock-in (lock-in A), which was locked to ω_1 and its integral time was set to $\frac{2\pi}{\omega_1} < t_1 \ll \frac{2\pi}{\omega_2}$. The output was then fed through the second lock-in (lock-in B), which was locked to ω_2 and its integral time was set to $t_2 > \frac{2\pi}{\omega_2}$. The output of the second lock-in was $\sin \omega_1 t \sin \omega_2 t$. The measurement setups for $\sigma_{yxx}^{2\omega}$ and $\sigma_{xyx}^{2\omega}$ are shown in Fig. S18.

II.2.2 Scaling property

For the second-order Drude effect, its nonlinear conductivity is expected to be quadratic with respect to the scattering time, $\sigma^{2\omega} \propto \tau^2$ [4, 22, 63]. For quantum metric dipole, we have $\sigma^{2\omega} \propto \tau^0$. Our data (Fig. 3A and Fig. S19) is consistent with the latter.

II.2.3 $\sigma_{yyy}^{2\omega}$ vs. $\sigma_{yxx}^{2\omega}$

The second-order Drude effect can lead to a second-order longitudinal current $\sigma_{yyy}^{2\omega}$ whereas the nonlinear Hall effect cannot generate $\sigma_{yyy}^{2\omega}$. Therefore, $\sigma_{yyy}^{2\omega}$ provides a way to isolate the nonlinear Drude contribution. Figure S34 shows $\sigma_{yyy}^{2\omega}$ vs. $\sigma_{yxx}^{2\omega}$ as a function of carrier density and temperature, which shows that $\sigma_{yxx}^{2\omega}$ is dominant. Therefore, based on these data, the nonlinear Hall effect is dominant, and there is no observable sign of transition from a non-linear Hall effect dominated regime to a nonlinear Drude dominated regime.

II.3. Joule heating induced Anomalous Nernst effect and overall temperature increase

We now consider a Joule heating induced anomalous Nernst effect: (1) Joule heating leads to a temperature gradient $\Delta T \propto I^2 R$; (2) Let us assume that our even-layered MnBi_2Te_4 is not fully-compensated, so there is a small M . The combination of the ΔT , and the magnetization M can lead to an anomalous Nernst current, $J = \sigma^{\text{Nernst}} \Delta T \propto \sigma^{\text{Nernst}}(I^2 R)$.

- **Mirror symmetry:** The Joule heating induced anomalous Nernst effect is insensitive to crystalline symmetry. By contrast, as described above, our data in Device-BMB1 shows clear dependence with respect to the mirror plane $\mathcal{M}_y$. Specifically, we observed $\sigma_{yxx} \neq 0$ and $\sigma_{xyy} \equiv 0$.
- **Fermi level dependence:** For the anomalous Nernst effect, in the presence of a fixed M , we expect the electrons and holes to deflect toward opposite directions. By contrast, in our data, the signals from electrons and holes have the same sign.
- **Scaling property:** The Joule heating induced anomalous Nernst effect is $J = \sigma^{\text{Nernst}} \Delta T \propto \sigma^{\text{Nernst}}(I^2 R)$. Because $\sigma^{\text{Nernst}} \propto \tau^0$ and $I^2 R \propto \tau^1$, this effect is expected to be proportional to τ . By contrast, our data is independent of τ .
- **Full compensated AFM:** The magnetization is necessary to generate anomalous Nernst effect. However, the BP/ MnBi_2Te_4 heterostructure is a fully compensate AFM system. The linear magnetotransport data of the BP/ MnBi_2Te_4 (Fig. S8) shows a zero Hall resistance at $B = 0$ which confirms that there is no global magnetization without external magnetic field.

Apart from the Nernst effect (which requires a temperature gradient), it is also important to consider the Joule heating induced overall temperature increase, which we carefully studied as follows. Figure S20 shows the R_{yx} vs. B at different temperatures and different currents. First, at $T = 2$ K, 12 K and 18 K, we measure R_{yx} using $I = 200$ nA (Figs. S20A-C), the plateau in R_{yx} are similar at 2K and 12K and becomes significantly shortened at 18K (we wish to note that here the AFM phase boundary at ± 2 T is not visible due to the choice of gate voltage, just like what we explained in Fig. S9. We have checked that the ± 2 T boundary can be seen in this device if we change to a different gate voltage). In Figs. S20D-F, we repeat the same measurements but using $I = 10$ μA . The plateau at corresponding temperature is similar between 200 nA and 10 μA . These data suggest that the current-induced heating effect is unlikely to explain the temperature dependence.

We have further checked this by another independent measurements. In Fig. S20G, we measure $R_{xx} - T$, where the AFM to paramagnetic (PM) transition is clearly seen in R_{xx} . Notably, the transition temperature

is the same for $I = 200$ nA and $I = 10$ μ A. Moreover, because R_{xx} varies with T , the R_{xx} value itself can represent temperature. If there is large current-induced temperature increase, the $R_{xx} - T$ curves with $I = 200$ nA and $I = 10$ μ A should strongly deviate. By contrast, our data show that the two curves almost overlap. Therefore, our $R_{xx} - T$ demonstrates that the current-induced overall temperature increase is insignificant.

II.4. Other extrinsic effects

Finally, we consider various extrinsic effects such as accidental contact junctions, flake shape, etc.

- **Contact junction:** An accidental contact junction can lead to a nonlinear effect. (1) The nonlinear signals induced by contact junction should not relate to AFM states. (2) The nonlinear signals induced by contact junction should not be sensitive to Neel temperature. (3) The nonlinear signals induced by contact junction should not show Hall dominance. (4) The nonlinear signals induced by contact junction should not relate to the mirror symmetry of BP/MnBi₂Te₄/BP heterostructure.

The nonlinear signals induced by contact junction can be excluded by the following observations: (1) The nonlinear signals have opposite sign for the different AFM states (Fig. 2, G and H). (2) The nonlinear signals decrease to zero when the temperature is higher than Neel temperature (Fig. 2, I and J). (3) The nonlinear signals show clear Hall dominance (Fig. 2B). (4) The nonlinear signals are significantly enhanced after covering BP on the same MnBi₂Te₄ sample (Fig. 1I). (5) For a careful aligned BP/MnBi₂Te₄/BP device, the nonlinear Hall signals can only be observed when the current applied in mirror plane $\mathcal{M}_a$ (Fig. S16).

- **Flake shape:** The asymmetric global shape of the sample can lead directional movement of the carriers by colliding against the asymmetric sample boundaries, which can also induce nonlinear signals. The nonlinear signals induced by flake shape should highly depend on flake shape and should not depend on AFM states, temperature and carrier density.

The flake shape induced nonlinear signals can be excluded by following observations: (1) Most MnBi₂Te₄ in the Hall bar devices are shaped into rectangular-like shape (Fig. S16A). As shown in Fig. S16, the asymmetry of the MnBi₂Te₄ is similar in x and y directions, but the nonlinear Hall signal can only be observed when current is in the mirror plane $\mathcal{M}_a$. (2) The nonlinear signals are highly dependent on AFM states, temperature and carrier density.

Therefore, our systematic data as a function of temperature, AFM states, crystalline direction, doping, etc. allow us to show that these extrinsic effects are not important.

II.5. Addressing the nonlinear Hall signals induced by skew scattering

By now, it has been firmly established that the linear AHE consists of the intrinsic AHE due to Berry curvature and the extrinsic AHE from defect scattering [35]. The defect scattering induced linear anomalous Hall effect has further been categorized into two kinds, skew scattering and side jump. Prior studies have made an important conclusion that scaling law can be used to differentiate the skew scattering induced AHE from intrinsic AHE. Skew scattering induced AHE is proportional to longitudinal conductivity σ_{xx}^2 [64], while the Berry curvature induced AHE only depends on Berry curvature. Because the side jump induced AHE has the same scaling properties as Berry curvature induced AHE, there is currently no effective way to differentiate the side jump from Berry curvature contribution. The working principle adopted by the community so far is to combine the Berry curvature and side jump contributions and consider them as the intrinsic Berry phase contribution [35]. Therefore, we only talk about the skew scattering induced AHE here.

Recent theory has extended the studies from linear AHE to the nonlinear AHE. Specifically, for the intrinsic nonlinear Hall effect studied here, the scaling is predicted to be the same as the linear AHE [22]: The intrinsic component induced by quantum metric scales as τ^0 , while the extrinsic component induced by skew scattering scales as τ^2 . Therefore, we use scaling laws to exclude the skew scattering induced nonlinear Hall current.

- **Temperature dependence:** As shown in Fig. 3A, the $\sigma_{yxx}^{2\omega}$ does not depend on σ_{xx} in the low temperature range (<15 K). In addition, Fig. S19 shows more details about temperature dependence

of σ_{xx}^2 and $\sigma_{yxx}^{2\omega}$. It is notable that σ_{xx}^2 and $\sigma_{yxx}^{2\omega}$ show different temperature dependence. σ_{xx}^2 decreases with temperature increasing when temperature is lower than 21 K. However, $\sigma_{yxx}^{2\omega}$ does not depend on temperature when temperature is lower than 15 K but quickly decrease to zero when temperature is higher than 15 K. To be more clear, we plot $\sigma_{yxx}^{2\omega}$ versus σ_{xx}^2 (Fig. S19C) which clear show that $\sigma_{yxx}^{2\omega}$ does not depend on σ_{xx}^2 when temperature is lower than 15 K.

- **Carrier density n dependence:** In BP/MnBi₂Te₄/BP device, both $\sigma_{yxx}^{2\omega}$ and σ_{xx} depend on n . Therefore, it is a good way to check the scaling properties of $\sigma_{yxx}^{2\omega}$ with σ_{xx} by tuning n . As shown in Fig. S21, A and B, σ_{xx}^2 keeps increasing with n increasing. By contrast, the $\sigma_{yxx}^{2\omega}$ first increase with n but eventually decrease to negative values. The contrasting behaviors between $\sigma_{yxx}^{2\omega}$ and σ_{xx}^2 suggest that skew scattering is not the main contribution to nonlinear Hall signals. Moreover, the sign reversal of $\sigma_{yxx}^{2\omega}$ in higher n doped regime can be explained by quantum metric dipole contribution.

All scaling laws related to temperature and carrier density show that the nonlinear Hall signals observed here are independent of τ (σ_{xx}). The result of the scaling law is consistent with the prediction of the quantum metric dipole induced intrinsic nonlinear Hall effect. Therefore, we can conclude that the nonlinear Hall signals observed here are not induced by skew scattering.

III. Additional data

In this section, we present additional experimental data that did not appear in the main text. These data help us to further confirm the major conclusions.

1. Figure S22 shows the summary of the measured 30 MnBi₂Te₄ heterostructure and 7 MnBi₂Te₄ devices. The nonlinear Hall signals in BP/MnBi₂Te₄ heterostructures are highly reproducible. The nonlinear Hall signals in MnBi₂Te₄ are induced by nonlinear Drude conductivity which is two orders smaller than the nonlinear Hall signals induced by quantum metric dipole. This further confirms that nonlinear Drude conductivity is not the main contribution to the nonlinear Hall signals. This result is consistent with the data in Fig. 1I.
2. Figure S23 shows the frequency dependence of the nonlinear Hall signal. In Fig. S23A, we have tested the entire $\sigma_{yxx}^{2\omega} - n_e$ curve at different frequencies. In Figs. S23B-E, we have also chosen specific charge density values and measured continuous frequency dependence. Our data show that the nonlinear Hall signal is indeed observed to be independent of frequency for the tested range (we have tested up to ~ 3 kHz because the electrical circuits in our cryostat will start to have frequency dependence for higher frequencies). Because there is no dependence of the scattering time τ , we expect the nonlinear Hall effect to persist without frequency dependence, all the way until the $\hbar\omega$ is large enough to induce an inter-band transition (i.e., comparable to the energy spacing between bands). This would be in the terahertz or far-infrared regime. When there is inter-band transition, we expect the nonlinear Hall effect to start to fail, because it is an intra-band process.
3. Figure S24 shows the nonlinear Hall signals as a function of out of plane magnetic field. The vertical electric field E_z is -0.17 V/nm. With a finite E_z , we can choose AFM states by sweeping magnetic field [44]. The nonlinear Hall signals $V_{yxx}^{2\omega}$ have opposite signals for the different AFM states.
4. Figure S25 shows the transport data for different AFM states in another device (device-BMB2). The linear longitudinal signals, R_{xx} and V_{xx}^ω , are identical for the two AFM states. However, the nonlinear Hall signals are opposite for two AFM states. The experiment data on the new device reproduces the main data features, and further confirms our major conclusion.
5. Figure S26C shows the measured nonlinear Hall signal at 0 T (A-type AFM) and at ± 8 T (forced FM). The nonlinear Hall signal at ± 8 T is negligibly small compared to at 0 T. This is consistent with the fact that the FM state preserves inversion symmetry (Figs. S26A,B). Therefore, the nonlinear Hall effect is expected to vanish.
6. Figure S27 shows the additional data related to Fig. 3A. Figures S27, A and B show the nonlinear Hall voltage $V_{yxx}^{2\omega}$ and conductivity $\sigma_{yxx}^{2\omega}$ as a function of temperature. When temperature is lower than 15 K, $\sigma_{yxx}^{2\omega}$ is constant and independent of temperature.

7. Figure S32 extends the $\sigma_{yxx}^{2\omega} - \sigma_{xx}^{\omega}$ in Fig. 3A of the main text up to 30 K. We see that, in the regime corresponding to 15 – 30 K, $\sigma_{yxx}^{2\omega}$ quickly drop to zero. This is because the nonlinear Hall effect only exists in the AFM phase. Hence, near the Néel temperature T_N , the change of $\sigma_{yxx}^{2\omega}$ is strongly dominated by the fact that the AFM order quickly vanishes across T_N . There, in this regime, it is difficult to judge whether $\sigma_{yxx}^{2\omega}$ also varies with the scattering time τ .
8. Figure S33 presents an attempt to extract the τ dependence in the higher temperature regime. We provide independent measurements of the AFM order parameter in 6SL MnBi₂Te₄ as a function of temperature. The even-layered MnBi₂Te₄ is known to host the magnetoelectric (ME) coupling [65], which is proportional to the AFM order. The ME coupling can be detected by measuring the RMCD signal generated by a vertical displacement electric field $\frac{d\text{RMCD}}{dD_{\perp}}$. Figure S33B shows $\frac{d\text{RMCD}}{dD_{\perp}}$ vs. T , which indeed shows that the AFM order quickly decays to zero above 15 K. Going a step further, we can assume that the nonlinear Hall conductivity depends linearly on the AFM order L , $\sigma_{yxx}^{2\omega} = (C_0\tau^0 + C_1\tau^1 + \dots)L$. Under this assumption, one can isolate the scattering time τ dependence by taking the ratio by $\sigma_{yxx}^{2\omega}/L$, which in turn becomes $\sigma_{yxx}^{2\omega}/(\frac{d\text{RMCD}}{dD_{\perp}})$. Indeed, $\sigma_{yxx}^{2\omega}/(\frac{d\text{RMCD}}{dD_{\perp}})$ obtained from our data (Fig. S33C) is observed to stay roughly constant at all temperature below T_N . In this way, we managed to extract the dependence on σ_{xx}^{ω} for 15 – 20 K, which is roughly flat (Fig. S33D).
9. Figure S31 shows the scaling relationship by varying temperature in 5 devices. We have reproduced the τ independent result in 5 samples. Figure S30 shows the scaling relationship by varying temperature in the same device at different carrier density. We have also observed the τ independent result in both electron-doped and hole-doped regimes. Figure S31 shows the $\sigma_{yxx}^{2\omega}$ and σ_{xx}^{ω} for different samples (at a fixed temperature and fixed charge density). We see that although their σ_{xx}^{ω} varies by a factor of 5, their $\sigma_{yxx}^{2\omega}$ are all similar. Therefore, by varying sample, we have further substantiated $\sigma_{yxx}^{2\omega}$ being independent of τ conclusion. These new measurements significantly improve the statistics, which therefore strengthen the scaling relation.
10. Figure S28 shows the nonlinear Hall signals in a BP/4SL MnBi₂Te₄ heterostructure (Device-BM3). The nonlinear Hall signals as a function of carrier density shows the similar behavior as the main text.
11. Figure S35 shows the microwave rectification in another BP/MnBi₂Te₄ heterostructure (Device-BM23). The DC signals as a function of carrier density n_e and RF power P are similar as the nonlinear signals in the main text, which confirms that the DC rectification effect is also induce by quantum metric dipole. The DC rectification effect can work at very broad frequency range. The frequency range is limited by the antenna and source generator rather than the BP/MnBi₂Te₄ device.
12. Figure S36 shows that the microwave rectification signal flips sign as we reverse the AFM state. This additional data suggests that the observed rectification is an intrinsic response of the AFM. Further systematic measurements are needed to unambiguously exclude the extrinsic origins.
13. Figure S40C shows the nonlinear Hall signal of the BP/7SL MnBi₂Te₄ is negligibly small compared to BP/6SL MnBi₂Te₄. This is consistent with the intrinsic symmetry of the AFM state. The even-layered AFM state breaks inversion symmetry, whereas the odd-layered AFM state preserves inversion symmetry (see Figs. S40A,B). Therefore, the nonlinear Hall effect (which requires the breaking of inversion symmetry) is only expected in even-layered samples.
14. Figure S40C shows the nonlinear Hall signal of the BP/7SL MnBi₂Te₄ is negligibly small compared to BP/6SL MnBi₂Te₄. This is consistent with the intrinsic symmetry of the AFM state. The even-layered AFM state breaks inversion symmetry, whereas the odd-layered AFM state preserves inversion symmetry (see Figs. S40A,B). Therefore, the nonlinear Hall effect (which requires the breaking of inversion symmetry) is only expected in even-layered samples.
15. Figures S38B,D show the distribution of the quantum metric g_{xx} around the Fermi contour for the two AFM domains. The quantum metric dipole D_{metric} is a vector that connects from the side with smaller g_{xx} to that with larger g_{xx} . Therefore, from Figs. S38B,D, it is clear that the opposite AFM domains have opposite quantum metric dipole D_{metric} .
16. Figures S39A,B reproduce Fig. 4D of the main text, which show the kernel $(v_y g_{xx} - v_x g_{yx})$ by the color on Fermi contours. The quantum metric dipole is computed by integrating this kernel around

the Fermi contour $D_{\text{Metric}} = \int_{\text{FS}} (v_y g_{xx} - v_x g_{yx}) dl$. However, comparing this kernel between the C_{3z} -symmetric and C_{3z} -breaking cases is difficult, because the two contours clearly have different lengths (dl is different in the two cases). Therefore, we turn the integral into an integral of the polar angle θ , $D_{\text{Metric}} = \int_{\text{FS}} (v_y g_{xx} - v_x g_{yx}) \frac{dl}{d\theta} d\theta$. In this way, we can compare $(v_y g_{xx} - v_x g_{yx}) \frac{dl}{d\theta}$ consistently, because θ goes from -180° to $+180^\circ$ in both cases. Figure S39C presents $(v_y g_{xx} - v_x g_{yx}) \frac{dl}{d\theta}$ for the two cases, which shows that the C_{3z} -breaking case develops more negative signal. We take the difference shown Fig. S39D, which clearly shows that the integral becomes nonzero when C_{3z} is broken.

17. Figure S41 shows the DFT-calculated Fermi contours of the 6SL AFM MnBi_2Te_4 without and with SOC. The breaking of time-reversal $\mathcal{T}$ and space-inversion $\mathcal{P}$ is provided by AFM order (Mn spins). However, without SOC, the electrons at the Fermi level (p orbitals of Bi and Te) cannot feel the symmetry breaking provided by the AFM order. As an example, the AFM order makes the MnBi_2Te_4 bands asymmetric between $+k$ and $-k$. This can be seen from Fig. S41B, where the contours with SOC are clearly asymmetric between $+k_y$ and $-k_y$. By contrast, when we turn off the SOC (Fig. S41A), the asymmetry disappears, showing that without SOC the low-energy electrons cannot feel the symmetry breaking provided by the AFM.
18. Figure S42 shows the DFT computed band structure of the BP/2SL MnBi_2Te_4 /BP heterostructure. Due to the large lattice mismatch between the two materials, DFT needs a super cell. To build a 1×2 supercell, we had to artificially distort the BP unit cell by a lot ($\sim 13\%$ elongation along b). Even then, we could only do BP/2SL MnBi_2Te_4 /BP, because BP/6SL MnBi_2Te_4 /BP with the supercell was too heavy for DFT. Given the large distortion and the fact that it is only for BP/2SL MnBi_2Te_4 /BP, we do not use it to fit our model parameters of BP/6SL MnBi_2Te_4 /BP. We only showed this DFT result here for completeness.

IV. Theoretical studies

IV.1 An intuitive picture of quantum metric

Here, we give an intuitive picture to understand the quantum metric. The Bloch wavefunction $\psi_{\mathbf{k}}(\mathbf{r})$ is the product of two terms, $\psi_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}}$, where $e^{i\mathbf{k}\mathbf{r}}$ describes how the Bloch wave propagates from one unit cell to the next, and $u_{\mathbf{k}}$ captures the physics within a unit cell. The quantum metric (g) describes the difference between neighboring $u_{\mathbf{k}}$, $|u_{\mathbf{k}+\delta\mathbf{k}} - u_{\mathbf{k}}|^2$, which becomes the distance between two points on a Bloch sphere when we map $u_{\mathbf{k}}$ to a specific point on a Bloch sphere.

Specifically, let us consider a two-band system, which arises from two atomic positions α and β in the unit cell. Within a unit cell, an electron has probability to occupy both positions. So the two atomic positions serve as two orthogonal quantum bases $|\alpha\rangle$ and $|\beta\rangle$, whose linear combination constructs the $|u_{\mathbf{k}}\rangle$, $|u_{\mathbf{k}}\rangle = C_{\alpha}(\mathbf{k})|\alpha\rangle + C_{\beta}(\mathbf{k})|\beta\rangle$. $|u_{\mathbf{k}}\rangle$ can be represented by a specific point on the Bloch sphere (Fig. S37A). The simplest scenario is that the electrons of the conduction band (CB) only occupy α ($C_{\alpha}^{\text{CB}}(\mathbf{k}) \equiv 1$ and $C_{\beta}^{\text{CB}}(\mathbf{k}) \equiv 0$, so, $|u_{\mathbf{k}}^{\text{CB}}\rangle \equiv |\alpha\rangle$) and the electrons of the valence band (VB) only occupy β ($C_{\alpha}^{\text{VB}}(\mathbf{k}) \equiv 0$ and $C_{\beta}^{\text{VB}}(\mathbf{k}) \equiv 1$, so, $|u_{\mathbf{k}}^{\text{VB}}\rangle \equiv |\beta\rangle$). Taking the CB as an example, according to the definition above, its quantum metric is the difference of $|u_{\mathbf{k}}^{\text{CB}}\rangle$ at $\mathbf{k}$ and $\mathbf{k} + \delta\mathbf{k}$. Clearly, because $|u_{\mathbf{k}}^{\text{CB}}\rangle \equiv |\alpha\rangle$ at every $\mathbf{k}$ point, the quantum metric is zero. Equivalently, $|u_{\mathbf{k}}^{\text{CB}}\rangle$ at every $\mathbf{k}$ point is mapped onto the north pole of the Bloch sphere. So the distance on the Bloch sphere between $u_{\mathbf{k}}$ and $u_{\mathbf{k}+\delta\mathbf{k}}$ is zero (Fig. S37B).

A nontrivial case is realized when CB and VB go through a band inversion. In this case, each band becomes a $\mathbf{k}$ -dependent superposition of α and β , i.e., both $C_{\alpha}(\mathbf{k})$ and $C_{\beta}(\mathbf{k})$ depend strongly on $\mathbf{k}$. The change of $|u\rangle$ is dramatic near $\mathbf{k} = 0$ where band inversion occurs but weak at large $\mathbf{k}$. Equivalently, $u_{\mathbf{k}}$ and $u_{\mathbf{k}+\delta\mathbf{k}}$ are mapped onto different points on the Bloch sphere. So the distance on the Bloch sphere between $u_{\mathbf{k}}$ and $u_{\mathbf{k}+\delta\mathbf{k}}$ (i.e., the quantum metric) is nonzero (Fig. S37C). Interestingly, if we correspond α and β to Bi and Te, the nontrivial case above can help us to understand the large quantum metric in the low-energy electronic states of MnBi_2Te_4 .

IV.2 Symmetry analysis

IV.2.1 Symmetric nonlinear Hall conductivity as a function of E_z

In the main text and SM Sec. II.1 our data showed that nonlinear Hall signals $\sigma_{yxx}^{2\omega}$ are symmetric as a function of vertical electric field E_z . Here we explain this behavior by symmetry analysis. We start from the generic expression of our nonlinear Hall effect:

$$J_{yxx}^{2\omega} = \sigma_{yxx}^{2\omega} E_x^\omega E_x^\omega \quad (S4)$$

Now, in order to understand the E_z dependence of the nonlinear Hall effect, we can expand $\sigma_{yxx}^{2\omega}$ as a function of E_z^n (n is an integer.)

$$\sigma_{yxx}^{2\omega} = A_0 E_z^0 + A_1 E_z^1 + A_2 E_z^2 + A_3 E_z^3 + \dots \quad (S5)$$

Substituting Eq.(S5) into Eq.(S4), $J_{yxx}^{2\omega}$ can be wrote:

$$J_{yxx}^{2\omega} = A_0 E_z^0 (E_x^\omega)^2 + A_1 E_z^1 (E_x^\omega)^2 + A_2 E_z^2 (E_x^\omega)^2 + A_3 E_z^3 (E_x^\omega)^2 + \dots \quad (S6)$$

The $\mathcal{PT}$ symmetry dictates that all odd-power terms of E_z to vanish and only even-power terms of the E_z are allowed, i.e.,

$$J_{yxx}^{2\omega} = A_0 E_z^0 (E_x^\omega)^2 + A_2 E_z^2 (E_x^\omega)^2 + A_4 E_z^4 (E_x^\omega)^2 + A_6 E_z^6 (E_x^\omega)^2 + \dots \quad (S7)$$

$$\sigma_{yxx}^{2\omega} = A_0 E_z^0 + A_2 E_z^2 + A_4 E_z^4 + A_6 E_z^6 + \dots \quad (S8)$$

Equation (S8) clearly shows that the nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ is symmetric as a function of E_z which is consistent with our observation in Fig. 3D.

IV.2.2 Symmetry comparison of second-order Drude and quantum metric nonlinear Hall conductivities

The second-order Drude conductivity can be express as [22]:

$$\sigma_{\alpha\beta\gamma}^{\text{Drude}} = -\frac{e^3 \tau^2}{h^3} \sum_n \int_{\mathbf{k}} (\partial_{k_\alpha} \partial_{k_\beta} \partial_{k_\gamma} \varepsilon_n) f(\varepsilon_n) \quad (S9)$$

One can see that the second-order Drude conductivity requires the breaking both $\mathcal{P}$ and $\mathcal{T}$, because the momentum derivative ∂_{k_i} is odd under $\mathcal{P}$ and $\mathcal{T}$ and there are three of them in the expression. In addition, exchanging α and β changes the sequence of derivatives. Therefore, the second-order Drude conductivity $\sigma_{\alpha\beta\gamma}^{\text{Drude}}$ is symmetric, i.e., $\sigma_{\alpha\beta\gamma}^{\text{Drude}} = \sigma_{\beta\alpha\gamma}^{\text{Drude}}$.

The quantum metric nonlinear Hall conductivity can be express as [22]:

$$\sigma_{\alpha\beta\gamma}^{\text{Metric}} = 2e^3 \text{Re} \int_{\mathbf{k}} \left(\frac{v_\alpha \langle u | i \partial_{k_\beta} \nu \rangle \langle \nu | i \partial_{k_\gamma} u \rangle - v_\beta \langle u | i \partial_{k_\alpha} \nu \rangle \langle \nu | i \partial_{k_\gamma} u \rangle}{\varepsilon_u - \varepsilon_\nu} \right) \delta(\varepsilon_u - \varepsilon_F) \quad (S10)$$

Here, exchanging α and β flips the signs of the quantum metric nonlinear Hall conductivity. Therefore, the quantum metric nonlinear Hall conductivity is antisymmetric, i.e., $\sigma_{\alpha\beta\gamma}^{\text{Metric}} = -\sigma_{\beta\alpha\gamma}^{\text{Metric}}$. Therefore, we can use electrical sum frequency generation measurements to differentiate the quantum metric nonlinear Hall conductivity and second-order Drude conductivity.

IV.3 Quantum metric dipole contribution dominated nonlinear Hall signal

In the main text we already showed that the nonlinear Hall signals in BP/MnBi₂Te₄/BP heterostructure could be exactly calculated using the general expression. This general expression contains the quantum metric dipole D_{Metric} contribution plus additional inter-band contributions (AIC).

$$\sigma_{yxx}^{2\omega} = 2e^3 \sum_n \int_{\mathbf{k}} \frac{v_y^n g_{xx}^n - v_x^n g_{yx}^n}{\varepsilon_n - \varepsilon_{\bar{n}}} \delta(\varepsilon_n - \varepsilon_F) + \text{AIC} \quad (S11)$$

$$\text{AIC} = 2e^3 \sum_{n,m}^{\varepsilon_m \neq \varepsilon_n, \varepsilon_{\bar{n}}} \text{Re} \int_{\mathbf{k}} \left(\frac{v_y^n \langle u_n | i\partial_{k_x} u_m \rangle \langle u_m | i\partial_{k_x} u_n \rangle}{\varepsilon_n - \varepsilon_m} - \frac{v_x^n \langle u_n | i\partial_{k_y} u_m \rangle \langle u_m | i\partial_{k_y} u_n \rangle}{\varepsilon_n - \varepsilon_m} \right) \frac{\varepsilon_m - \varepsilon_{\bar{n}}}{\varepsilon_n - \varepsilon_{\bar{n}}} \delta(\varepsilon_n - \varepsilon_F) \quad (\text{S12})$$

Where the first term in Eq. (S11) is the quantum metric dipole contribution, the second term is the additional inter-band contributions. n and m are band indices. $\bar{n}$ is the partner band of band n . Because the energy difference term $\varepsilon_n - \varepsilon_{\bar{n}}$ is in the denominator, to maximize the D_{Metric} contribution, we always choose the band whose energy is the closest to n as the partner band.

Figure S43 shows the nonlinear Hall signals calculated with general expression (blue) and D_{Metric} contribution (red). Interestingly, the nonlinear Hall signals calculated with D_{Metric} contribution are approximate to the result calculated with general expression. Therefore, the nonlinear Hall signals observed in the BP/MnBi₂Te₄ heterostructure are dominated by the quantum metric dipole contribution.

IV.4 Low-energy model for BP/MnBi₂Te₄/BP heterostructure

In this section, we give an overview of the structure of our minimal model for the heterostructure composed by 6SL MnBi₂Te₄ (MBT) encapsulated by monolayers of black phosphorus (BP). The following Secs. IV.5 – IV.8 contain the details of the different parts (or blocks) of the model. The MBT-BP Hamiltonian we use to model the heterostructure in Fig. S44(a) consists of a 40-band model,

$$\hat{H}(\mathbf{k}) = \begin{pmatrix} \hat{h}_{\text{MBT}}(\mathbf{k}) & \hat{U}_t(\mathbf{k}) & \hat{U}_b(\mathbf{k}) \\ \hat{U}_t^\dagger(\mathbf{k}) & \hat{h}_{\text{BP},t}(\mathbf{k}) & 0 \\ \hat{U}_b^\dagger(\mathbf{k}) & 0 & \hat{h}_{\text{BP},b}(\mathbf{k}) \end{pmatrix}, \quad (\text{S13})$$

where $\mathbf{k} = (k_x, k_y)$. We now discuss the structure of each of the building blocks of Eq.(S13), and we refer to the corresponding later section about the details of each block.

IV.4.1 MnBi₂Te₄ septuple Hamiltonian

We start with the 24×24 block Hamiltonian $\hat{h}_{\text{MBT}}(\mathbf{k})$, which describes the set of six MBT SLs:

$$\hat{h}_{\text{MBT}}(\mathbf{k}) = \begin{pmatrix} \hat{h}_{\text{SL},1}(\mathbf{k}) & \hat{T}_0(\mathbf{k}) & 0 & 0 & 0 & 0 \\ \hat{T}_0^\dagger(\mathbf{k}) & \hat{h}_{\text{SL},2}(\mathbf{k}) & \hat{T}_0(\mathbf{k}) & 0 & 0 & 0 \\ 0 & \hat{T}_0^\dagger(\mathbf{k}) & \hat{h}_{\text{SL},3}(\mathbf{k}) & \hat{T}_0(\mathbf{k}) & 0 & 0 \\ 0 & 0 & \hat{T}_0^\dagger(\mathbf{k}) & \hat{h}_{\text{SL},4}(\mathbf{k}) & \hat{T}_0(\mathbf{k}) & 0 \\ 0 & 0 & 0 & \hat{T}_0^\dagger(\mathbf{k}) & \hat{h}_{\text{SL},5}(\mathbf{k}) & \hat{T}_0(\mathbf{k}) \\ 0 & 0 & 0 & 0 & \hat{T}_0^\dagger(\mathbf{k}) & \hat{h}_{\text{SL},6}(\mathbf{k}) \end{pmatrix}. \quad (\text{S14})$$

Here, $\hat{h}_{\text{SL},i}$ is the Hamiltonian of the SL i and $\hat{T}_0$ denotes the hopping between nearest neighbor SLs. Both $\hat{h}_{\text{SL},i}$ and $\hat{T}_0$ are obtained from the bulk MBT Hamiltonian by taking explicitly into account the system's periodicity along the stacking direction $\hat{z}$ [57]. This procedure is highlighted in Sec. IV.5 and yields a SL Hamiltonian that is independent of k_z [57],

$$\hat{h}_{\text{SL},i}(\mathbf{k}) = \hat{h}_{\text{N}}(\mathbf{k}) - \gamma_{af} \hat{h}_{\text{AFM},i}(\mathbf{k}). \quad (\text{S15})$$

The term $\hat{h}_{\text{N}}$ describes the normal state dispersion close to the Γ point and $\hat{h}_{\text{AFM},i}$ accounts for terms that arise from AFM order. Their explicit $\mathbf{k}$ -dependence is shown in Sec. IV.5. Both are 4-band $k \cdot p$ Hamiltonians in the basis

$$\left\{ \left| p_{z,\text{Bi}}^+, \uparrow \right\rangle, \left| p_{z,\text{Te}}^-, \downarrow \right\rangle, \left| p_{z,\text{Te}}^-, \uparrow \right\rangle, \left| p_{z,\text{Bi}}^+, \downarrow \right\rangle \right\}. \quad (\text{S16})$$

This basis is formed by the symmetric (+) superposition of the p_z orbitals of the two Bi layers within a given SL and by the anti-symmetric superposition of the p_z orbitals of the Te atoms in the top and bottom layers of a given SL (see Fig. S44). These orbitals carry the largest weight in the low-energy bands of MBT close to the Γ point [37]. The constant γ_{af} controls the strength of the magnetic order.

The inter-SL hopping matrix $\hat{T}_0(\mathbf{k})$, on the other hand, only contains contributions from the normal state since it is difficult to obtain from first principles the momentum-dependence of the exchange fields,

which are thus set to constants [57] (see also Sec. IV.5). Importantly, in this work, we go beyond the $k \cdot p$ expansion considered in Ref. [57] and include cubic terms in $\mathbf{k}$. This is needed in order to capture the 3-fold symmetry of the Fermi surface of the six MBT layers as it appears fully isotropic to order $\mathcal{O}(k^2)$. The explicit $\mathbf{k}$ -dependences of $\hat{h}_{\text{SL},i}$ and $\hat{T}_0$ are shown in Sec. IV.5.

IV.4.2 BP Hamiltonian

The 8×8 blocks $\hat{h}_{\text{BP},t}$ and $\hat{h}_{\text{BP},b}$ in the heterostructure Hamiltonian in Eq.(S13) contain the tight-binding Hamiltonian for BP monolayer located on the top and bottom of the heterostructure, respectively (see Fig. S44). Note that top and bottom layer Hamiltonians are related by inversion and the two blocks are thus identical: $\hat{h}_{\text{BP},t} = \hat{h}_{\text{BP},b}$. We discuss the explicit momentum dependence of $\hat{h}_{\text{BP},t}$ in Sec. IV.7.

IV.4.3 MnBi₂Te₄ and BP band hybridization coupling Hamiltonian

The coupling between MBT and BP can lead to two consequences. (1) It can lead to a structural strain on MBT, which we will model in the diagonal MBT Hamiltonian term $\hat{h}_{\text{MBT}}$ in Eq.(S13) (see IV.6 below). (2) It can also lead to a direction hybridization between the MBT and BP bands. Such an electronic band coupling between MBT and BP are described by the off-diagonal blocks in the heterostructure Hamiltonian in Eq.(S13). These couplings are obtained from the Fourier transform of the hopping integral between the p_z orbitals of the P atoms in the BP layers, and the p_z orbitals of Bi and Te atoms in the MBT SLs. Since the hopping amplitude decays exponentially as a function of the distance between the orbitals, the coupling is truncated beyond the nearest neighbors layers (see Fig. S44A). We have checked explicitly that this is an extremely accurate approximation. If desired it is straightforward to include couplings to further neighbor layers within our formalism. Within this approximation, $\hat{U}_t(\mathbf{k})$ denotes the coupling between the top BP monolayer and the top-most MBT SL, while $\hat{U}_b(\mathbf{k})$ accounts for the coupling between the bottom BP layer and the bottom-most MBT SL. The detailed derivation of the coupling terms and their explicit momentum dependence is discussed in Sec. IV.8.

IV.5 Modeling MnBi₂Te₄ septuple layers

In this section, we derive the MBT SL Hamiltonian $\hat{h}_{\text{SL},i}(\mathbf{k})$ in Eq.(S15) and the inter-SL hopping $\hat{T}_0(\mathbf{k})$. As mentioned previously, they are obtained from the bulk MBT Hamiltonian. This procedure is explained in Ref. [57], but for completeness we briefly describe it here as well. To be consistent with the notation introduced in Sec. IV.4, we denote $\mathbf{k} \equiv (k_x, k_y)$ and keep the k_z dependence explicit in the bulk Hamiltonian.

The first step in obtaining $\hat{h}_{\text{SL},i}$ and $\hat{T}_0$ is to write a $k \cdot p$ Hamiltonian for bulk MBT centered at the Γ point, which is where the low-energy bands are located [37]. The form of the bulk $k \cdot p$ Hamiltonian is determined by symmetry. In its paramagnetic phase, MBT belongs to the space group $R\bar{3}m$ (No. 166). Therefore, its Hamiltonian is invariant under all operations of the point group $\bar{3}m1'$, which is generated by time-reversal ($\mathcal{T}$), spatial inversion ($\mathcal{I}$), three-fold rotation around $\hat{z}$ (C_{3z}), and two-fold rotation around $\hat{x}$ (C_{2x}), where $\hat{x}$ is aligned with one of the in-plane primitive vectors of the underlying triangular lattice (see Fig. S44E). In the basis $\left\{ \left| p_{z,\text{Bi}}^+, \uparrow \right\rangle, \left| p_{z,\text{Te}}^-, \downarrow \right\rangle, \left| p_{z,\text{Te}}^-, \uparrow \right\rangle, \left| p_{z,\text{Bi}}^+, \downarrow \right\rangle \right\}$ these symmetry operations are represented by

$$\hat{U}_{\mathcal{T}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \hat{U}_{\mathcal{I}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \hat{U}_{3z} = \begin{pmatrix} e^{-i\frac{\pi}{3}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\pi}{3}} & 0 & 0 \\ 0 & 0 & e^{-i\frac{\pi}{3}} & 0 \\ 0 & 0 & 0 & e^{i\frac{\pi}{3}} \end{pmatrix}, \hat{U}_{2x} = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}. \quad (\text{S17})$$

We derive the generic form of the Hamiltonian up to order k^3 that is consistent with these symmetries using the Python toolkit QSymm [66]:

$$\hat{H}_0(\mathbf{k}, k_z) = E_0(\mathbf{k}, k_z) + \begin{pmatrix} M(\mathbf{k}, k_z) & iA_2k_- & iA_1k_z & 0 \\ -iA_2k_+ & -M(\mathbf{k}, k_z) & 0 & -iA_1k_z \\ -iA_1k_z & 0 & -M(\mathbf{k}, k_z) & iA_2k_- \\ 0 & iA_1k_z & -iA_2k_+ & M(\mathbf{k}, k_z) \end{pmatrix} + \begin{pmatrix} 0 & A_3k_zk_+^2 & -P(\mathbf{k}) & 0 \\ A_3k_zk_-^2 & 0 & 0 & P(\mathbf{k}) \\ -P(\mathbf{k}) & 0 & 0 & A_3k_zk_+^2 \\ 0 & P(\mathbf{k}) & A_3k_zk_-^2 & 0 \end{pmatrix}, \quad (\text{S18})$$

where we define $k_{\pm} \equiv k_x \pm ik_y$ and

$$E_0(\mathbf{k}, k_z) = C + D_1k_z^2 + D_2(k_x^2 + k_y^2), \quad (\text{S19})$$

$$M(\mathbf{k}, k_z) = M_0 + B_1k_z^2 + B_2(k_x^2 + k_y^2). \quad (\text{S20})$$

$$P(\mathbf{k}) = \beta_1k_x(k_x^2 - 3k_y^2). \quad (\text{S21})$$

The constants $A_1, A_2, A_3, B_1, B_2, C, D_1, D_2, M_0$, and β_1 are obtained by fitting the $k \cdot p$ model to a DFT band structure of MBT. We discuss the numerical values of these parameters in Sec. IV.9. The first two terms in the right-hand-side of Eq.(S18) are equivalent to the bulk MBT Hamiltonian shown in Refs. [37, 57] after a unitary transformation using the matrix $\tilde{U} = \text{diag}(i, 1, 1, -i)$. The third term on the right-hand-side of Eq.(S18) is of order k^3 and is called the *warping term*.

In bulk MBT the SLs are stacked along the $\hat{z}$ direction with lattice constant $c_0 = 40.91 \text{ \AA}$ [57]. Therefore, by replacing $k_z \rightarrow \frac{1}{c_0} \sin(c_0k_z)$ and $k_z^2 \rightarrow \frac{2}{c_0^2} [1 - \cos(c_0k_z)]$ in Eq.(S18), we can reinterpret this compactified model as a one-dimensional (1D) hopping model along $\hat{z}$,

$$\hat{H}_0(\mathbf{k}, k_z) = \hat{h}_N(\mathbf{k}) + \cos(c_0k_z) [\hat{T}_0(\mathbf{k}) + \hat{T}_0^\dagger(\mathbf{k})] + i \sin(c_0k_z) [\hat{T}_0^\dagger(\mathbf{k}) - \hat{T}_0(\mathbf{k})]. \quad (\text{S22})$$

In this scenario, terms that are independent of k_z give the normal state contribution to a given SL

$$\hat{h}_N(\mathbf{k}) = \varepsilon_0(\mathbf{k}) + \begin{pmatrix} m(\mathbf{k}) & i\alpha k_- & -P(\mathbf{k}) & 0 \\ -i\alpha k_+ & -m(\mathbf{k}) & 0 & P(\mathbf{k}) \\ -P(\mathbf{k}) & 0 & -m(\mathbf{k}) & i\alpha k_- \\ 0 & P(\mathbf{k}) & -i\alpha k_+ & m(\mathbf{k}) \end{pmatrix}. \quad (\text{S23})$$

Here, $P(\mathbf{k})$ is defined in Eq.(S21) and

$$\varepsilon_0(\mathbf{k}) = \gamma_0 + \gamma(k_x^2 + k_y^2), \quad (\text{S24})$$

$$m(\mathbf{k}) = m_0 + \beta_0(k_x^2 + k_y^2), \quad (\text{S25})$$

with $\gamma_0 = C + \frac{2D_1}{c_0^2}$, $\gamma = D_2$, $m_0 = M_0 + \frac{2B_1}{c_0^2}$, $\beta_0 = B_2$, and $\alpha = A_2$. The remaining terms that are proportional to $\sin(ck_z)$ and $\cos(ck_z)$ define the hopping between two nearest-neighbour SLs,

$$\hat{T}_0 = \begin{pmatrix} t_1 & i\frac{\alpha_3}{2}k_+^2 & -\lambda & 0 \\ i\frac{\alpha_3}{2}k_-^2 & t_2 & 0 & \lambda \\ \lambda & 0 & t_2 & i\frac{\alpha_3}{2}k_+^2 \\ 0 & -\lambda & i\frac{\alpha_3}{2}k_-^2 & t_1 \end{pmatrix} \quad (\text{S26})$$

with $t_1 = -\frac{D_1+B_1}{c_0^2}$, $t_2 = -\frac{D_1-B_1}{c_0^2}$, $\lambda = \frac{A_1}{2c_0}$, and $\alpha_3 = \frac{A_3}{c_0}$. We note that other cubic terms are allowed by symmetry in Eq.(S18), but we verified explicitly that the leading terms promoting the formation of a threefold symmetric Fermi surface are those proportional to β_1 and α_3 that were kept in Eqs.(S23) and (S26).

At low temperatures MBT develops A-type AFM order, where the Mn local moments order ferromagnetically along $\pm\hat{z}$ within each SL, while moments in nearest-neighbor SLs point antiparallel (see Fig. S44A). To find the magnetic contribution to the SL Hamiltonian, we repeat the previous steps starting from a FM bulk MBT, which has point group $\bar{3}m'$. This point group is generated by the elements C_{3z} , inversion $\mathcal{I}$ and the combination of time-reversal and two-fold rotation $\mathcal{TC}_{2x}$. A generic Hamiltonian up to order $\mathcal{O}(k^0)$ that obeys these symmetry constraints reads

$$\hat{H}_{\text{mag}} = \begin{pmatrix} m_1 & 0 & 0 & 0 \\ 0 & -m_2 & 0 & 0 \\ 0 & 0 & m_2 & 0 \\ 0 & 0 & 0 & -m_1 \end{pmatrix}, \quad (\text{S27})$$

Here we kept only the constant terms in the $k \cdot p$ expansion because the $\mathbf{k}$ dependence of the exchange fields is difficult to reliably calculate from first principles [57]. We note that Eq.(S27) acquires a global minus sign if the FM order points towards $-\hat{z}$. Since consecutive nearest-neighbors SLs are ordered AFM, the magnetic contribution for the Hamiltonian of the SL i defined in Eq.(S15) is simply

$$\hat{h}_{\text{AFM},i} = (-1)^i \hat{H}_{\text{mag}}. \quad (\text{S28})$$

As we approximate $\hat{H}_{\text{mag}}$ to be independent of $\mathbf{k}$, there is no magnetic contribution to the interlayer hopping $\hat{T}_0$.

IV.6 Effects of strain on low-energy model of MnBi_2Te_4

Encapsulating the MBT SLs by BP monolayer introduces strain due to the large mismatch between their lattice constants. The effects of strain can be straightforwardly incorporated in our model using symmetry arguments similar to those in Sec. IV.5.

In this section, we focus on the effects of strain on the MBT part of the heterostructure. We show here that strain modifies the SL Hamiltonian and the inter-SL hopping according to

$$\hat{h}_{\text{SL},i}(\mathbf{k}) = \hat{h}_{\text{N}}(\mathbf{k}) + \gamma_s \hat{h}_{\text{N},s}(\mathbf{k}) - \gamma_{af} \left[\hat{h}_{\text{AFM},i}(\mathbf{k}) + \gamma_s \hat{h}_{\text{AFM},s}(\mathbf{k}) \right]. \quad (\text{S29})$$

$$\hat{T}(\mathbf{k}) = \hat{T}_0(\mathbf{k}) + \gamma_s \hat{T}_s(\mathbf{k}). \quad (\text{S30})$$

Here, $\hat{h}_{\text{N}}$, $\hat{h}_{\text{AFM},i}$ and $\hat{T}_0$ are the same as defined in Eq.(S23), Eq.(S28) and Eq.(S26), respectively. The constant $\gamma_s > 0$ tunes the strength of strain. The new terms $\hat{h}_{\text{N},s}$, $\hat{h}_{\text{AFM},s}$ and $\hat{T}_s$ are identical for all SLs and are obtained from compactifying the $k \cdot p$ MBT bulk Hamiltonian along $\hat{z}$ in the presence of strain following the same steps shown in Sec. IV.5.

We focus on the case of uniaxial strain along $\hat{x} = [100]$, which breaks the threefold rotation around $\hat{z} = [001]$ (C_{3z}) as well as the twofold rotations around $[010]$ and $[110]$ originally present in the unstrained $R\bar{3}m$ space group of the material. Note that this strain preserves the two-fold rotation around $[100]$. We obtain a generic term consistent with these symmetries using Qsymm [66]:

$$\hat{H}_s(\mathbf{k}) = \begin{pmatrix} s_1 k_x^2 + s_2 k_y k_z & i s_5 k_x + s_6 k_z & s_7 k_x + i s_8 k_y & 0 \\ -i s_5 k_x + s_6 k_z & s_3 k_x^2 + s_4 k_y k_z & 0 & -s_7 k_x - i s_8 k_y \\ s_7 k_x - i s_8 k_y & 0 & s_3 k_x^2 + s_4 k_y k_z & i s_5 k_x + s_6 k_z \\ 0 & -s_7 k_x + i s_8 k_y & -i s_5 k_x + s_6 k_z & s_1 k_x^2 + s_2 k_y k_z \end{pmatrix}. \quad (\text{S31})$$

This term is added to the bulk MBT Hamiltonian in Eq.(S18). Similarly, the magnetic part of the Hamiltonian (Eq.(S27)) gains a term of the form

$$\hat{H}_{\text{mag},s} = s_9 \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}. \quad (\text{S32})$$

For simplicity, we carried the $k \cdot p$ expansion up to k^2 order in Eqs.(S31) and (S32). The quantities a_j are real parameters. Replacing $k_z \rightarrow \frac{1}{c_0} \sin c_0 k_z$ in Eqs.(S31) and (S32) and separating the terms proportional to $\sin(c_0 k_z)$ from those independent of k_z as in Eq.(S22), we obtain the strain correction to the normal state Hamiltonian of the SL,

$$\hat{h}_{\text{N},s}(\mathbf{k}) = \begin{pmatrix} s_1 k_x^2 & i s_5 k_x & s_7 k_x + i s_8 k_y & 0 \\ -i s_5 k_x & s_3 k_x^2 & 0 & -s_7 k_x - i s_8 k_y \\ s_7 k_x - i s_8 k_y & 0 & s_3 k_x^2 & i s_5 k_x \\ 0 & -s_7 k_x + i s_8 k_y & -i s_5 k_x & s_1 k_x^2 \end{pmatrix}. \quad (\text{S33})$$

The strain contribution to the magnetic part of the SL Hamiltonian $\hat{h}_{\text{AFM},s}(\mathbf{k})$ is identical to $\hat{H}_{\text{mag},s}$ defined in Eq.(S32), and the modification to the inter-SL hopping reads

$$\hat{T}_s(\mathbf{k}) = \frac{i}{2c_0} \begin{pmatrix} s_2 k_y & s_6 & 0 & 0 \\ s_6 & s_4 k_y & 0 & 0 \\ 0 & 0 & s_4 k_y & s_6 \\ 0 & 0 & s_6 & s_2 k_y \end{pmatrix}. \quad (\text{S34})$$

The low-energy bands for a single MBT SL modeled by Eq.(S29) and for a set of six SL described by Eq.(S14) are shown in Fig.S45. In the paramagnetic state, the SL is a topological insulator with protected Dirac cones in the top and bottom surfaces normal to $\hat{z}$. As a result of the small thickness of the SL, these surface states hybridize and a energy gap develops at the Γ point, as shown in Fig. S45A. When multiple SLs are stacked along $\hat{z}$, the hybridization of the surface states suppressed and the stacking of six SLs is already enough to recover the surface Dirac cone at Γ . At the onset of the AFM order, however, time-reversal symmetry is lost, which culminates in the emergence of a gap in the surface states.

The magnetic order preserves C_{3z} , which is reflected in a threefold symmetric Fermi surface of six SL system shown in Fig. S45D, as long as warping terms discussed in Sec. IV.5 are included in the low energy model. This 3-fold symmetry of the Fermi surface is further lowered in the presence of uniaxial strain as highlighted in Fig. S45E and Fig. S45F.

IV.7 Modeling the BP monolayers

In this section we derive the minimal model for the BP monolayer, $\hat{h}_{\text{BP},t}$ and $\hat{h}_{\text{BP},b}$ in Eq.(S13). Since the BP layers on the top and bottom of the heterostructure are identical, $\hat{h}_{\text{BP},t} = \hat{h}_{\text{BP},b}$, we hereafter drop the layer subindex.

Following Refs. [58,67], the low-energy features of the material is captured by a tight-binding model where the hopping processes involve only the p_z orbitals of the P atoms. The BP layer is corrugated rectangular lattice with primitive vectors $\mathbf{a}_1^{(\text{BP})} = a\hat{x}$ and $\mathbf{a}_2^{(\text{BP})} = b\hat{y}$, with lattice constants $a = 4.43 \text{ \AA}$, $b = 3.27 \text{ \AA}$. There are four nonequivalent P atoms per unit cell, and its low-energy tight-binding model is thus a 4-band Hamiltonian,

$$\hat{h}_{\text{BP},\sigma}(\mathbf{k}) = \begin{pmatrix} 0 & f_1(k_x, k_y) & 0 & f_2(k_x, k_y) \\ f_1^*(k_x, k_y) & 0 & f_2(k_x, k_y) & 0 \\ 0 & f_2^*(k_x, k_y) & 0 & f_1(k_x, k_y) \\ f_2^*(k_x, k_y) & 0 & f_1^*(k_x, k_y) & 0 \end{pmatrix}. \quad (\text{S35})$$

This matrix is written in the basis $\{|p_{z,P_1}\rangle, |p_{z,P_2}\rangle, |p_{z,P_3}\rangle, |p_{z,P_4}\rangle\}$, which is ordered according to the fractional positions $\boldsymbol{\rho}_i$ of the phosphorus atoms in the unit cell (see Table S2). Since the hopping processes are spin independent, the Hamiltonian in Eq.(S35) is identical for spin up ($\sigma = \uparrow$) and spin down ($\sigma = \downarrow$). The full BP Hamiltonian that enters in Eq.(S13) is block-diagonal,

$$\hat{h}_{\text{BP},t}(\mathbf{k}) = \hat{h}_{\text{BP},b}(\mathbf{k}) = \begin{pmatrix} \hat{h}_{\text{BP},\uparrow}(\mathbf{k}) & 0 \\ 0 & \hat{h}_{\text{BP},\downarrow}(\mathbf{k}) \end{pmatrix}. \quad (\text{S36})$$

In Eq.(S35) we kept only two distinct hopping parameters $\tilde{t}_1$ and $\tilde{t}_2$ defined in the functions

$$f_1(k_x, k_y) = 2\tilde{t}_1 e^{i\frac{ak_x}{2\sqrt{3}}} \cos\left(\frac{bk_y}{2}\right), \quad (\text{S37})$$

$$f_2(k_x, k_y) = \tilde{t}_2 e^{-i\frac{ak_x}{\sqrt{3}}}, \quad (\text{S38})$$

and thus our BP model is a simplified version of the tight-binding Hamiltonian considered in Ref. [58]. This simplification based on the fact that our minimal model for MBT, and by construction also the model for the restructure, is defined around the Γ point ($\mathbf{k} = 0$). In this region of the Brillouin zone, $\tilde{t}_1$ and $\tilde{t}_2$ are the dominant contributions the band gap and energy dispersion of BP [58]. Note that the Γ point in BP Brillouin zone coincides with the Γ point in MBT Brillouin zone since we assume that the origin of these two lattices are aligned.

To account for the experimentally observed electronic gap of the BP monolayers in our system, we choose values for the hopping amplitudes that are slightly different from Ref. [58]. The BP band gap at Γ is given by

$$\Delta = 4 \left| \tilde{t}_1 \right| \left| 1 - \frac{1}{2} \left| \frac{\tilde{t}_2}{\tilde{t}_1} \right| \right|, \quad (\text{S39})$$

and we choose $\tilde{t}_1 = -1.7575$ eV and $\tilde{t}_2 = 3.665$ eV to obtain $\Delta = 0.3$ eV in agreement with experimental observation. The energy bands of our minimal model for BP [Eq.(S35)] is shown in Fig. S46.

IV.8 Modeling the band hybridization between MnBi_2Te_4 and BP

In this section we derive the expressions for the band hybridization between the MBT SLs and the BP monolayer, which are denoted as $\hat{U}_t(\mathbf{k})$ and $\hat{U}_b(\mathbf{k})$ in Eq. (S13). These off-diagonal blocks results from the electronic hopping between the p_z orbitals of the BP and MBT layers. Due to the exponential decrease of the electronic coupling with distance between orbitals, we restrict the coupling to nearest-neighbor layers. The BP monolayer at the bottom is coupled only to the lowest MBT layer (SL index $l = 6$ in Fig. S44A) and the BP monolayer at the top is only coupled to the top SL (SL index $l = 1$).

IV.8.1 Construction of the MnBi_2Te_4 and BP band coupling Hamiltonian

Let us first consider the hopping involving the bottom BP layer. In the site basis it reads

$$\hat{H}_{\text{int}}^{(b)} = \sum_{i \in \text{SL}} \sum_{j \in \text{BP}} \sum_{\sigma} \left(\hat{\psi}_{i,\sigma}^{\dagger} \hat{t}_{ij}^{(b)} \hat{\phi}_{j,\sigma} + \text{h.c.} \right). \quad (\text{S40})$$

Here, the sum over i runs over the unit cells of the bottommost MBT SL ($l = 6$) with position $\mathbf{R}_i \equiv n_1 \mathbf{a}_1^{(\text{MBT})} + n_2 \mathbf{a}_2^{(\text{MBT})}$ (n_1, n_2 integers). The sum over j runs over the unit cells of the bottom BP layer, with $\mathbf{R}_j = m_1 \mathbf{a}_1^{(\text{BP})} + m_2 \mathbf{a}_2^{(\text{BP})}$ (m_1, m_2 integers) and $\mathbf{a}_i^{(\text{MBT})} \neq \mathbf{a}_j^{(\text{BP})}$. Besides, $\hat{\phi}_{j,\sigma}^{\dagger}$ is a 4-dimensional spinor whose ν th component corresponds to creating an electron with spin σ in the p_z orbital of the P atom located at position $\boldsymbol{\rho}_{\nu}$ in the j -th unit cell of the BP lattice (see Table S2). Similarly, $\hat{\psi}_{i,\sigma}^{\dagger}$ ($l = 1, 2$) is a 7-dimensional spinor whose μ th component creates an electron with spin σ on Mn (if $\mu = 7$), Bi (if $\mu = 5, 6$) or Te (if $\mu = 1, 2, 3, 4$) atoms that are located at position $\boldsymbol{\tau}_{\mu}$ in the i -th unit cell of the bottom most MBT SL.

The quantity $\hat{t}_{ij}^{(b)}$ is a 7×4 dimensional hopping matrix, whose matrix elements

$$\left(\hat{t}_{ij}^{(b)} \right)_{\mu,\nu} \equiv t_b(\mathbf{R}_i - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu}) \quad (\text{S41})$$

depend only on the relative distance between the sites involved in the hopping process. It is important to note that while $\mathbf{R}_i$ are two-dimensional vectors denoting the unit cell position in the layers, the basis vectors $\boldsymbol{\tau}_{\mu}$ and $\boldsymbol{\rho}_{\mu}$ are three-dimensional. We can decompose the basis vectors as $\boldsymbol{\tau}_{\mu} = \boldsymbol{\tau}_{\mu}^{\parallel} + \boldsymbol{\tau}_{\mu}^{\perp}$ and $\boldsymbol{\rho}_{\nu} = \boldsymbol{\rho}_{\nu}^{\parallel} + \boldsymbol{\rho}_{\nu}^{\perp}$, where the subscript $\parallel$ denotes the component in the layer and $\perp$ denotes the component perpendicular to the layer, i.e., along the stacking direction $\hat{\mathbf{z}}$. In this work we calculate the hopping elements t_b using the Slater-Koster approach (see Sec. IV.8.2). The Fourier transform of Eq.(S40) gives $\hat{U}_b(\mathbf{k})$. This procedure has subtleties because of the lattice mismatch between BP and MBT and is thoroughly presented in the remainder of this section.

To go from site to momentum space, we use convention to include the basis atom positions in the Fourier transformation

$$\psi_{i\mu,\sigma}^{\dagger} = \frac{1}{\sqrt{N_{\text{SL}}}} \sum_{\mathbf{k} \in \text{BZ}_{\text{SL}}} e^{-i\mathbf{k} \cdot (\mathbf{R}_i + \boldsymbol{\tau}_{\mu}^{\parallel})} \psi_{\mathbf{k}\mu,\sigma}^{\dagger}, \quad (\text{S42})$$

$$\phi_{j\nu,\sigma}^{\dagger} = \frac{1}{\sqrt{N_{\text{BP}}}} \sum_{\mathbf{q} \in \text{BZ}_{\text{BP}}} e^{-i\mathbf{q} \cdot (\mathbf{R}_j + \boldsymbol{\rho}_{\nu}^{\parallel})} \phi_{\mathbf{q}\nu,\sigma}^{\dagger}. \quad (\text{S43})$$

Here, N_{SL} (N_{BP}) corresponds to the number of unit cells in the MBT SL (BP layer), and $\boldsymbol{\tau}_{\mu}^{\parallel}$ and $\boldsymbol{\rho}_{\nu}^{\parallel}$ are the components of $\boldsymbol{\tau}_{\mu}$ and $\boldsymbol{\rho}_{\nu}$ normal to the stacking direction $\hat{\mathbf{z}}$.

Substituting Eqs.(S42) and (S43) into Eq.(S40), we find

$$\hat{H}_{\text{int}}^{(b)} = \sum_{\substack{\mathbf{k} \in \text{BZ}_{\text{SL}} \\ \mathbf{q} \in \text{BZ}_{\text{BP}}}} \sum_{\sigma} \left[\frac{e^{-i(\mathbf{k} \cdot \boldsymbol{\tau}_{\mu}^{\parallel} - \mathbf{q} \cdot \boldsymbol{\rho}_{\nu}^{\parallel})}}{\sqrt{N_{\text{SL}} N_{\text{BP}}}} \psi_{\mathbf{k}\mu, \sigma}^{\dagger} \left\{ \sum_{\substack{i \in \text{SL}_6 \\ j \in \text{BP}}} t_b(\mathbf{R}_i - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu}) e^{-i\mathbf{k} \cdot \mathbf{R}_i} e^{i\mathbf{q} \cdot \mathbf{R}_j} \right\} \phi_{\mathbf{q}\nu, \sigma} + \text{h.c.} \right]. \quad (\text{S44})$$

The $\mathbf{k}$ and $\mathbf{q}$ sums run over the Brillouin zones of the MBT SL and BP layer, respectively. To deal with the remaining sums over i and j , we note that

$$\begin{aligned} \sum_{i \in \text{SL}_6} t_b(\mathbf{R}_i - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu}) e^{-i\mathbf{k} \cdot \mathbf{R}_i} &= \sum_{(n_1, n_2) \in \mathbb{Z}^2} t_b \left(n_1 \mathbf{a}_1^{(\text{MBT})} + n_2 \mathbf{a}_2^{(\text{MBT})} - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu} \right) \\ &\quad \times e^{-i\mathbf{k} \cdot (n_1 \mathbf{a}_1^{(\text{MBT})} + n_2 \mathbf{a}_2^{(\text{MBT})})} \\ &= \sum_{(n_1, n_2) \in \mathbb{Z}^2} \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \delta(x_1 - n_1) \delta(x_2 - n_2) t_b \left(x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})} - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu} \right) \\ &\quad \times e^{-i\mathbf{k} \cdot (x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})})}, \end{aligned} \quad (\text{S45})$$

where x_1 and x_2 are real dimensionless variables. Using the Poisson summation formula, we can rewrite

$$\sum_{n=-\infty}^{\infty} \delta(x - n) = \sum_{m=-\infty}^{\infty} e^{-i2\pi m x}, \quad (\text{S46})$$

and then

$$\begin{aligned} \sum_{(n_1, n_2) \in \mathbb{Z}^2} \delta(x_1 - n_1) \delta(x_2 - n_2) &= \sum_{(m_1, m_2) \in \mathbb{Z}^2} e^{-i2\pi(m_1 x_1 + m_2 x_2)} \\ &= \sum_{(m_1, m_2) \in \mathbb{Z}^2} e^{-i(x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})}) \cdot (m_1 \mathbf{b}_1^{(\text{MBT})} + m_2 \mathbf{b}_2^{(\text{MBT})})} \\ &= \sum_i e^{-i(x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})}) \cdot \mathbf{G}_i^{(\text{MBT})}}. \end{aligned} \quad (\text{S47})$$

Here, $\mathbf{G}_i^{(\text{MBT})} = i_1 \mathbf{b}_1^{(\text{MBT})} + i_2 \mathbf{b}_2^{(\text{MBT})}$ denotes the unit cell position of the reciprocal MBT SL lattice, and the sum over m runs over the reciprocal MBT SL lattice unit cells.

Substituting Eq.(S47) into Eq.(S45), we obtain

$$\begin{aligned} \sum_{i \in \text{SL}_6} t_b(\mathbf{R}_i - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu}) e^{-i\mathbf{k} \cdot \mathbf{R}_i} &= \sum_i \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 t_b \left(x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})} - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu} \right) \\ &\quad \times e^{-i(\mathbf{k} + \mathbf{G}_i^{(\text{MBT})}) \cdot (x_1 \mathbf{a}_1^{(\text{MBT})} + x_2 \mathbf{a}_2^{(\text{MBT})})} \end{aligned} \quad (\text{S48})$$

Similarly

$$\begin{aligned} \sum_{j \in \text{BP}} t_b(\mathbf{R}_i - \mathbf{R}_j + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu}) e^{i\mathbf{q} \cdot \mathbf{R}_j} &= \sum_j \int_{-\infty}^{\infty} dx_3 \int_{-\infty}^{\infty} dx_4 t_b \left(\mathbf{R}_i - x_3 \mathbf{a}_1^{(\text{BP})} - x_4 \mathbf{a}_2^{(\text{BP})} + \boldsymbol{\tau}_{\mu} - \boldsymbol{\rho}_{\nu} \right) \\ &\quad \times e^{i(\mathbf{q} - \mathbf{G}_j^{(\text{BP})}) \cdot (x_3 \mathbf{a}_1^{(\text{BP})} + x_4 \mathbf{a}_2^{(\text{BP})})} \end{aligned} \quad (\text{S49})$$

where again x_3 and x_4 are real dimensionless variables and the j sum runs over the unit cells of the reciprocal BP lattice $\mathbf{G}_j^{(\text{BP})} = j_1 \mathbf{b}_1^{(\text{BP})} + j_2 \mathbf{b}_2^{(\text{BP})}$.

Combining Eqs. (S48) and (S49) and substituting them into Eq. (S44) yields

$$\hat{H}_{\text{int}}^{(b)} = \frac{1}{\sqrt{N_{\text{SL}}N_{\text{BP}}}} \frac{1}{\Omega_{\text{SL}}\Omega_{\text{BP}}} \sum_{\mu,\nu,\sigma} \sum_{\mathbf{k},i} \sum_{\mathbf{q},j} \left[\int d^2r \int d^2r' t_b(\mathbf{r} - \mathbf{r}' + \boldsymbol{\tau}_\mu - \boldsymbol{\rho}_\nu) \right. \\ \left. \times e^{i(-\mathbf{k}\cdot\boldsymbol{\tau}_\mu^\parallel + \mathbf{q}\cdot\boldsymbol{\rho}_\nu^\parallel)} e^{-i(\mathbf{k} + \mathbf{G}_i^{(\text{MBT})})\cdot\mathbf{r}} e^{i(\mathbf{q} - \mathbf{G}_j^{(\text{BP})})\cdot\mathbf{r}'} \psi_{\mathbf{k}\mu,\sigma}^\dagger \phi_{\mathbf{q}\nu,\sigma} + \text{h.c.} \right], \quad (\text{S50})$$

where we applied the change of variables $\mathbf{r} = x_1\mathbf{a}_1^{(\text{MBT})} + x_2\mathbf{a}_2^{(\text{MBT})}$ and $\mathbf{r}' = x_3\mathbf{a}_1^{(\text{BP})} + x_4\mathbf{a}_2^{(\text{BP})}$. Recall that here the sum over i (j) runs over the reciprocal unit cells of MBT SL (BP), while the sum over $\mathbf{k}$ ($\mathbf{q}$) are restricted to the first BZ of MBT SL (BP). Besides, Ω_{SL} (Ω_{BP}) denotes the area of the unit cell of the SL (BP) Bravais lattices.

We focus now on the arguments of the hopping amplitude t_b . While $\mathbf{r}$ and $\mathbf{r}'$ are 2D vectors, $\boldsymbol{\tau}_\mu$ and $\boldsymbol{\rho}_\nu$ are in general 3D, as mentioned before. Defining

$$s_{\mu\nu} \equiv (\boldsymbol{\tau}_\mu - \boldsymbol{\rho}_\nu) \cdot \hat{\mathbf{z}} = \tau_\mu^\perp - \rho_\nu^\perp, \quad (\text{S51})$$

we can rewrite $\boldsymbol{\tau}_\mu - \boldsymbol{\rho}_\nu = \boldsymbol{\tau}_\mu^\parallel - \boldsymbol{\rho}_\nu^\parallel + s_{\mu\nu}\hat{\mathbf{z}}$. Performing the change of variables $\mathbf{y} = \mathbf{r} - \mathbf{r}' + \boldsymbol{\tau}_\mu^\parallel - \boldsymbol{\rho}_\nu^\parallel$, we obtain

$$\hat{H}_{\text{int}}^{(b)} = \sqrt{\frac{N_{\text{SL}}}{N_{\text{BP}}}} \frac{1}{\Omega_{\text{BP}}} \sum_{\mu,\nu,\sigma} \sum_{i,\mathbf{k}} \sum_{j,\mathbf{q}} \left[t_{\mu\nu}(\mathbf{k} + \mathbf{G}_i^{(\text{MBT})}) e^{i(\mathbf{G}_i^{(\text{MBT})}\cdot\boldsymbol{\tau}_\mu^\parallel + \mathbf{G}_j^{(\text{BP})}\cdot\boldsymbol{\rho}_\nu^\parallel)} \right. \\ \left. \times \psi_{\mathbf{k}\mu,\sigma}^\dagger \phi_{\mathbf{q}\nu,\sigma} \delta_{\mathbf{q},\mathbf{k} + \mathbf{G}_i^{(\text{MBT})} + \mathbf{G}_j^{(\text{BP})}} + \text{h.c.} \right]. \quad (\text{S52})$$

Here,

$$t_{\mu\nu}^{(b)}(\mathbf{q}) = \int d^2y t_b(\mathbf{y} + s_{\mu\nu}\hat{\mathbf{z}}) e^{-i\mathbf{q}\cdot\mathbf{y}}, \quad (\text{S53})$$

is the Fourier transform of the hopping amplitude, which we explicitly calculate in Sec. IV.8.2, and we used

$$\int d^2r e^{i\mathbf{r}\cdot(-\mathbf{k} - \mathbf{G}_i^{(\text{MBT})} - \mathbf{G}_j^{(\text{BP})} + \mathbf{q})} = N_{\text{SL}}\Omega_{\text{SL}} \delta_{\mathbf{q} - \mathbf{k}, \mathbf{G}_i^{(\text{MBT})} + \mathbf{G}_j^{(\text{BP})}}. \quad (\text{S54})$$

Since we here focus on a minimal model for the MBT-BP heterostructure that is valid around the Γ point and $t_{\mu\nu}^{(b)}(\mathbf{k} + \mathbf{G}_i)$ decays rapidly with $|\mathbf{k} + \mathbf{G}_i|$ (on a scale determined by the size of the orbitals), only $\mathbf{G}_i^{(\text{MBT})} = 0$ contributes significantly in the summation over i in Eq. (S52). Momentum conservation in Eq.(S52) then enforces $\mathbf{q} = \mathbf{G}_j^{(\text{BP})} + \mathbf{k}$, which can only be satisfied for $\mathbf{G}_j^{(\text{BP})} = 0$ as $\mathbf{q}$ and $\mathbf{k}$ lie in the first Brillouin zone. Therefore, the contribution to the continuum model of the heterostructure around Γ due to MBT-BP coupling takes the final form

$$\hat{H}_{\text{int}}^{(b)} = \sqrt{\frac{N_{\text{SL}}}{N_{\text{BP}}}} \frac{1}{\Omega_{\text{BP}}} \sum_{\mu,\nu,\sigma} \sum_{\mathbf{k}} \left[t_{\mu\nu}^{(b)}(\mathbf{k}) \psi_{\mathbf{k}\mu,\sigma}^\dagger \phi_{\mathbf{k}\nu,\sigma} + \text{h.c.} \right]. \quad (\text{S55})$$

In the following, we set $N_{\text{SL}} = N_{\text{BP}}$ and we recall that Ω_{BP} denotes the are of the BP unit cell. This interaction Hamiltonian involves the full 7×4 dimensional hopping matrix $t_{\mu\nu}^{(b)}$. However, since the minimum low-energy model for MBT SL contains only four specific even and odd combinations of selected Bi and Te orbitals [see low-energy MBT basis in Eq. (S16)], we only need to consider a part of the matrix $t_{\mu\nu}^{(b)}(\mathbf{k})$ to construct $\hat{U}_b(\mathbf{k})$. Projecting $\hat{H}_{\text{int}}^{(b)}$ into the low-energy basis in Eq. (S16) yields

$$\mathcal{P} \hat{H}_{\text{int}}^{(b)} \mathcal{P}^{-1} = \sum_{\mathbf{k}} \left[\hat{\varphi}_{\mathbf{k}}^\dagger \hat{V}_{b,\uparrow}(\mathbf{k}) \hat{\phi}_{\mathbf{k},\uparrow} + \hat{\varphi}_{\mathbf{k}}^\dagger \hat{V}_{b,\downarrow}(\mathbf{k}) \hat{\phi}_{\mathbf{k},\downarrow} + \text{h.c.} \right]. \quad (\text{S56})$$

Here, $\hat{\varphi}_{\mathbf{k}}$ is a 4-dimensional spinor that creates electrons in the MBT basis states in Eq. (S16) and

$$\hat{V}_{b,\uparrow}(\mathbf{k}) = \frac{1}{\sqrt{2}\Omega_{\text{BP}}} \begin{pmatrix} t_{51}^{(b)}(\mathbf{k}) + t_{61}^{(b)}(\mathbf{k}), & t_{52}^{(b)}(\mathbf{k}) + t_{62}^{(b)}(\mathbf{k}), & t_{53}^{(b)}(\mathbf{k}) + t_{63}^{(b)}(\mathbf{k}), & t_{54}^{(b)}(\mathbf{k}) + t_{64}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \\ t_{11}^{(b)}(\mathbf{k}) - t_{41}^{(b)}(\mathbf{k}), & t_{12}^{(b)}(\mathbf{k}) - t_{42}^{(b)}(\mathbf{k}), & t_{13}^{(b)}(\mathbf{k}) - t_{43}^{(b)}(\mathbf{k}), & t_{14}^{(b)}(\mathbf{k}) - t_{44}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{S57})$$

$$\hat{V}_{b,\downarrow}(\mathbf{k}) = \frac{1}{\sqrt{2}\Omega_{\text{BP}}} \begin{pmatrix} 0 & 0 & 0 & 0 \\ t_{11}^{(b)}(\mathbf{k}) - t_{41}^{(b)}(\mathbf{k}), & t_{12}^{(b)}(\mathbf{k}) - t_{42}^{(b)}(\mathbf{k}), & t_{13}^{(b)}(\mathbf{k}) - t_{43}^{(b)}(\mathbf{k}), & t_{14}^{(b)}(\mathbf{k}) - t_{44}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \\ t_{51}^{(b)}(\mathbf{k}) + t_{61}^{(b)}(\mathbf{k}), & t_{52}^{(b)}(\mathbf{k}) + t_{62}^{(b)}(\mathbf{k}), & t_{53}^{(b)}(\mathbf{k}) + t_{63}^{(b)}(\mathbf{k}), & t_{54}^{(b)}(\mathbf{k}) + t_{64}^{(b)}(\mathbf{k}) \end{pmatrix}. \quad (\text{S58})$$

Here, we have set $N_{\text{SL}} = N_{\text{BP}}$. The relation between the matrices $\hat{V}_{b,\sigma}$ and $\hat{U}_b$ becomes evident when connecting Eq.(S56) with Eq.(S13): $\hat{U}_b$ is a 24×8 matrix with zero matrix elements everywhere other than its last four lines, which are given by the 4×8 matrix $(\hat{V}_{b,\uparrow}, \hat{V}_{b,\downarrow})$. Note that $t_{\mu,1}^{(b)} = t_{\mu,2}^{(b)}$ because the z -component of the fractional positions of phosphorus 1 and 2 are the same, $\rho_{1,z} = \rho_{2,z}$ (see Table S2 and Fig. S44). Similarly, $t_{\mu,3}^{(b)} = t_{\mu,4}^{(b)}$. Moreover, the dominant matrix element of $\hat{V}_{b,\sigma}$ is $t_{13}^{(b)} - t_{43}^{(b)}$, as shown in Fig. S47.

Similar steps apply for calculating the coupling between the top BP layer and the top-most MBT SL (SL number one in, which is described by a Hamiltonian as Eq.(S56) with $b \rightarrow t$). Due to inversion symmetry, the hopping amplitudes $t_{\mu\nu}^{(t)}$ are related to $t_{\mu\nu}^{(b)}$. More specifically, $t_{\mu\nu}^{(t)} = t_{\mu'\nu'}^{(b)}$, where $\mu' = 4, 1, 6, 5$ if $\mu = 1, 2, 5, 6$, respectively and $\nu' = 3, 4, 1, 2$ if $\nu = 1, 2, 3, 4$. We thus obtain

$$\hat{V}_{t,\uparrow}(\mathbf{k}) = \frac{1}{\sqrt{2}\Omega_{\text{BP}}} \begin{pmatrix} t_{53}^{(b)}(\mathbf{k}) + t_{63}^{(b)}(\mathbf{k}) & t_{54}^{(b)}(\mathbf{k}) + t_{64}^{(b)}(\mathbf{k}) & t_{51}^{(b)}(\mathbf{k}) + t_{61}^{(b)}(\mathbf{k}) & t_{52}^{(b)}(\mathbf{k}) + t_{62}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \\ -t_{13}^{(b)}(\mathbf{k}) + t_{43}^{(b)}(\mathbf{k}) & -t_{14}^{(b)}(\mathbf{k}) + t_{44}^{(b)}(\mathbf{k}) & -t_{11}^{(b)}(\mathbf{k}) + t_{41}^{(b)}(\mathbf{k}) & -t_{12}^{(b)}(\mathbf{k}) + t_{42}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{S59})$$

$$\hat{V}_{t,\downarrow}(\mathbf{k}) = \frac{1}{\sqrt{2}\Omega_{\text{BP}}} \begin{pmatrix} 0 & 0 & 0 & 0 \\ -t_{13}^{(b)}(\mathbf{k}) + t_{43}^{(b)}(\mathbf{k}) & -t_{14}^{(b)}(\mathbf{k}) + t_{44}^{(b)}(\mathbf{k}) & -t_{11}^{(b)}(\mathbf{k}) + t_{41}^{(b)}(\mathbf{k}) & -t_{12}^{(b)}(\mathbf{k}) + t_{42}^{(b)}(\mathbf{k}) \\ 0 & 0 & 0 & 0 \\ t_{53}^{(b)}(\mathbf{k}) + t_{63}^{(b)}(\mathbf{k}) & t_{54}^{(b)}(\mathbf{k}) + t_{64}^{(b)}(\mathbf{k}) & t_{51}^{(b)}(\mathbf{k}) + t_{61}^{(b)}(\mathbf{k}) & t_{52}^{(b)}(\mathbf{k}) + t_{62}^{(b)}(\mathbf{k}) \end{pmatrix}. \quad (\text{S60})$$

Here, we have set $N_{\text{SL}} = N_{\text{BP}}$. The matrices $\hat{V}_{t,\uparrow}$ and $\hat{V}_{t,\downarrow}$ define the blocks $\hat{U}_t$ in Eq. (S13). Like $\hat{U}_b$, the matrix $\hat{U}_t$ is also a 24×8 matrix, but here the only non-zero elements are in the first four lines of this matrix, which are formed by the 4×8 matrix $(\hat{V}_{t,\uparrow}, \hat{V}_{t,\downarrow})$.

IV.8.2 MnBi₂Te₄-BP hopping integrals using Slater-Koster approach

In this section we determine the hopping amplitude $t_{\mu\nu}^{(b)}(\mathbf{q})$ that enters the final expression for the coupling Hamiltonian in Eq. (S55). We obtain the real-space hopping amplitudes between the p_z orbitals of BP and its nearest-neighbor MBT SL using the Slater-Koster approach, and then Fourier transform to obtain the desired expression for $t_b(\mathbf{q})$ in Eq.(S53).

The Slater-Koster approach [68] consist of parametrizing the real-space hopping amplitude between two p_z orbitals separated by a distance vector $\mathbf{r} = r(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ as

$$t_b(\mathbf{r}) = V_\sigma(\mathbf{r}) \cos^2 \theta + V_\pi(\mathbf{r}) \sin^2 \theta. \quad (\text{S61})$$

Here, the subscripts σ and π denote the two orthogonal projections of the p_z orbitals along the direction of $\mathbf{r}$ (σ and π bonding). The radial functions $V_\sigma(\mathbf{r}) = A_\sigma e^{-r/a_\sigma}$ and $V_\pi(\mathbf{r}) = A_\pi e^{-r/a_\pi}$ decay exponentially with the distance r between the orbitals. The parameters A_σ , A_π , a_σ , and a_π are adjustable parameters in the Slater-Koster approach. While a_σ and a_π play the role of a effective Bohr radii that set the characteristic radial decay range of the MBT-BP coupling: larger a_σ and a_π results in a slower $t(\mathbf{r})$ decays and coupling functions $\hat{U}_b(\mathbf{k})$ and $\hat{U}_t(\mathbf{k})$ that are more localized in momentum space. In this work, we used $a_\sigma = a_\pi = 2 \text{ \AA}$.

The parameters A_σ and A_π have dimension of energy and determine the hopping strength, which are physically given by the overlaps of the orbital wavefunctions. Since the σ projection of two orbitals is generally larger than the π projection, we choose $A_\sigma \gtrsim A_\pi$. In this work, we used $A_\sigma = 0.34$ eV and $A_\pi = 0.0$.

In Eq. (S53), we take a partial Fourier transform of Eq.(S61) with respect to the in-plane components, leaving the z coordinate unchanged,

$$t(\mathbf{q}, z) = \int d^2y \left[A_\sigma \frac{z^2}{y^2 + z^2} e^{-a_\sigma^{-1} \sqrt{y^2 + z^2}} + A_\pi \frac{y^2}{y^2 + z^2} e^{-a_\pi^{-1} \sqrt{y^2 + z^2}} \right] e^{-i\mathbf{q} \cdot \mathbf{y}}. \quad (\text{S62})$$

We numerically calculate this integral and fit the result to a sum of two decaying exponentials of the form

$$t(\mathbf{q}, z) = z^2 \left[A_\sigma \gamma_1 e^{-\sqrt{\gamma_2 + \gamma_3 q^2 z^2}} + A_\pi \gamma_4 (1 - \gamma_5 q^2 z^2) e^{-\sqrt{\gamma_6 + \gamma_7 q^2 z^2}} \right]. \quad (\text{S63})$$

Here, γ_i ($i = 1, \dots, 6$) are real fitting constants that implicitly depend on the Bohr radii a_σ (for $i = 1, 2, 3$) and a_π (for $i = 4, 5, 6$). Substituting z by the z component distance of the orbitals $s_{\mu\nu} = \tau_\mu^\perp - \rho_\mu^\perp$ in Eq.(S63) finally yields $t_{\mu\nu}^{(b)}(\mathbf{q})$.

IV.9 Parameter set used in the main text

In Fig. S3, we provide the numerical parameter values that we have used in the main text to reproduce the experimental observations. The parameters in our model can be categorized into the following three types: (1) the bare MnBi_2Te_4 parameters, (2) the bare BP parameters, and (3) the parameters describing the interaction between MnBi_2Te_4 and BP.

The bare MnBi_2Te_4 parameters were fixed based on fitting to 6SL DFT MnBi_2Te_4 band structure, which is consistent with Ref. [57].

The bare BP parameters were chosen also based on fitting to DFT BP band structure.

The MnBi_2Te_4 coupling parameters can be further classified into the following two classes: the strain on MnBi_2Te_4 due to the introduction of BP and the direct hybridization between MnBi_2Te_4 and BP bands. Due to the large lattice mismatch between the two materials, there is no solid reliable DFT for BP/6SL MnBi_2Te_4 /BP. Nevertheless, we could partly constrain the parameters by fitting the R_{xx}/R_{yy} data. As shown in Fig. S3, there are many coupling parameters. Therefore, we first make simplifications, which reduce to only three independent coupling parameters. Within the nine strain parameters $s_1 - s_9$, five are important linear strain terms and two are quadratic strain terms. We set all linear terms to be equal ($s_2 = s_4 = s_5 = s_7 = s_8$) and we set all quadratic terms $s_1 = 0.1$ and $s_3 = 50$. The other two unimportant terms s_6 and s_9 are always zero. Within the four band hybridization terms, we only consider the $A_\sigma/\Omega_{\text{BP}}$ (i.e., the sigma bonding between orbitals). A_π/Ω_{BP} is always set to zero. The a_σ and a_π (effective bond length) are also fixed at 2 Å. In this way, we reduce to only three independent coupling parameters, a linear strain term, a quadratic strain term and a band hybridization term.

After reducing to three coupling parameters, we use the measured R_{xx}/R_{yy} ratio to fix the linear strain term. Experimentally, our data (Fig. S48) show $R_{xx}/R_{yy} \simeq 1.6$ at $n_e \simeq -0.8 \times 10^{12} \text{ cm}^{-2}$ (hole-doped). Theoretically, we can compute R_{xx}/R_{yy} at different values of the linear strain term (specifically, we compute the Drude conductivity σ_{xx} and σ_{yy} by the Kubo formula. The scattering time is not need as we only plot their ratio). As shown in Fig. S48, we found that, when the linear strain term is set to $s_2 = s_4 = s_5 = s_7 = s_8 = 0.1$, the calculated R_{xx}/R_{yy} is also about 1.6 at this specific carrier density. In this way, we independently set the value of the linear strain term.

Figure S49 shows the calculated nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ with only the linear strain term. We see that the overall magnitude already agrees with the data. Also, an important overall shape (conduction and valence bands having the same sign across charge neutrality) is also obtained. We then adjust the values of the other two parameters (the quadratic strain term and the band hybridization term) to achieve a good matching with the experimental data. In particular, the sign inversion in the high electron-doped regime can only be obtained once the band hybridization $A_\sigma/\Omega_{\text{BP}}$ is turned on. This suggests that the sign reversal comes from an avoided crossing between MnBi_2Te_4 and BP conduction bands.

V. Additional discussions

V.1 Extrinsic origins for the microwave rectification

We enumerate the possible extrinsic origins for the observed microwave rectification and then discuss possible future experiments to rule them out.

- The metal contact-MnBi₂Te₄ junction can form a Schottky diode, which can rectify microwave signals.
- The MnBi₂Te₄ may contain an unintentional PN diode due to spatial inhomogeneity, which can rectify microwave signals.
- The gap between the two gates can serve as a rectifying capacitor, which can rectify microwave signals.

While the correlation between the AFM domain and the rectification voltage (Fig. S36) suggest an intrinsic origin for the wireless rectification, future experiments are needed to unambiguously determine the origin.

- Hall nature: If the rectification is caused by the quantum metric nonlinear Hall effect, then the rectified DC voltage should be perpendicular to the incident microwave electric field.
- Polarization dependence: Measure the rectification as one rotates the direction of the polarization direction of the microwave. If the rectification arises from the quantum metric dipole, the polarization dependence should show the direction of the quantum metric dipole.
- Charge density dependence: Measure the rectification as a function of the charge density. If the rectification arises from the quantum metric dipole, the measured charge density dependence should be consistent with the Fermi level dependence of the quantum metric dipole.

V.2. Estimation of rectification efficiency and comparison with existing technology

In this subsection we carefully studied the energy harvesting efficiency. Before going into details, we wish to acknowledge that this estimation is quite tricky and the final number we provide should only serve as an order of magnitude estimate. The challenge arises from the fact that only a tiny fraction of the initially generated microwave is actually absorbed by the sample (there are many loss channels). This is because many components of our setup are not specialized for GHz rectification and hence are not well characterized (our setup is mainly used for NV center magnetometry). Also, our device does not have a professional receiving antenna to efficiently collect all the microwave that reaches the sample. In a commercial rectifier, these loss channels are eliminated by better engineering so that the absorbed power roughly equals the initial power. In future, it would be very interesting to achieve such a condition in our BP/6SL MnBi₂Te₄ sample so that we can more precisely obtain the efficiency. Below, we try our best to provide a thorough estimation.

We first briefly go through the setup, as shown in Fig. S50. The microwave was generated by a function generator, transmitted into the cryostat by coaxial cable, delivered into free-space by antenna, and shone onto the sample. The sample then absorbs a fraction of the microwave.

V.2.1. External efficiency

We can easily calculate the external efficiency: Initial power by the microwave generator was 100 mW. The final harvested power can be computed by $I_{\text{DC}}^2 R_{\text{load}} = (0.5 \mu\text{A})^2 \simeq 1 \text{ kohm} = 2.5 \times 10^{-10} \text{ W}$ (R_{load} is the load resistor connected to the device). The ratio is the external efficiency $\simeq 2.5 \times 10^{-9}$, which is really small. As explained above, this is because most of the microwave power was not delivered to the sample. Even for the microwave that reaches the sample, only a small fraction was absorbed.

V.2.2. Internal efficiency

To compute the internal efficiency, we carefully the loss and the absorption:

- We first estimate the fraction of microwave that could reach the antenna. The coaxial cables are known to be lossy (3 dB per meter). Also, a portion of the microwave may be reflected backward. Using the vector network analyzer that can directly measure the reflected microwave, we found that the reflected microwave is negligible in our experiment (< -30 dB). So we mainly consider the loss by the coaxial cable. The coaxial cable of our setup is about 3 meter. which means that 12.5% of the initial microwave power was transmitted by the coaxial cable to the antenna.
- The antenna is about 10 mm away from the sample surface. As a crude approximation, we treat the microwave radiation as an isotropic spherical wave. At the sample surface height, the area of the spherical wavefront is $4\pi(10\text{mm} \times 10\text{mm})$ (this is an overestimate: the actual wavefront should be more directional because the antenna is a 1D wire). Here, the consideration becomes a bit tricky. There are two possibilities. One possibility is to consider the microwave directly absorbed by the BP/6SL MnBi₂Te₄ flake. We note that the flake area is only $\sim 20 \mu\text{m} \times 10 \mu\text{m}$, which is many orders smaller than the microwave spot. Another possibility is to consider the metal contacts and pads around the flake absorbs the free space microwave and converts it to GHz current. The GHz current is then rectified into DC signal by the nonlinear Hall effect of the BP/6SL MnBi₂Te₄.

We believe that the second possibility is more reasonable because the gold contacts/pads have much bigger size ($\sim 100 \mu\text{m} \times 100$) than the flake and a much larger absorption. Using Comsol, we estimate the microwave absorption coefficient of the gold contacts/pads (50 nm thick) is about $\simeq 0.5\%$.

Therefore, the final power absorbed by the device can be estimated by

$$\text{Initial power} \times \text{transmission by cable} \times \frac{\text{Metal pad area}}{\text{Microwave spot size}} \times \text{Absorption of gold} \simeq 6.6 \times 10^{-10} \text{ W}$$

The internal efficiency is given by $\frac{\text{Harvested power}}{\text{Absorbed power}} \simeq 38\%$. We wish to caution again that this number should be treated as an order of magnitude estimate because of the challenge to obtain the final absorbed power. A better engineered device that can eliminate all the loss channels and the small absorption issue is an important future direction, which can allow us to obtain precise efficiency.

V.2.3. Comparison with existing technologies

Threshold input voltage and low-input-power inefficiency: As shown in Fig. 2 of Ref. [69], the efficiency as a function of input power for existing rectifiers (Note that in these commercial rectifiers, the input power and absorbed power are similar because all the losses during transmission and absorption are carefully eliminated by designing the transmission lines and receiving antenna). The efficiency can be really high (up to 80%) if the input power is large. However, at low input power, the efficiency decays quickly. This is a well-known problem of a semiconductor diode: rectification based on semiconductor diodes requires the input electric field to be greater than a threshold due to the finite energy barrier (see Ref. [69]). Below this threshold input voltage, the efficiency is exceedingly low at low input power.

By contrast, our rectification is based on the intrinsic quantum geometry properties in a homogeneous material. It does not require a diode, and hence it is not constrained by any threshold input voltage. According to our estimation above, in our case, the absorbed power is as low as $6.6 \times 10^{-10} \text{ W}$, but our estimated efficiency is $\sim 38\%$. Of course, this is still preliminary, which requires future verification with a better engineered device.

High frequency inefficiency: As shown in Fig. 13 of Ref. [70], the normalized efficiency as a function of input power for a representative existing rectifier. The efficiency quickly decays at high frequencies. Indeed, it is well-known that the most existing rectifiers become very inefficient at high frequencies.

By contrast, our rectification based on the quantum metric nonlinear Hall effect is expected to be independent of frequency all the way until $\hbar\omega$ can induce an inter-band transition. Hence it has the potential to enable efficient, high frequency rectification. The high frequency potential of our nonlinear Hall effect requires future verification.

V.2.4. Advantages and Disadvantages

From the discussions above, we conclude the following

Potential advantages of our rectification include (1) its potential for very low-power rectification and (2) its potential for high frequency rectification. These advantages suggest important future applications. For example, the power density of ambient wireless radiation (wifi, etc) are around nW/cm^2 . Realizing an efficient harvester of that can lead to novel sensors and self-powered microdevices. Higher frequency is future trend for next generation wireless communication. (3) The fact that our nonlinear Hall signal flips sign with AFM order suggest integration with spintronic devices.

The disadvantages of our rectification include (1) the current demonstration is only a proof of concept. One will need well-engineered transmission lines and antenna to avoid losses during transmission and absorption; (2) the Néel temperature of 6SL MnBi_2Te_4 is only ~ 21 K.

V.2.5. Future directions

The discussions above lead to a few important future directions, which are summarized as follows. (1) Achieve a well-engineered setup and BP/6SL MnBi_2Te_4 /BP device to eliminate the loss channels and the finite absorption issues. This will allow one to precisely determine the efficiency. This will also allow one to test its potential advantages at ultra-low input power and at high frequencies. (2) Demonstrating the wireless rectification based on quantum metric nonlinear Hall in a room temperature $\mathcal{PT}$ -symmetric AFM material would be an exciting breakthrough in future.

V.3. Possible nonlinear Hall effect due to Berry curvature higher order in M in canted AFM

Recent experiments have uncovered a very interesting (linear) AHE in the canted AFM phase of MnBi_2Te_4 due to Berry curvature higher order in M [38, 46, 51]. An interesting question is whether this Berry curvature higher order in M can also lead to a second-order nonlinear Hall effect, which would be a new nonlinear Hall effect that is quadratic in E and higher order to M . This is a very interesting future direction. One idea is as follows: because the system already has nonzero Berry curvature in this canted regime, if the Berry curvature distributes unevenly in k space, it will lead to a Berry curvature dipole and in turn a nonlinear Hall effect. Rigorous theory studies are needed to analyze this possibility. Important aspects include the source of inversion symmetry breaking as well as the magnitude of the effect.

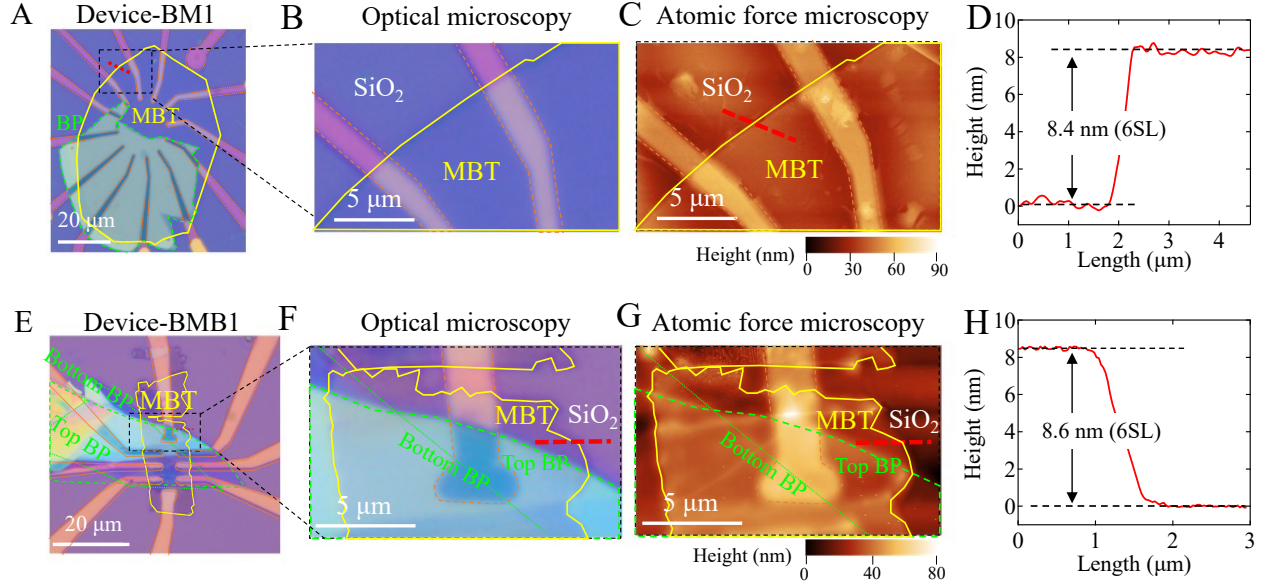


Fig. S1. Atomic force microscope data for the main devices of our paper. (A) Microscope image of the Device-BM1. Half of the BP flake was removed by blue-tape so that the MnBi₂Te₄ flake can be exposed for the atomic force microscopy (AFM) measurements. (B) Zoomed-in microscope image. (C) AFM image. (D) AFM thickness data along the red line in panel C. (E-H) Same as panels A-D but for Device-BMB1. In this case, we found that the top BP flake stuck really well to the MnBi₂Te₄ and could not be removed. Luckily, this MnBi₂Te₄ flake was exposed in the outer areas, allowing for the AFM measurement.

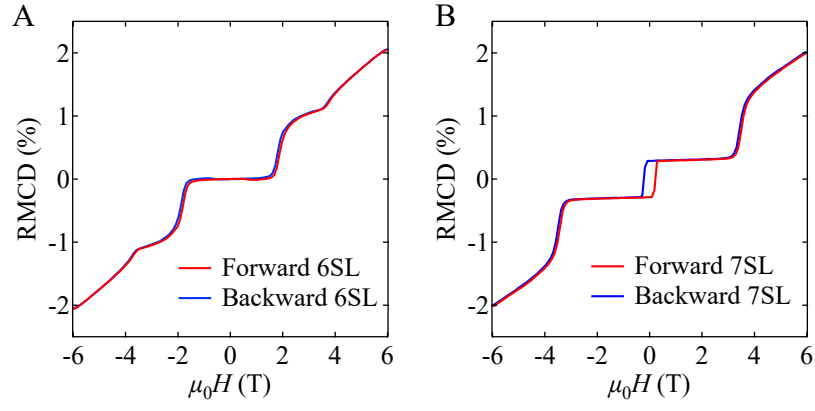


Fig. S2. (A) RMCD of our BP/6SL MnBi_2Te_4 sample. (B) RMCD of our BP/7SL MnBi_2Te_4 sample as a comparison.

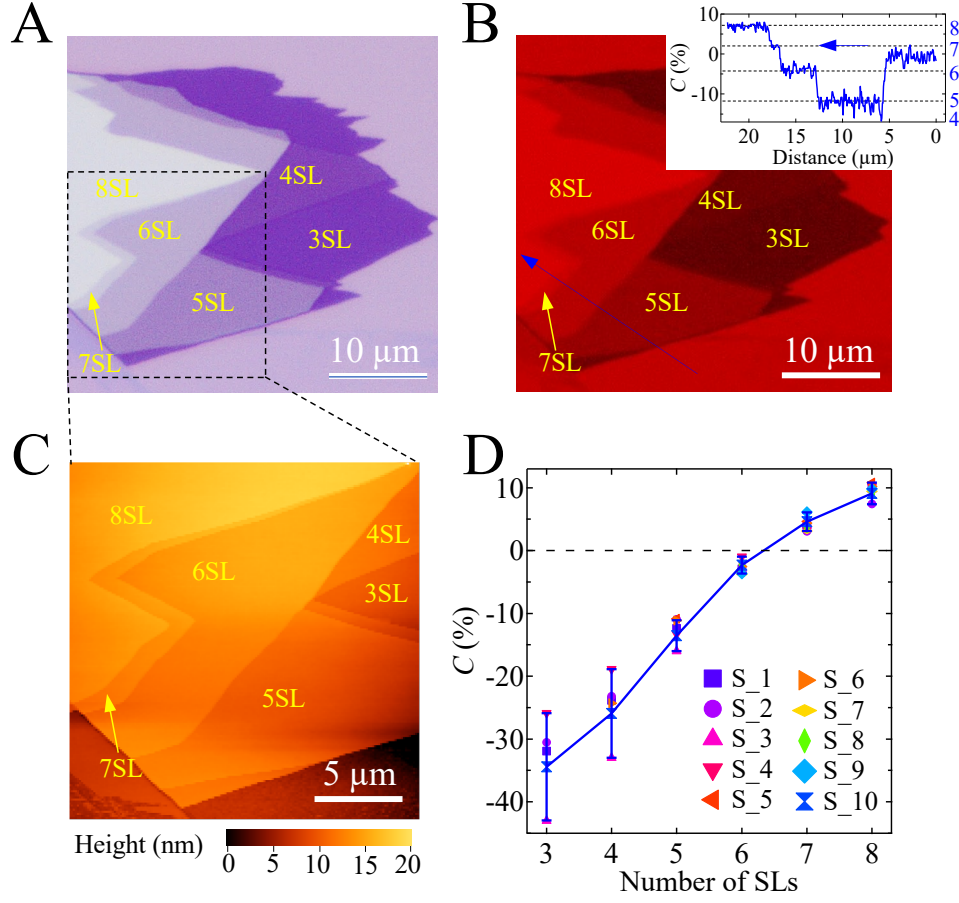
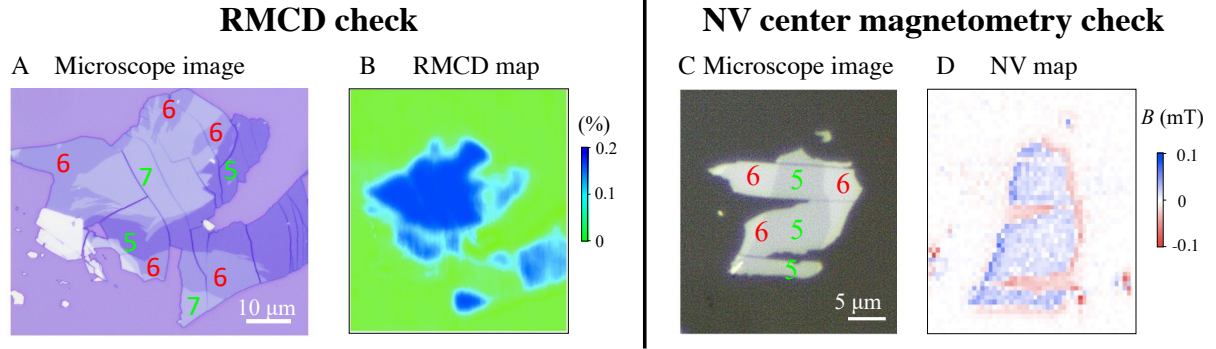


Fig. S3. Optical contrast of MnBi_2Te_4 flakes. (A) Optical image of an exfoliated MnBi_2Te_4 flake with different thicknesses. (B) Same as panel A but shown only based on the R (red) value of the RGB. Inset: optical contrast C along the line-cut defined by the blue arrow in the main figure. (C) Atomic force microscopic image of the same flake. (D) Optical contrast C as a function of number of layers. As described in the text, the correspondence between the optical contrast C and the layer number was checked many times on different exfoliated flakes to ensure reproducibility and reliability. Different data points represent results obtained in different measurements. The error bars represent the largest variation. Adapted from SI Figs. S13 of Ref. [44].



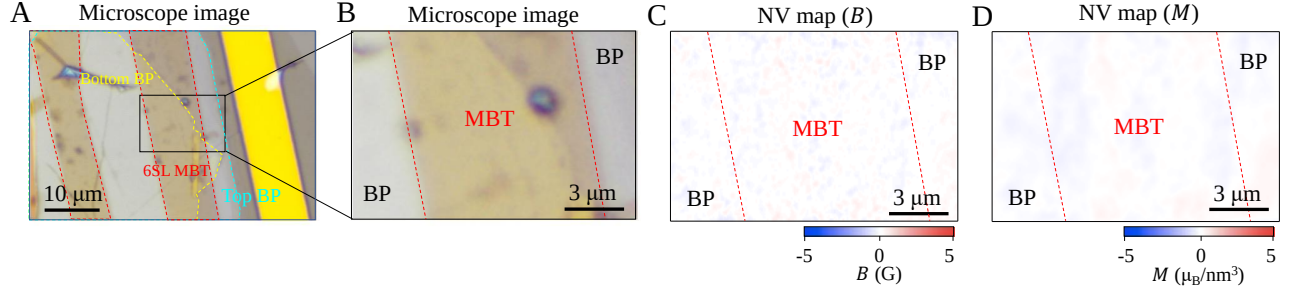


Fig. S5. (A) Microscope image of the BP/6SL MnBi₂Te₄/BP sample on a diamond substrate for NV measurements. (B) Zoomed-in image. (C) Corresponding NV map of the stray B field. (D) Magnetization M map (converted from the stray B field map, through an established reverse-propagation protocol shown in Ref. [71])

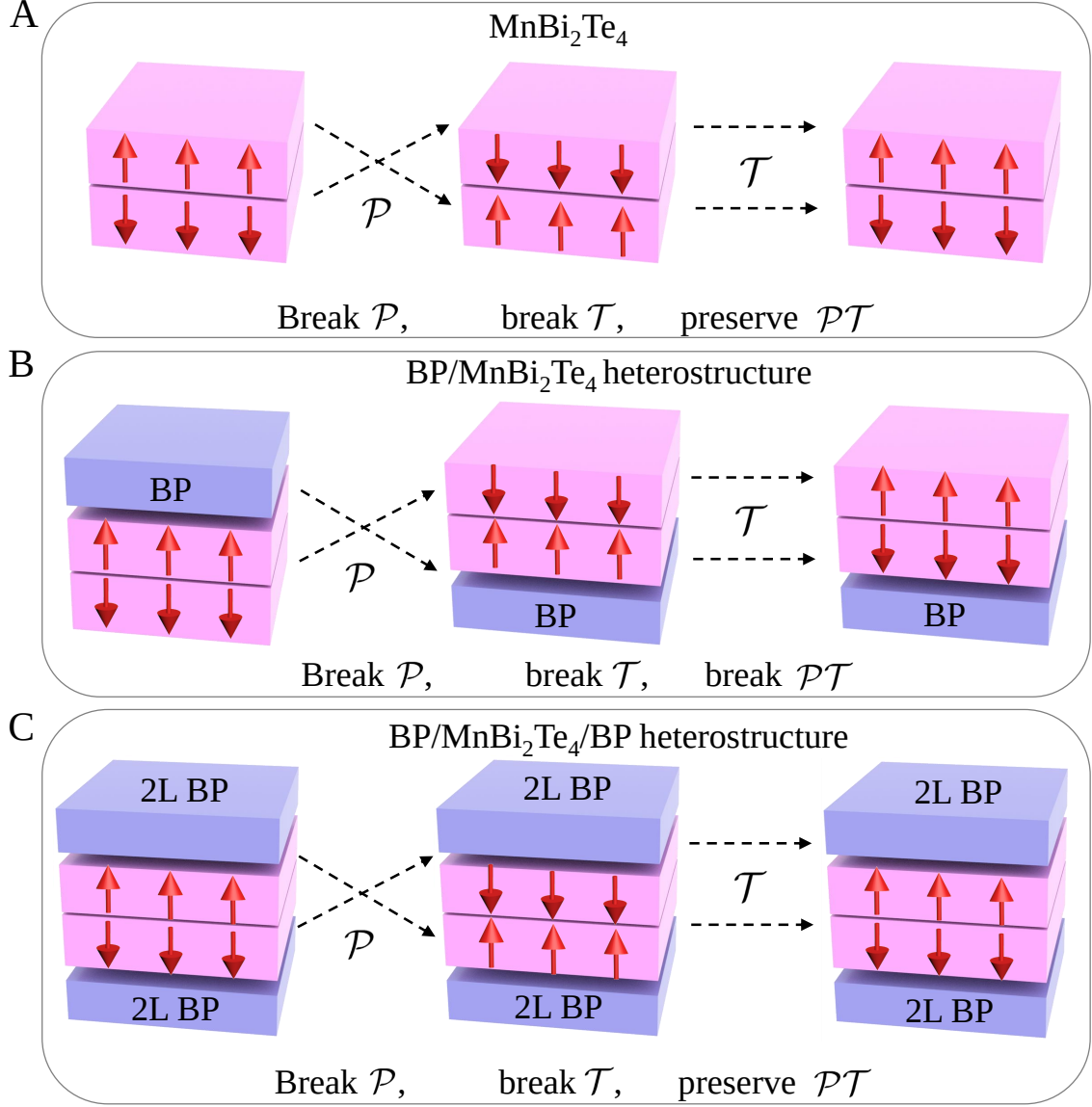


Fig. S6. The symmetry properties of even-layered MnBi_2Te_4 , BP/ MnBi_2Te_4 and BP/ MnBi_2Te_4 /BP heterostructures at low temperature. (A) The symmetry of even-layered MnBi_2Te_4 . (B) The symmetry of BP/ MnBi_2Te_4 heterostructure. (C) The symmetry of BP/ MnBi_2Te_4 /BP heterostructure. The mirror planes of top BP and bottom BP are aligned.

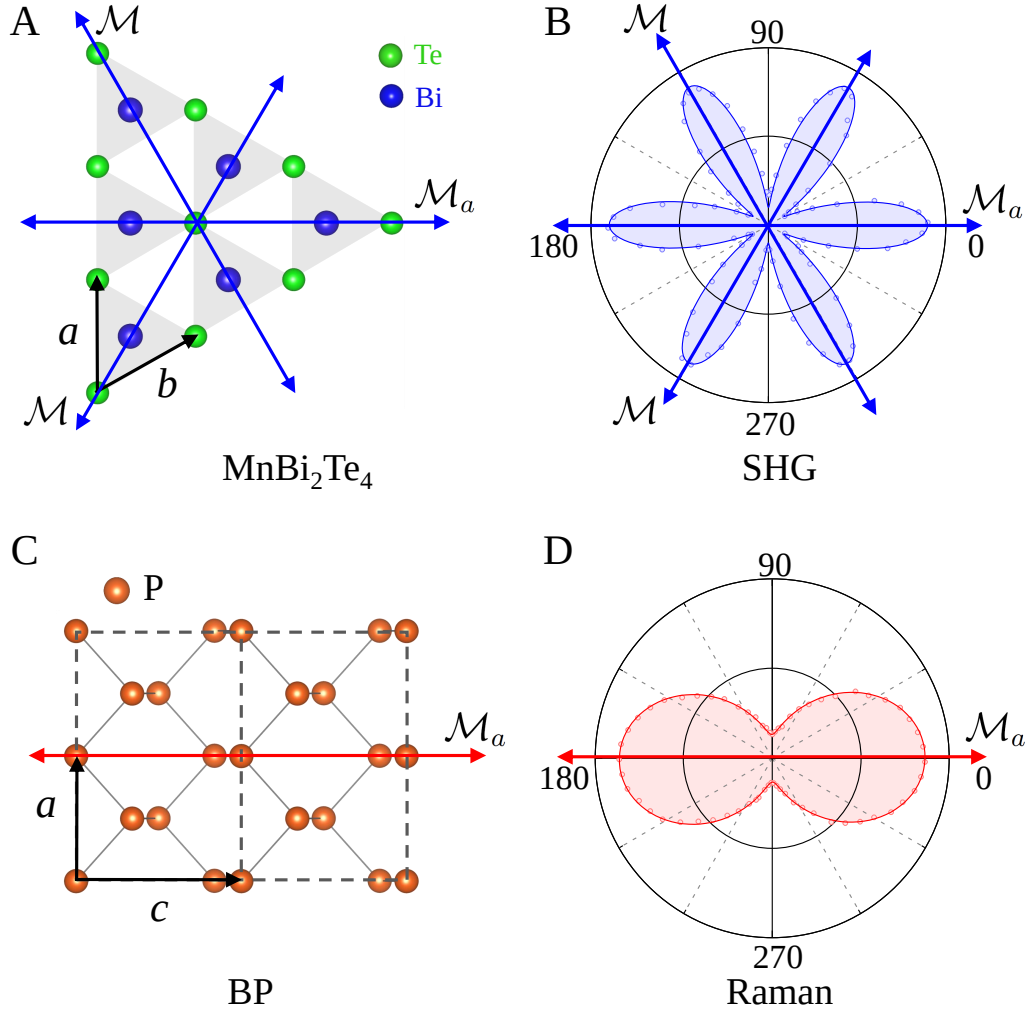


Fig. S7. The determination of crystalline directions of the MnBi_2Te_4 and BP. (A) Top view of the MnBi_2Te_4 lattice. It shows three-fold rotational symmetry C_{3z} . The three mirror planes are noted by the blue double arrow lines. (B) The SHG signals of the MnBi_2Te_4 measured at room temperature. The largest signals corresponding to the mirror planes of the MnBi_2Te_4 . (C) Top view of the BP lattice. It has a mirror plane $\mathcal{M}_a$ which is noted by the red double arrow line. (D) The angular-resolved Raman signals of the BP measured with a 532-nm laser. The largest signals corresponding to the mirror plane (armchair direction) of the BP.

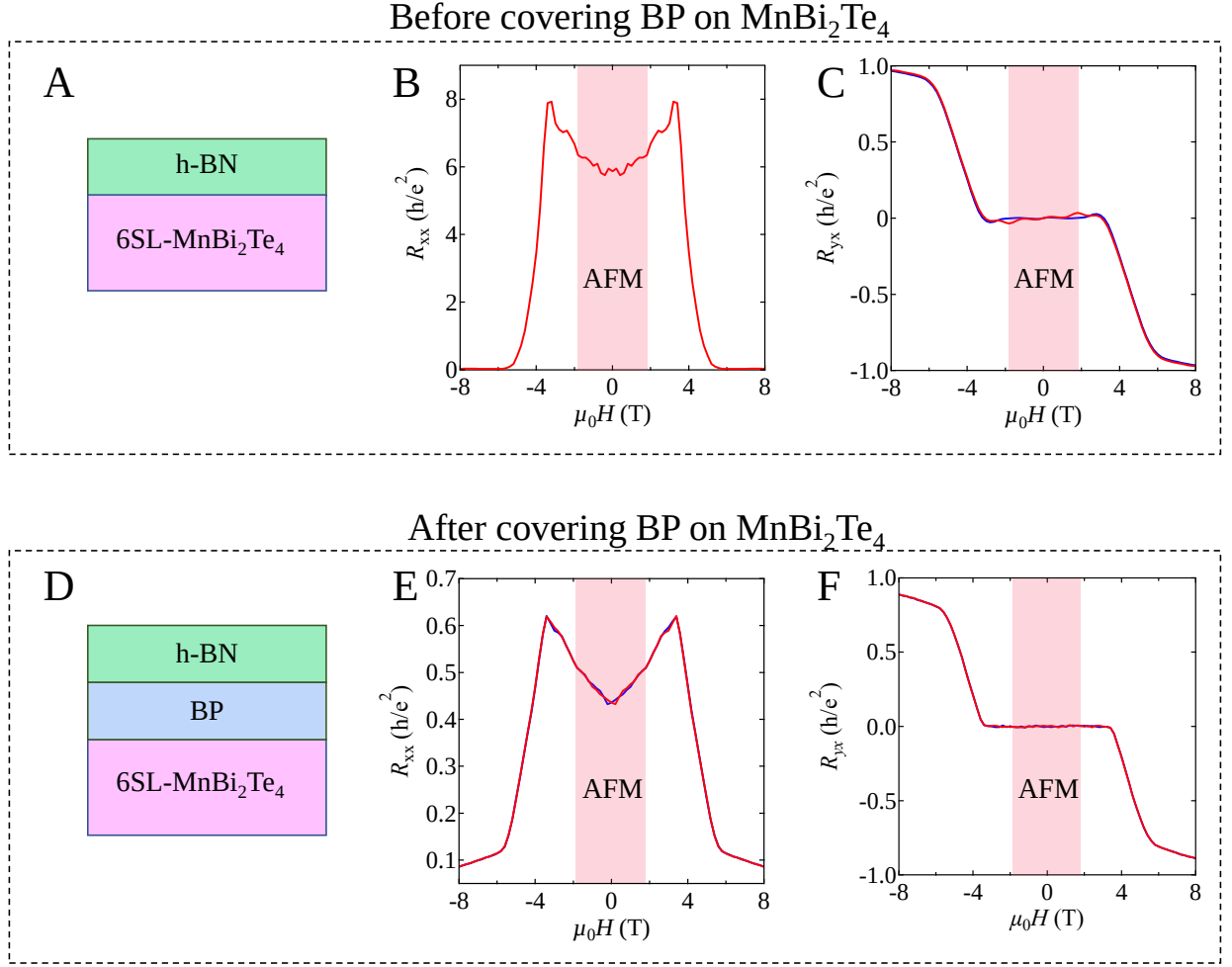


Fig. S8. The magneto-transport data of MnBi_2Te_4 with and without BP. (A to C) The magneto-transport data of the MnBi_2Te_4 at charge neutral point. (D to F) The magneto-transport data of the BP/ MnBi_2Te_4 heterostructure at charge neutral point. The BP/ MnBi_2Te_4 heterostructure exhibits the same topological phases as the MnBi_2Te_4 , i.e. Chern insulator in FM phase and Axion insulator in AFM phase.

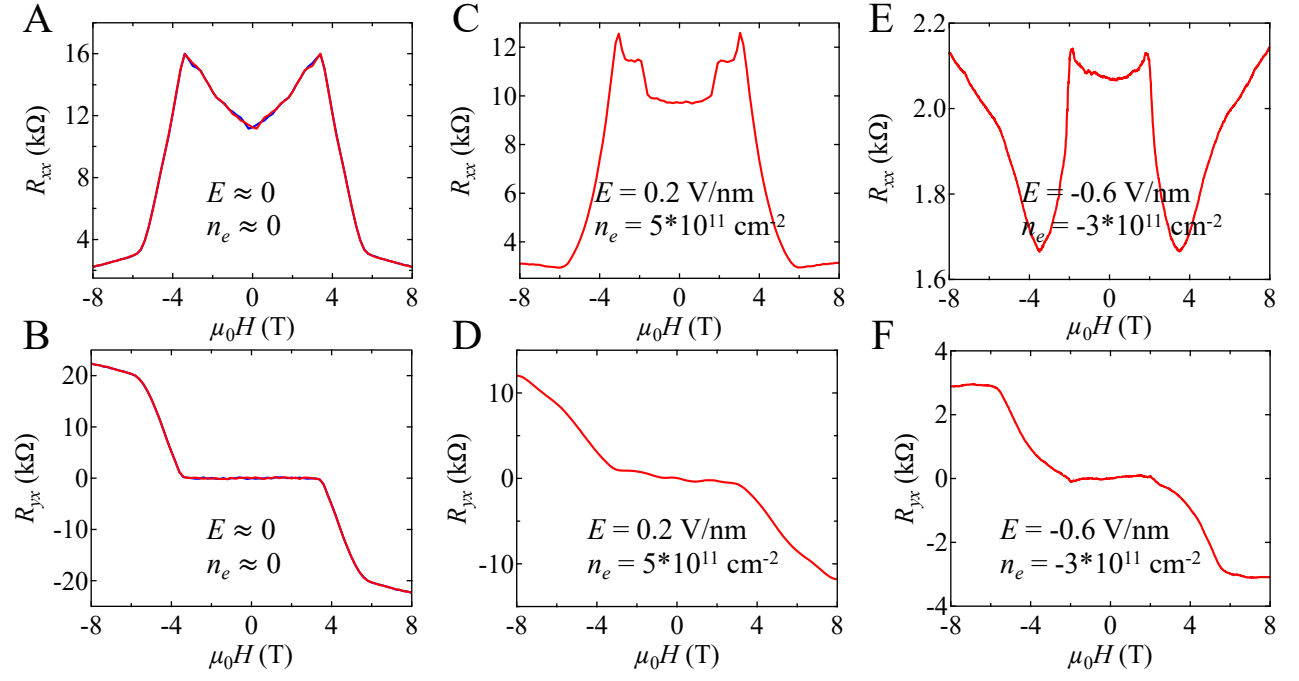


Fig. S9. (A-F) Gate-dependent transport data of our BP/6SL MnBi₂Te₄ device.

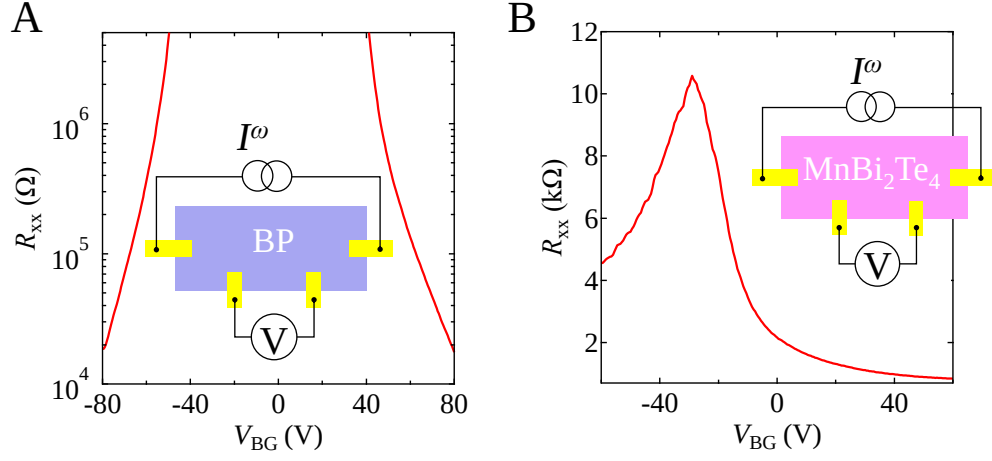


Fig. S10. The typical four-probe resistance of BP and MnBi_2Te_4 as a function of back gate voltages at 1.8 K. The thickness of BP is around 10 nm and MnBi_2Te_4 is 6SL. The contact geometry of BP and MnBi_2Te_4 is identical. The four-probe resistance of BP is two orders of magnitude larger than that of MnBi_2Te_4 .

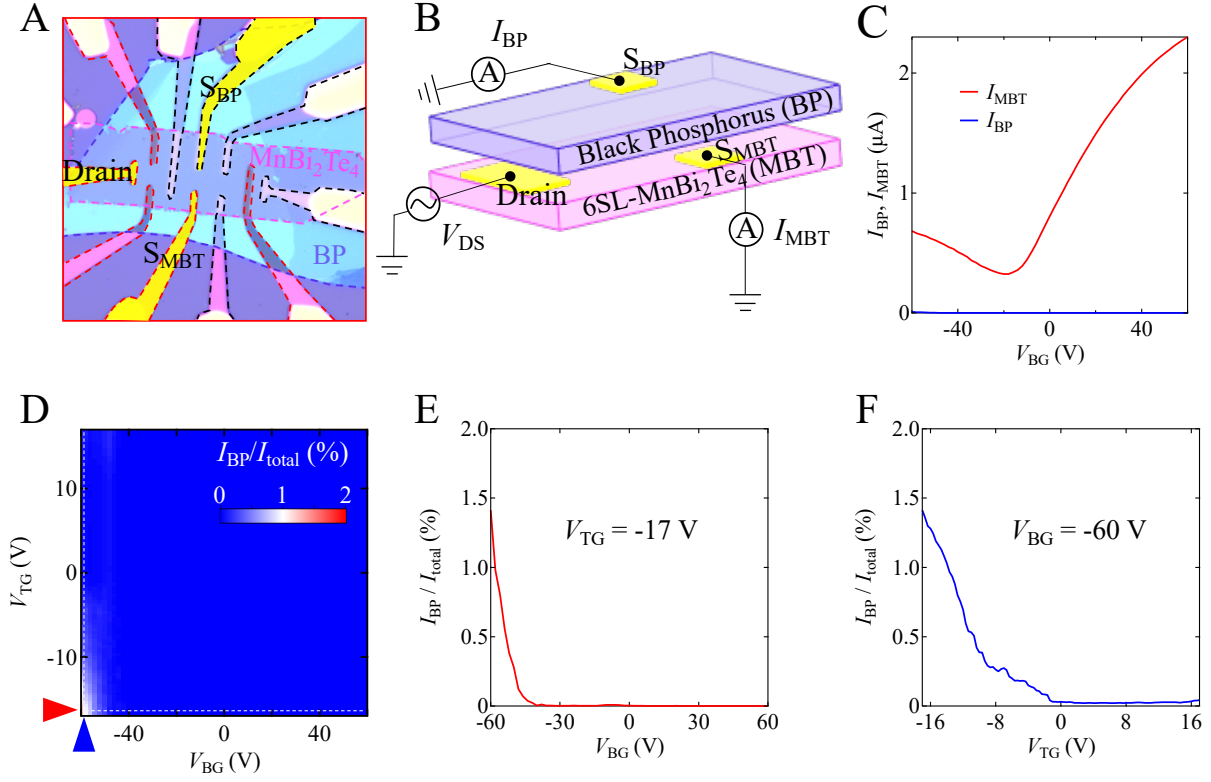


Fig. S11. Identifying the current flowing in the BP and MnBi₂Te₄ layers. (A and B) Optical microscopic image and schematic of the special BP/MnBi₂Te₄ device. The contacts outlined by red dashed lines are deposited on the MnBi₂Te₄ layer (which also contact with BP layer) while the contacts outlined by black dashed lines are deposited on the BP layer. The measurement electrodes are highlighted in yellow. The measurement circuit is shown in panel (B). A constant AC current was applied on the drain contact and was collected from two source electrodes S_{MBT} and S_{BP}, respectively. (C) I_{BP} and I_{MBT} as a function of V_{BG} . (D) The percentage of current flowing in BP layer (I_{BP}/I_{total}) as a function of V_{BG} and V_{TG} . (E and F) The line-cut data in panel (D). The current flowing in BP layer is always less than 2% of the total current.

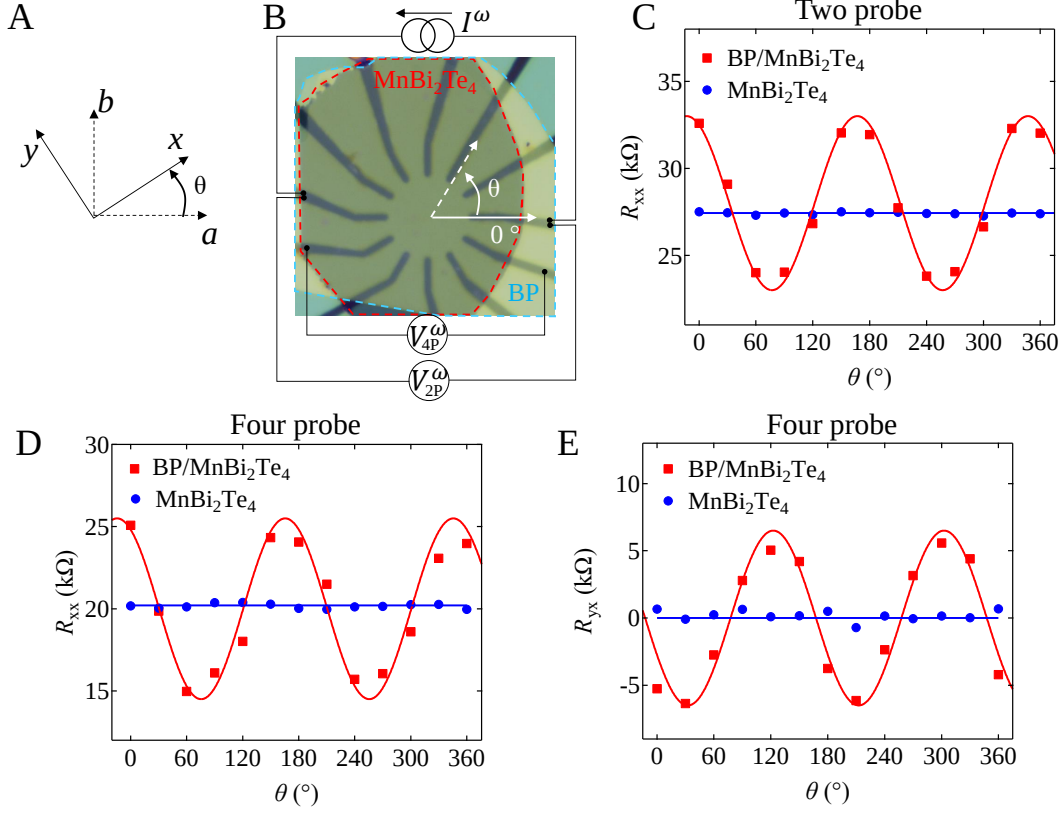


Fig. S12. The demonstration of C_{3z} breaking by electrical transport measurements. (A) The crystallographic bases are noted by a and b . The Cartesian bases are noted by x and y . The angle difference between crystallographic and Cartesian basis is noted by θ . (B) A BP/MnBi₂Te₄ heterostructure with the circular-disc contacts. MnBi₂Te₄ was roughly scratched into a circular shape with a tip in glovebox. Two-probe and four-probe measurements configuration are shown in the panel. (C and D) The longitudinal resistance (R_{xx}) of two-probe and four-probe measurements. (E) The transverse resistance (R_{yx}) of four-probe measurements. The blue and red curves correspond to the resistance of the same MnBi₂Te₄ sample before and after interfacing with BP.

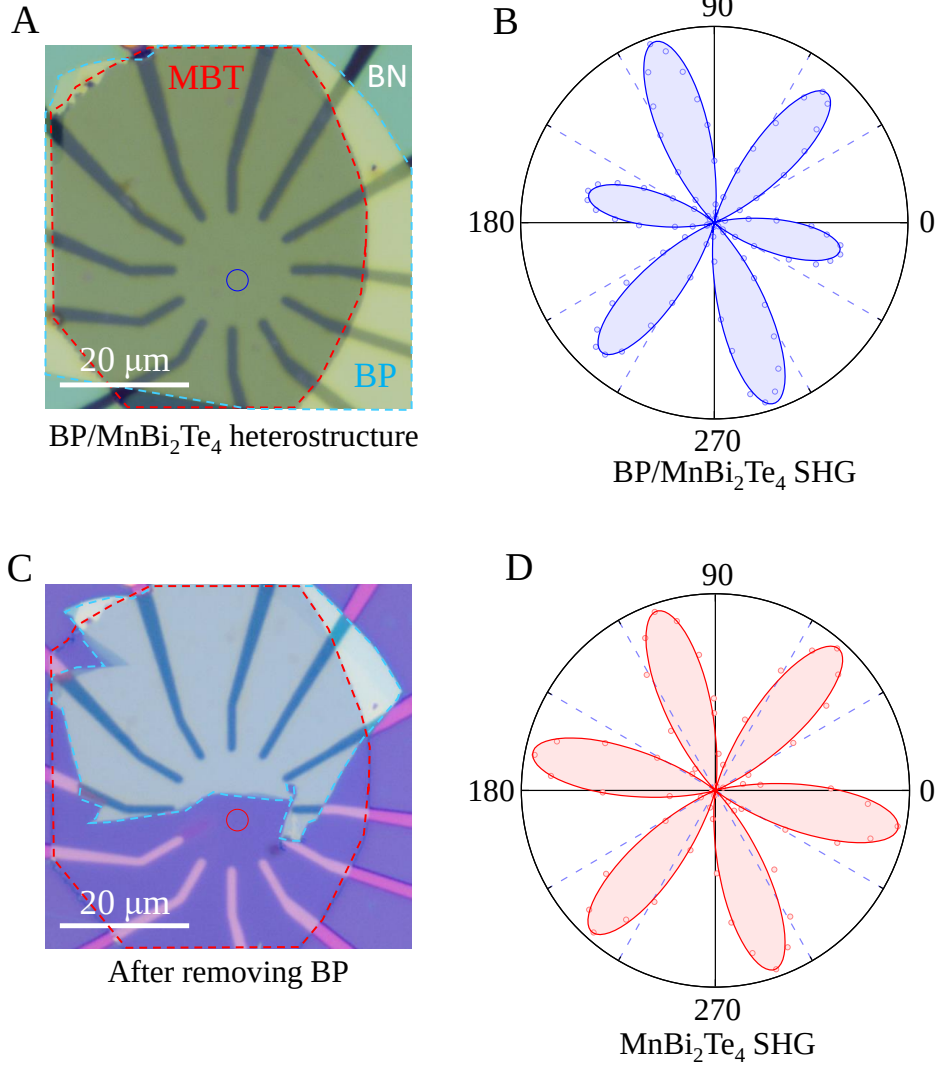


Fig. S13. The demonstration of C_{3z} breaking by optical SHG measurements at room temperature. (A) One of the BP/MnBi₂Te₄ heterostructures for optical SHG measurements. The MnBi₂Te₄ and BP are outline by red and cyan dashed line. The measured area is marked by a blue circle. (B) The optical SHG signals of BP/MnBi₂Te₄ heterostructure. (C) The same device as in (A) but removed the BN and part of BP with Scotch tape. The laser shone on the same area as in (A), noted by the red circle. (D) The optical SHG signals of the MnBi₂Te₄.

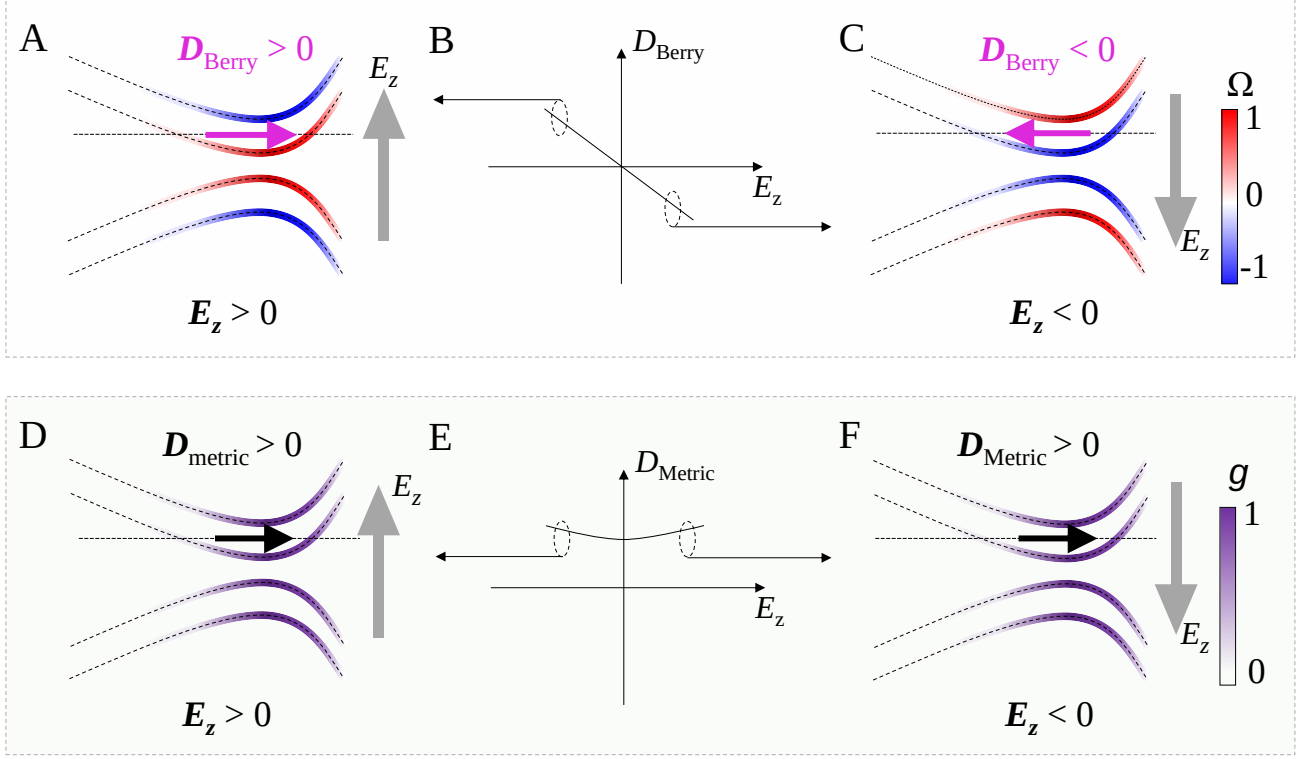


Fig. S14. The electric field dependence of Berry curvature dipole and quantum metric dipole. (A to C) The Berry curvature dipoles under the out of plane electrical field $E_z > 0$ and $E_z < 0$. The bands for top and bottom layers of MnBi_2Te_4 have the opposite Berry curvature. The electrical field $E_z > 0$ and $E_z < 0$ break the band degeneration in opposite ways [44]. Therefore, Berry curvature dipoles have different signs for opposite electrical fields. (D to F) The quantum metric dipoles under the out of plane electrical field $E_z > 0$ and $E_z < 0$. The bands for top and bottom layers of MnBi_2Te_4 have the same quantum metric. Although the opposite electrical field shift the bands of top and bottom layers of MnBi_2Te_4 in opposite ways, the quantum metric distribution is identical. Therefore, the quantum metric dipoles are symmetric as a function of E_z .

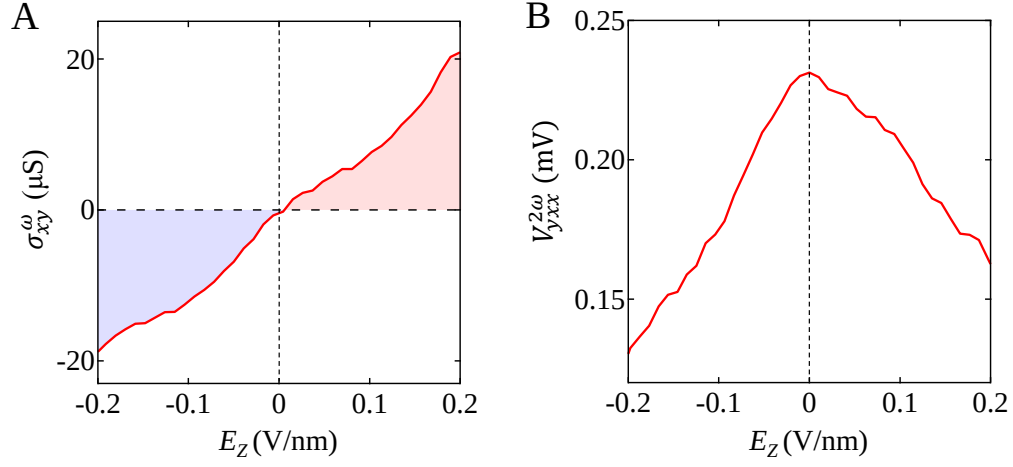


Fig. S15. The electric field dependence of linear and nonlinear Hall signals. (A) The linear Hall conductivity σ_{xy}^ω as a function of vertical electric field E_z . (B) The nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ as a function of E_z . σ_{xy}^ω is antisymmetric as a function of E_z , while $\sigma_{yxx}^{2\omega}$ is symmetric as a function of E_z . At $E_z = 0$, the $\mathcal{PT}$ confirms the $\sigma_{xy}^\omega = 0$ while $\sigma_{yxx}^{2\omega} \neq 0$.

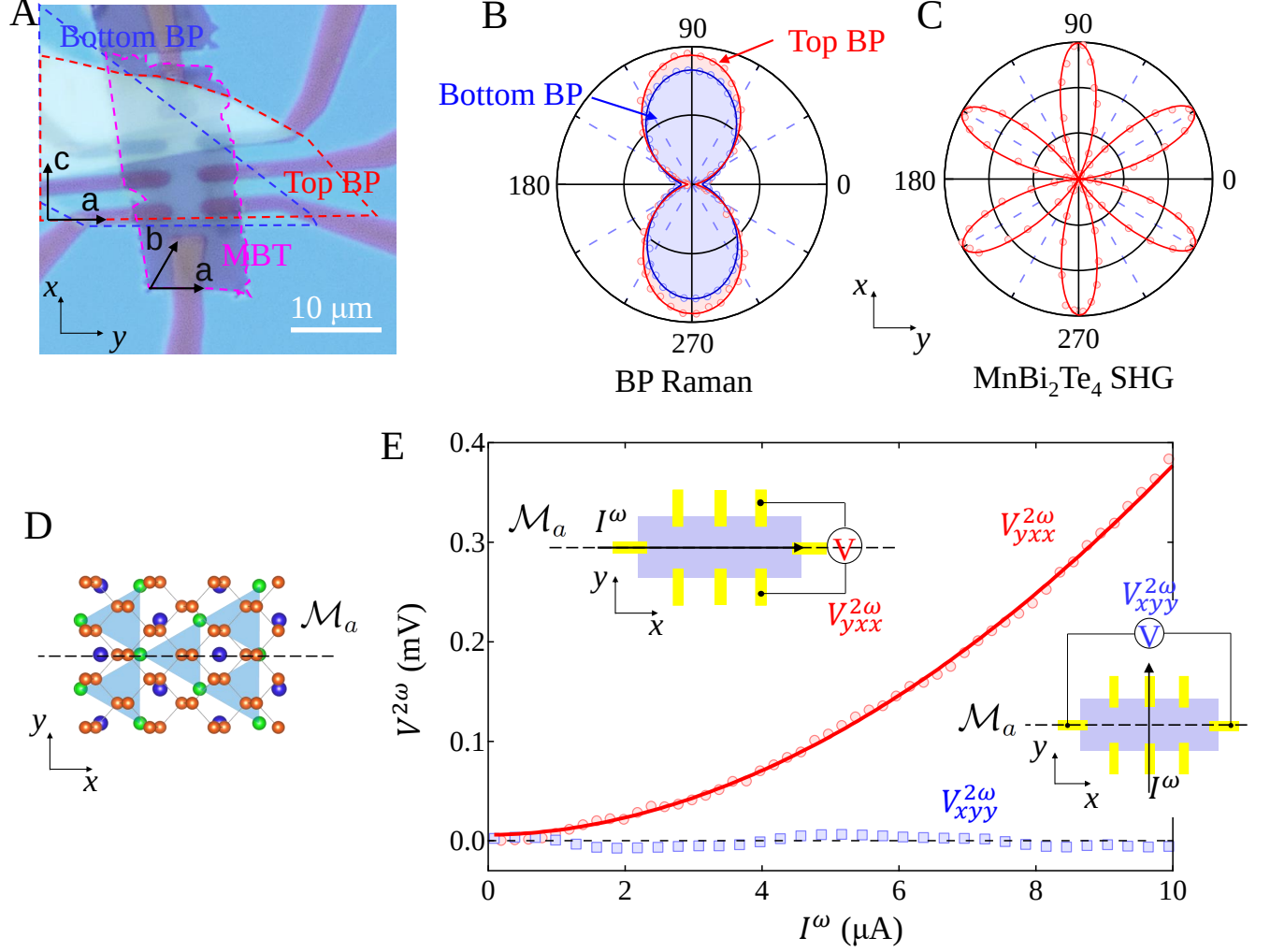


Fig. S16. A mirror plane aligned 2L BP/6SL MnBi₂Te₄/2L BP heterostructure. (A) The microscopic image of the BP/MnBi₂Te₄/BP heterostructure. The mirror planes of top BP, MnBi₂Te₄ and bottom BP are all aligned along x direction in Cartesian axis. MnBi₂Te₄, top and bottom BP are outlined by pink, red and blue dashed lines, respectively. The top and bottom BP are both bilayer. MnBi₂Te₄ is 6SL. (B) The polarized Raman signals of top (red curve) and bottom (blue curve) BP at room temperature. (C) Room-temperature SHG signals of the 6SL MnBi₂Te₄. (D) The schematic of aligned BP/MnBi₂Te₄/BP lattice structure. The mirror plane is along a (x). (E) The nonlinear Hall signals of the heterostructure for driving current along ($V_{yxx}^{2\omega}$) and vertical to ($V_{xyy}^{2\omega}$) the mirror direction.

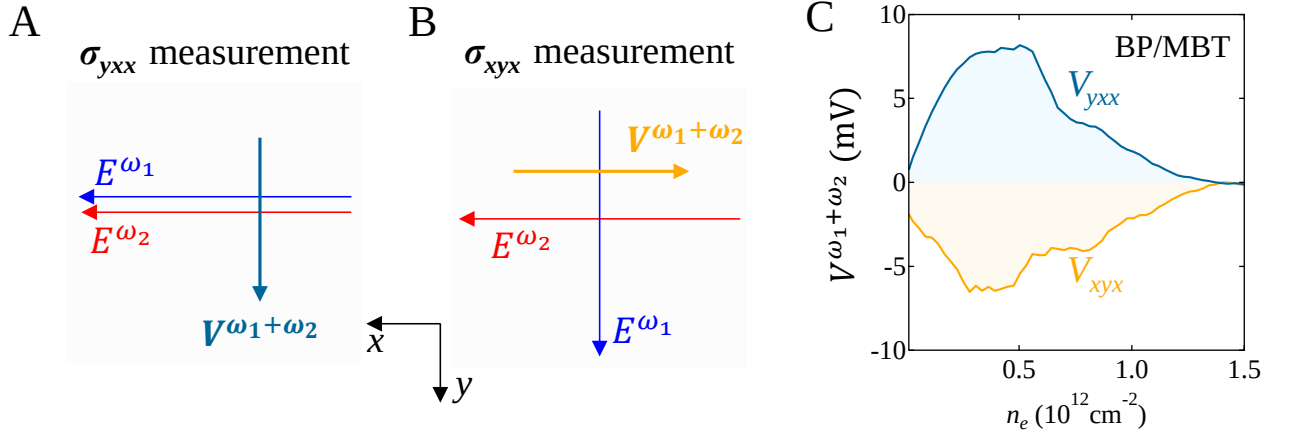


Fig. S17. The sum frequency generation measurements. (A and B) The schematic of σ_{yxx} and σ_{xyx} measurements. $\omega_1 \gg \omega_2$. (C) The experiment results of the σ_{yxx} and σ_{xyx} as a function of carrier density n_e on a BP/MnBi₂Te₄ device. It is notable that σ_{yxx} and σ_{xyx} are antisymmetric, i.e., $\sigma_{yxx} = -\sigma_{xyx}$.

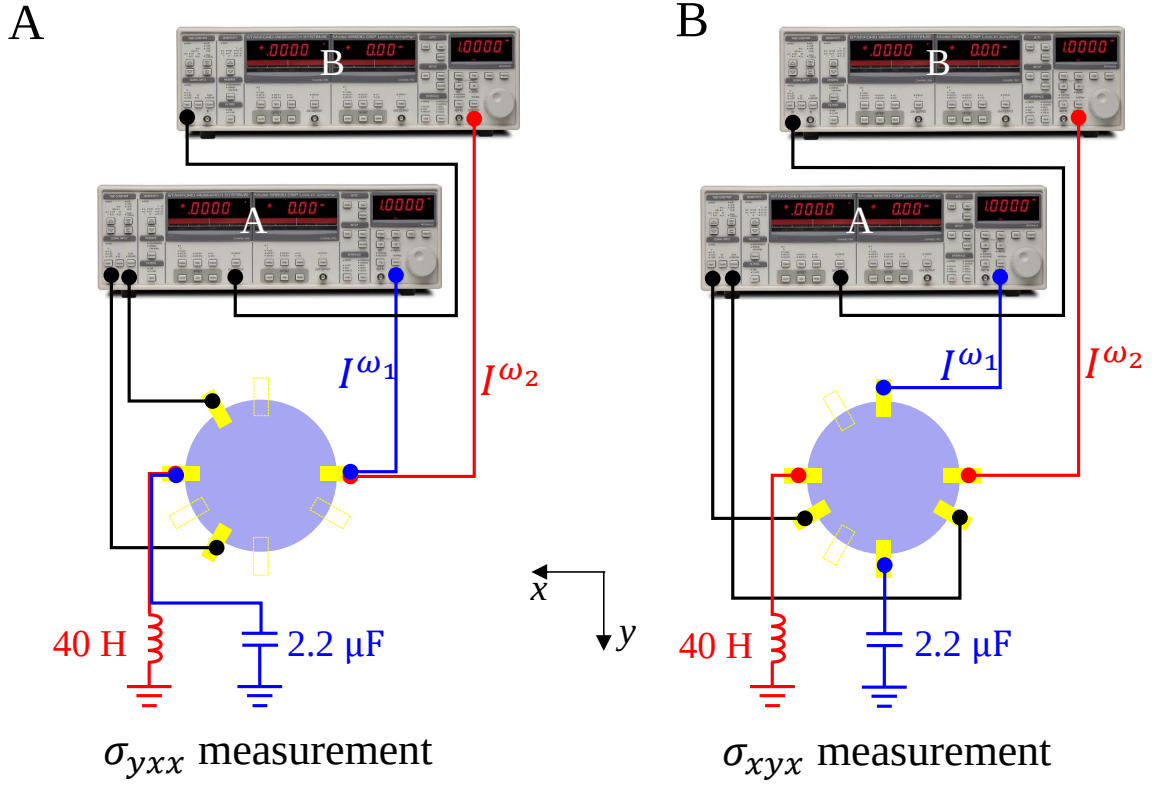


Fig. S18. The measurement setup of sum frequency generation. (A and B) The circuit of σ_{yxx} and σ_{xyx} measurements. The time constant t for lock-in A and B are 30 ms and 3 s, respectively. $\omega_1 = 547$ Hz, $\omega_2 = 1.37$ Hz. A 2.2 μF capacitor and 40 H inductor were chosen to confirm the current flow in target direction.

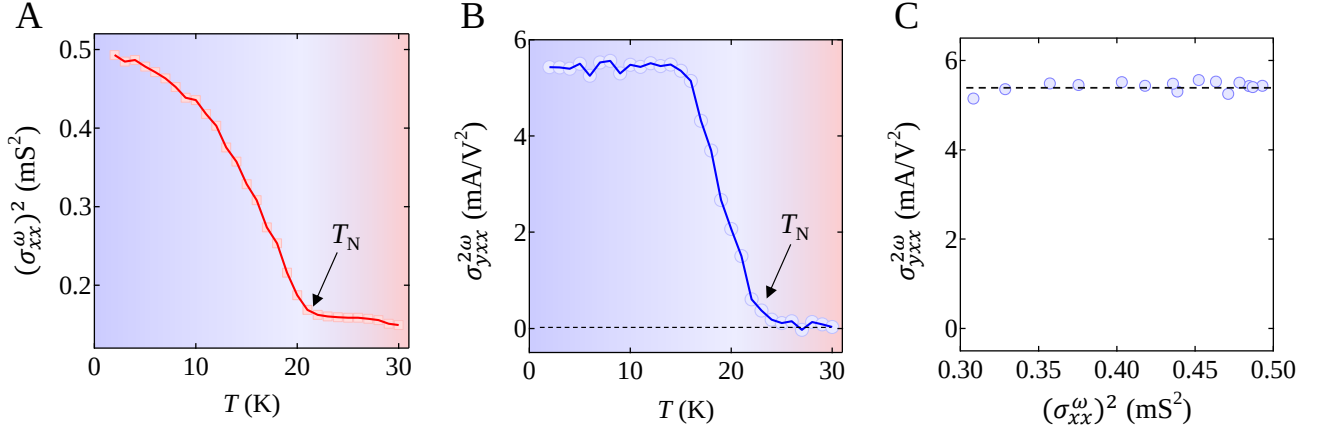


Fig. S19. The temperature dependence of nonlinear Hall signals. (A and B) The square of linear conductivity $(\sigma_{xx}^{\omega})^2$ and nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ as a function of temperature. The scaling properties of $\sigma_{yxx}^{2\omega}$ with $(\sigma_{xx}^{\omega})^2$ for temperature lower the 15 K.

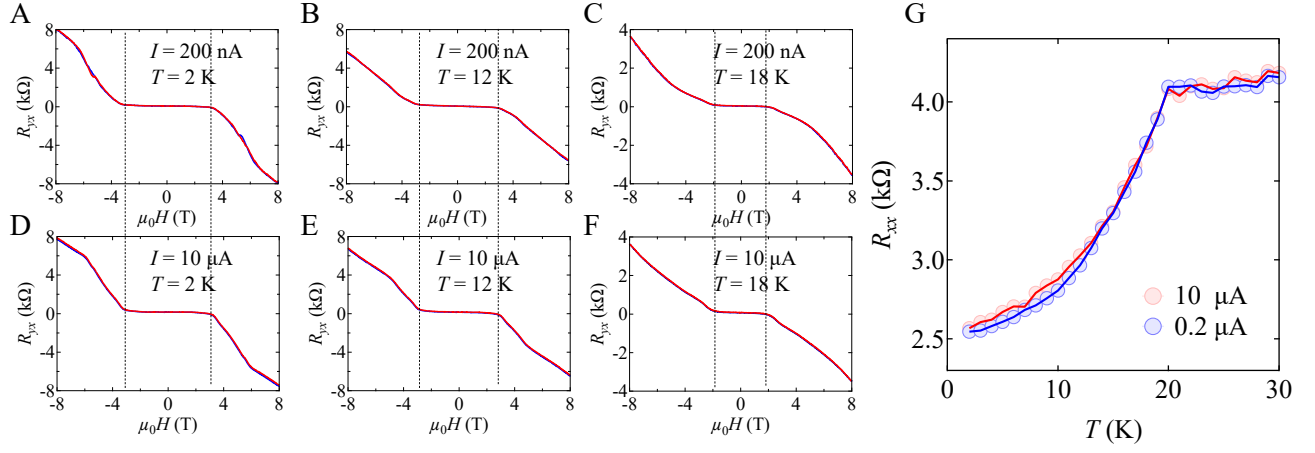


Fig. S20. (A-C) Temperature dependent $R_{yx} - B$ with 200 nA current. (D-F) Same as panels (A-C) but with 10 μ A current. (G) $R_{xx} - T$ with 200 nA and 10 μ A currents exhibits the same T_N , showing that the current-induced temperature increase is very small.

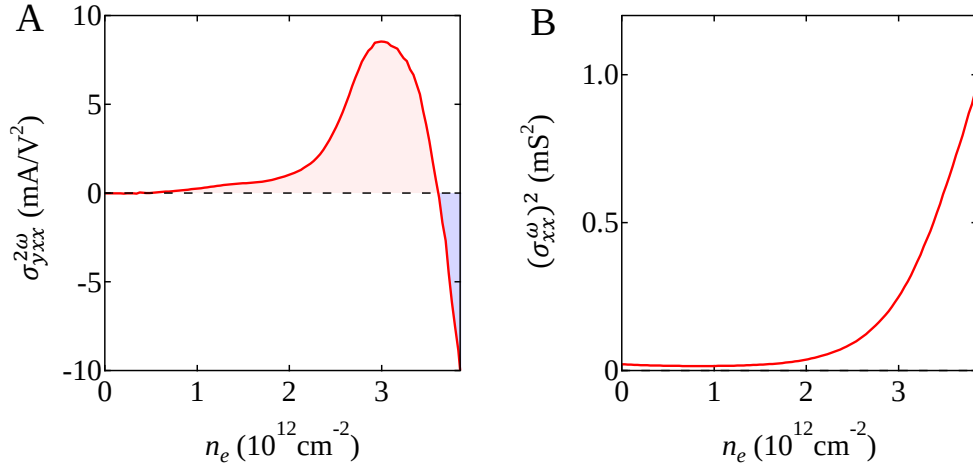


Fig. S21. The carrier density dependence of the nonlinear Hall signals. (A and B) The experimental results of $\sigma_{yxx}^{2\omega}$ and $(\sigma_{xx}^\omega)^2$ as a function of carrier density n_e .

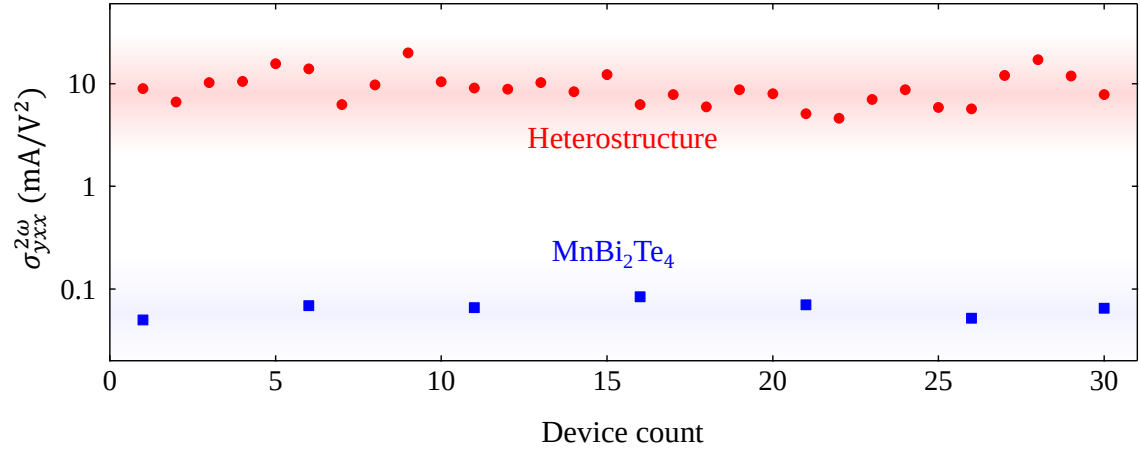


Fig. S22. The summary of $\sigma_{yxx}^{2\omega}$ of 30 MnBi₂Te₄ heterostructures and 7 MnBi₂Te₄ devices. The $\sigma_{yxx}^{2\omega}$ in MnBi₂Te₄ device is induced by nonlinear Drude conductivity which is two orders of amplitude smaller than that in MnBi₂Te₄ heterostructures.

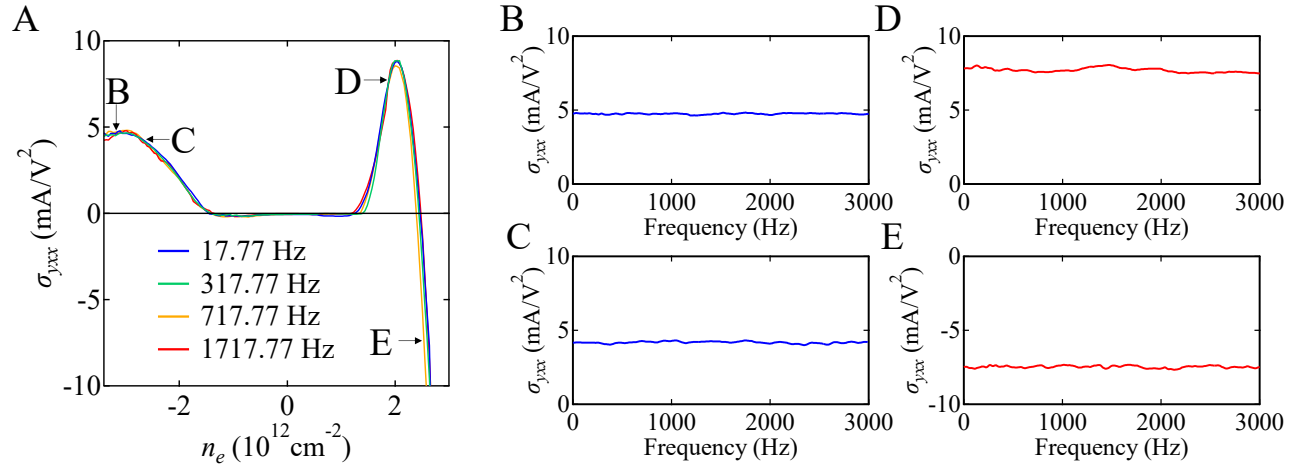


Fig. S23. Measured frequency dependence of the nonlinear Hall effect. (A) $\sigma_{yxx}^{2\omega} - n_e$ at discrete frequency ω values. (B-E) $\sigma_{yxx}^{2\omega} - \omega$ at four charge density n_e values (noted by the black arrows in panel A).

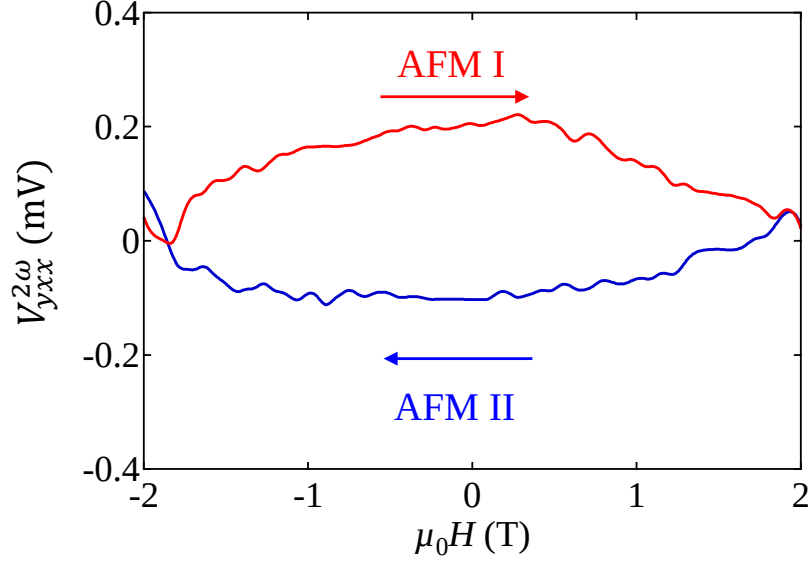


Fig. S24. The magnetic field dependence of nonlinear Hall signals in the BP/MnBi₂Te₄/BP heterostructure. The nonlinear transverse voltages $V_{yxx}^{2\omega}$ as a function of out of plane magnetic field $\mu_0 H$. At $\mu_0 H = 0$, the $V_{yxx}^{2\omega}$ have opposite signals for different AFM states. The forward and backward scans and corresponding AFM states are noted by the red and blue curves, respectively.

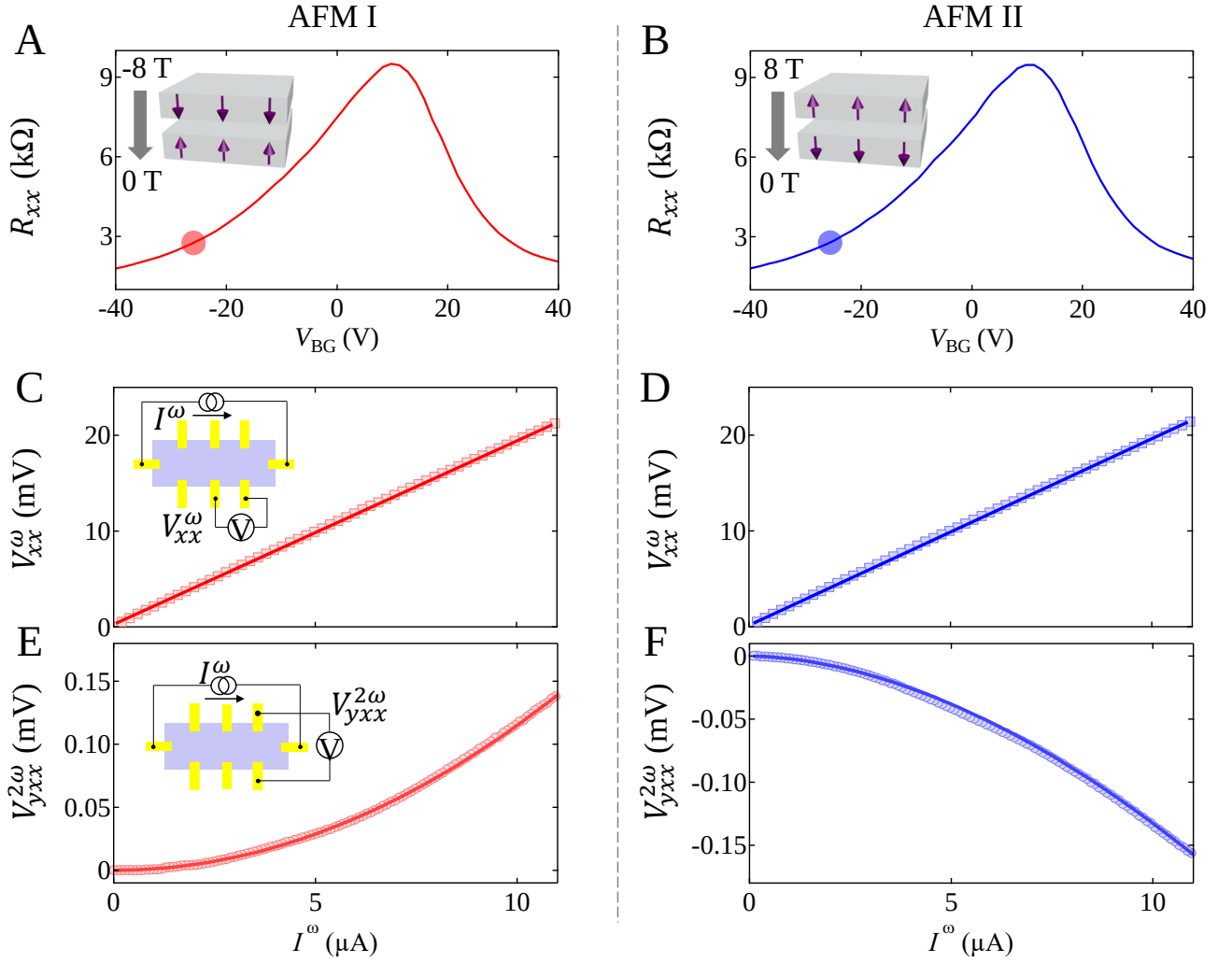


Fig. S25. The observation of antiferromagnetic nonlinear Hall in another BP/MnBi₂Te₄/BP device. (A and B) The linear longitudinal resistance R_{xx} as a function of back gate voltages for different AFM states. (C and E) The linear longitudinal voltages σ_{xx}^{ω} and nonlinear Hall voltages $\sigma_{yxx}^{2\omega}$ as a function of supply current for an AFM state. Insets show the corresponding measurement circuits. (D and F) The same as the panels (C) and (E) but for another AFM state.

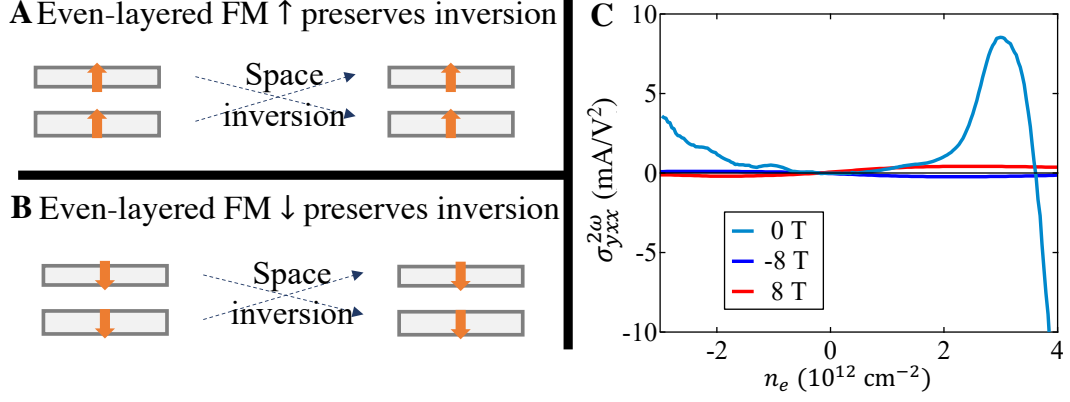


Fig. S26. (A,B) Schematic showing the FM states preserve inversion symmetry. (C) Measured nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ in the AFM (0 T) and FM (± 8 T) states. The nonlinear Hall signal in FM states are very small. The small signals (which are opposite for ± 8 T) may be due to a Nernst effect due to the large magnetization of the FM states

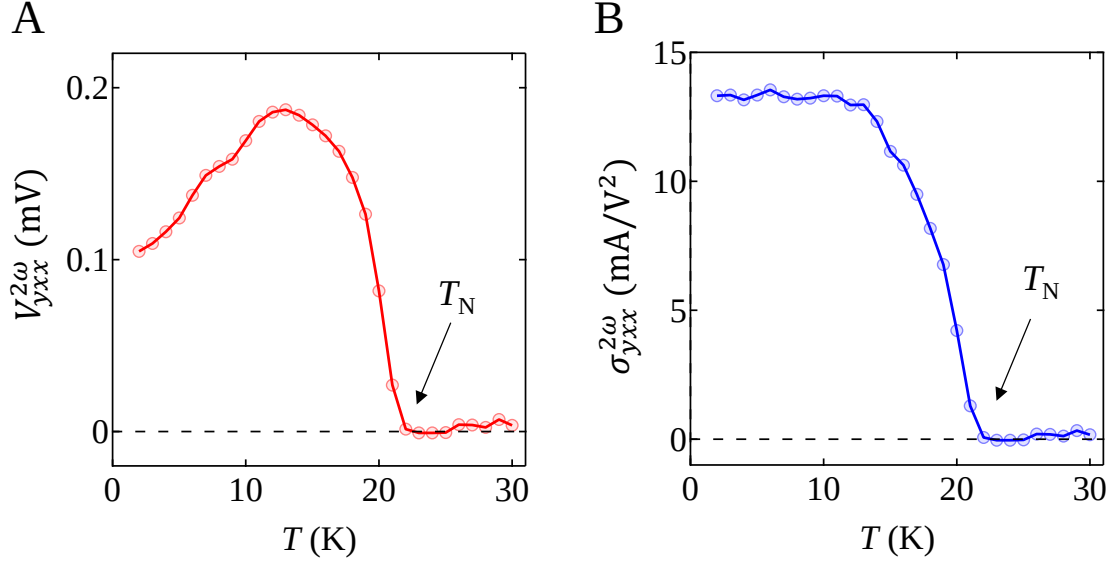


Fig. S27. Additional temperature dependence data corresponding to Fig. 3A. (A) The nonlinear Hall voltages $V_{yxx}^{2\omega}$ as a function of temperature. (B) The nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ as a function of temperature. When temperature is lower than 15 K, $\sigma_{yxx}^{2\omega}$ is almost independent of temperature.

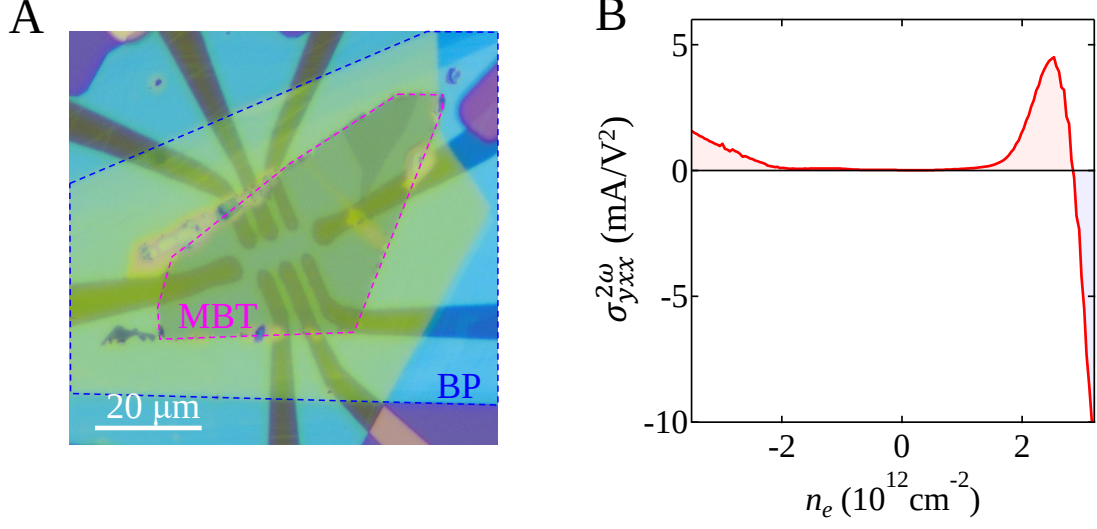


Fig. S28. Additional data of the nonlinear Hall signals on a BP/4SL MnBi₂Te₄ device. (A) The microscopic image of a BP/4SL MnBi₂Te₄ device. BP and 4SL MnBi₂Te₄ are outlined by blue and red dashed lines. (B) The nonlinear Hall conductivity $\sigma_{yxx}^{2\omega}$ as a function of carrier density n_e . The $\sigma_{yxx}^{2\omega}$ in the BP/4SL MnBi₂Te₄ device shows the same behavior as that in the BP/6SL MnBi₂Te₄ device (Fig. 4A).

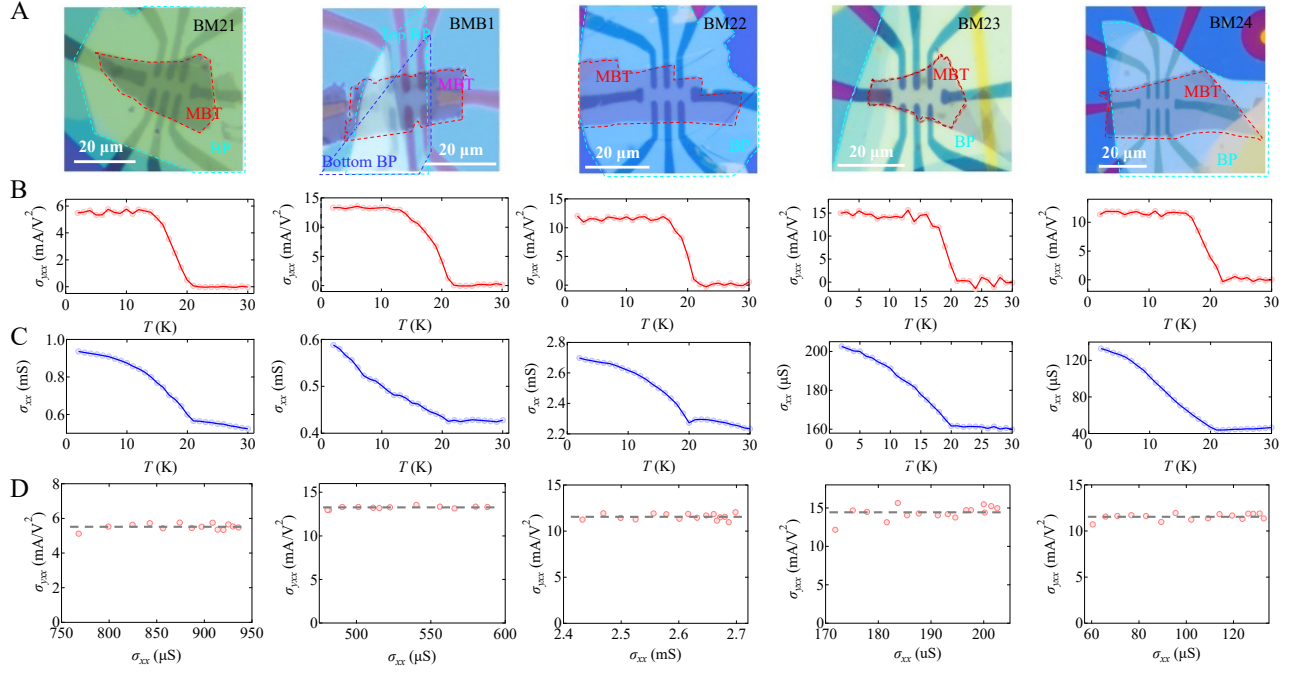


Fig. S29. Scaling performed by varying temperature is reproduced in the nonlinear Hall effect of 5 different devices.

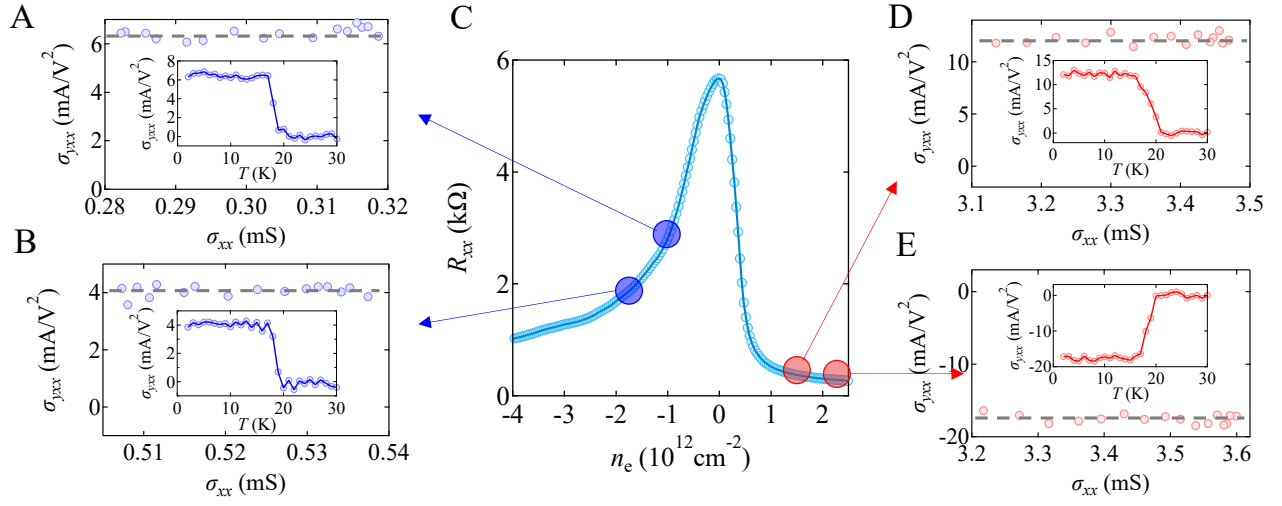


Fig. S30. Scaling performed by varying temperature at different charge densities in the same device.

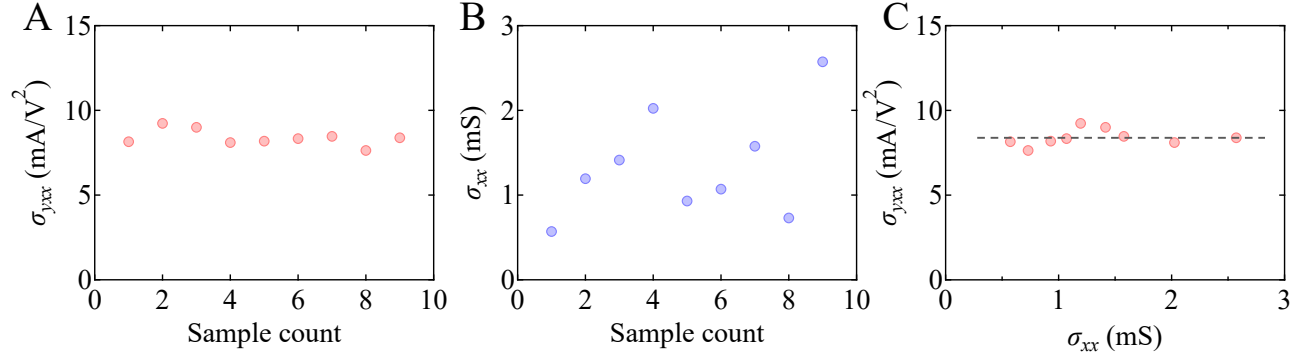


Fig. S31. Scaling performed by varying different samples. (A,B) $\sigma_{yxx}^{2\omega}$ and σ_{xx}^{ω} of different samples. $n_e = 2.8 \times 10^{12} \text{ cm}^{-2}$ and $T = 2 \text{ K}$ for all samples. (C) $\sigma_{yxx}^{2\omega} - \sigma_{xx}^{\omega}$ based on panels A,B.

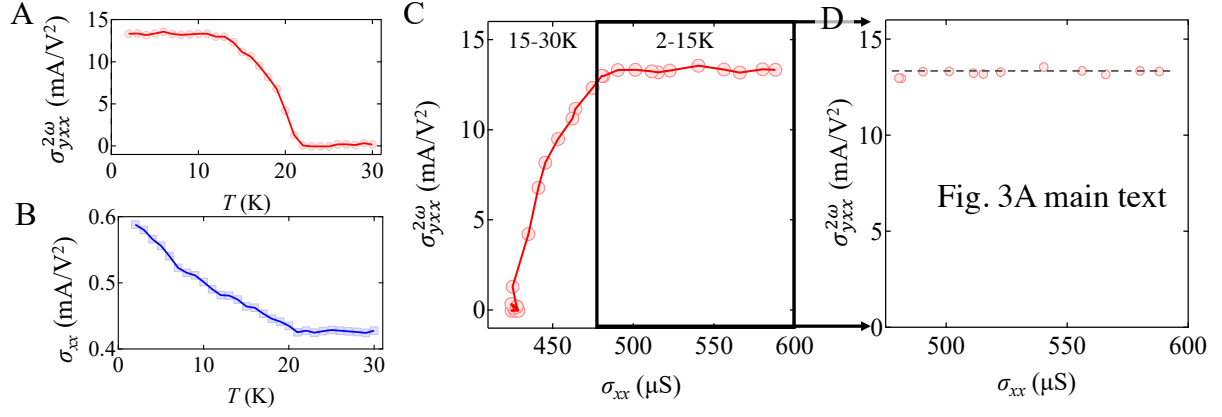


Fig. S32. (A) $\sigma_{yxx}^{2\omega} - T$. (B) $\sigma_{xx}^{\omega} - T$. (C) Scaling $\sigma_{yxx}^{2\omega} - \sigma_{xx}^{\omega}$ data for the entire temperature range 2 – 30 K. (D) Scaling $\sigma_{yxx}^{2\omega} - \sigma_{xx}^{\omega}$ data up to 15 K, where $\sigma_{yxx}^{2\omega}$ is independent of σ_{xx}^{ω} (Fig. 3A of main text).

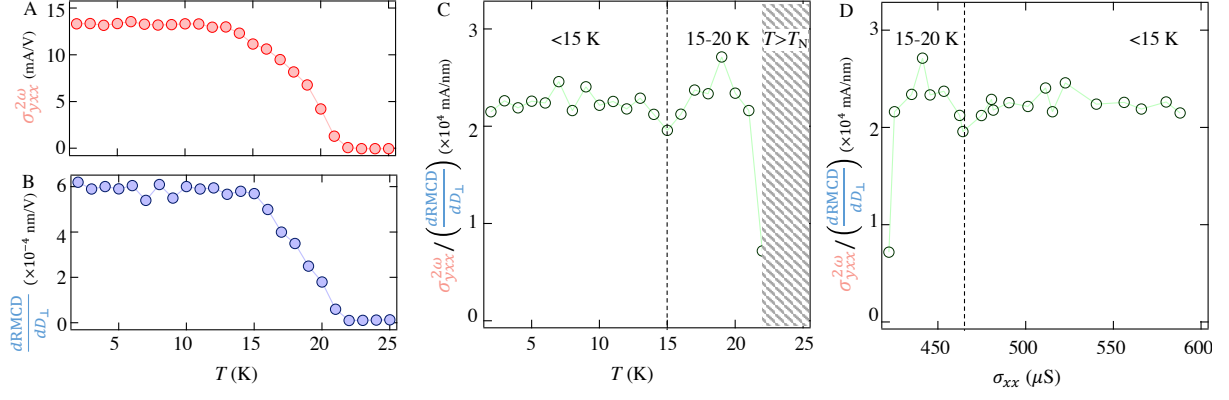


Fig. S33. (A) $\sigma_{yxx}^{2\omega} - T$. (B) vertical displacement field $D_{\perp}$ generated RMCD ($\frac{d\text{RMCD}}{dD_{\perp}}$) as a measure of the AFM order. (C) The ratio between panels (A) and (B), which can take account into the change of AFM order parameter, therefore isolating the scattering time τ dependence even above 15K. (D) $\sigma_{yxx}^{2\omega} / \frac{d\text{RMCD}}{dD_{\perp}}$ as a function of σ_{xx}^{ω} .

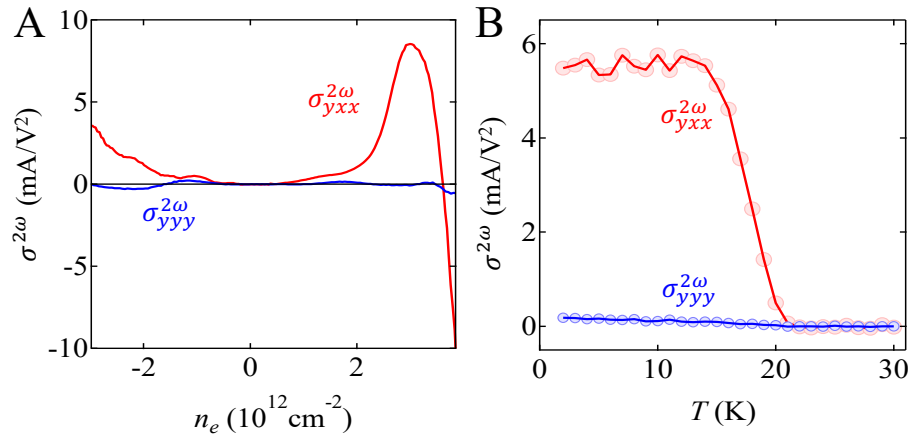


Fig. S34. Measured $\sigma_{yxx}^{2\omega}$ (predominantly nonlinear Hall) and $\sigma_{yyy}^{2\omega}$ (purely nonlinear Drude) as a function of charge density and temperature.

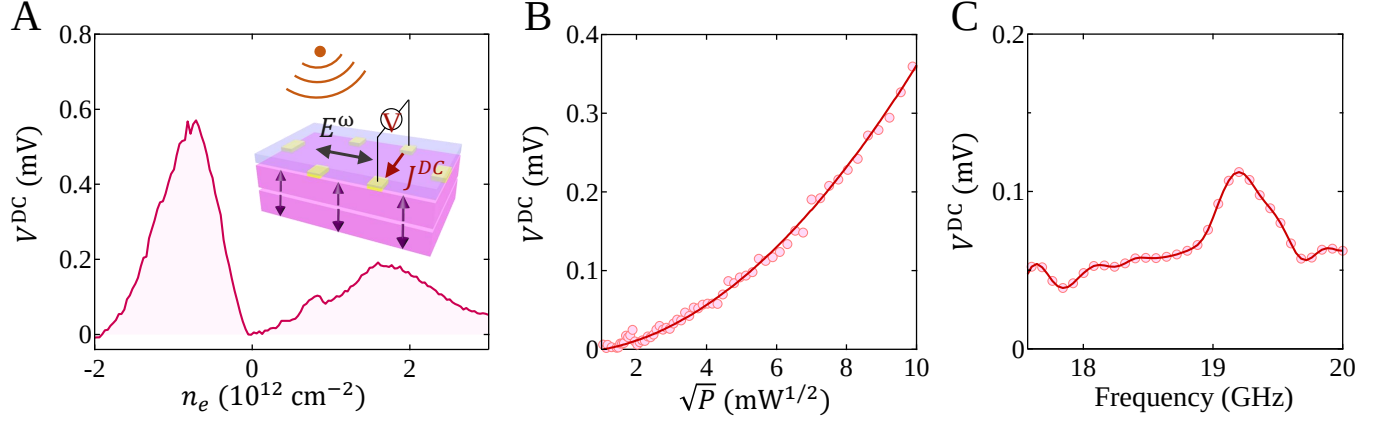


Fig. S35. Microwave rectification measurements on a BP/6SL MnBi₂Te₄ device at 2 K. (A) The DC signals V^{DC} as a function of carrier density n_e at 2.8 GHz. The inset shows the DC microwave rectification measurement circuit. The electrical field direction of the microwave is not well defined. Therefore, the recorded DC signals are only part of the the microwave generated signals. (B) The power dependence of the V^{DC} . The solid line is the fit of the V^{DC} . It scales quadratically with $\sqrt{P}$ ($\sqrt{P} \propto I$). (C) The V^{DC} as a function of Frequency. The fluctuation is caused by the transmission losses of the RF signal.

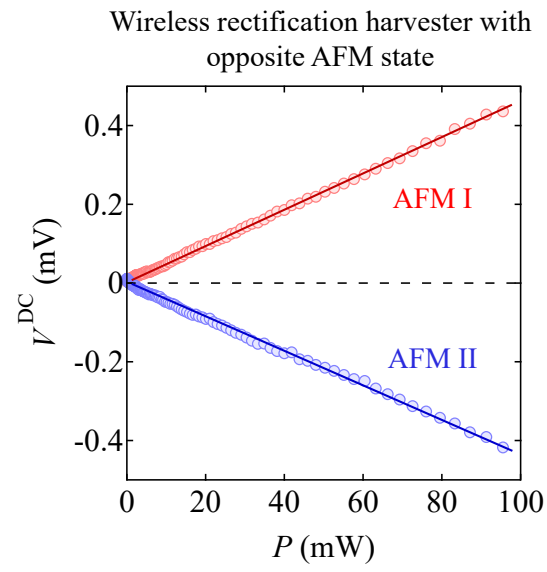


Fig. S36. Wireless rectification signal for the two opposite AFM states.

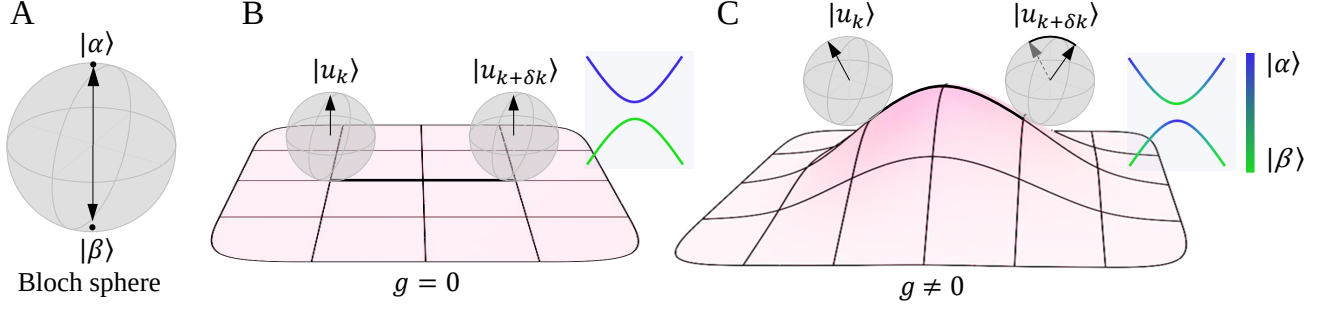


Fig. S37. An intuitive picture of quantum metric. (A) The Bloch sphere. The north and south poles of the Bloch sphere correspond to the basis vectors of $|\alpha\rangle$ and $|\beta\rangle$, respectively. (B) A trivial case ($g = 0$) without band inversion. When electrons momenta change from k to $k + \delta k$, they occupy the same state ($|\alpha\rangle$) and map on the same point on the Bloch sphere. Therefore, the distance between the two states ($|u_k\rangle$ and $|u_{k+\delta k}\rangle$) on the Bloch sphere is zero. (C) A nontrivial case ($g \neq 0$) in which conducting band and valance band go through a band inversion. In this case, each band becomes a $\mathbf{k}$ -dependent superposition of $|\alpha\rangle$ and $|\beta\rangle$. The change of $|u\rangle$ is dramatic near $\mathbf{k} = 0$ where band inversion occurs but weak at large $\mathbf{k}$. When electrons momenta change from k to $k + \delta k$, the state changes from one to another. The nonzero distance between the two states is drawn on the Bloch sphere.

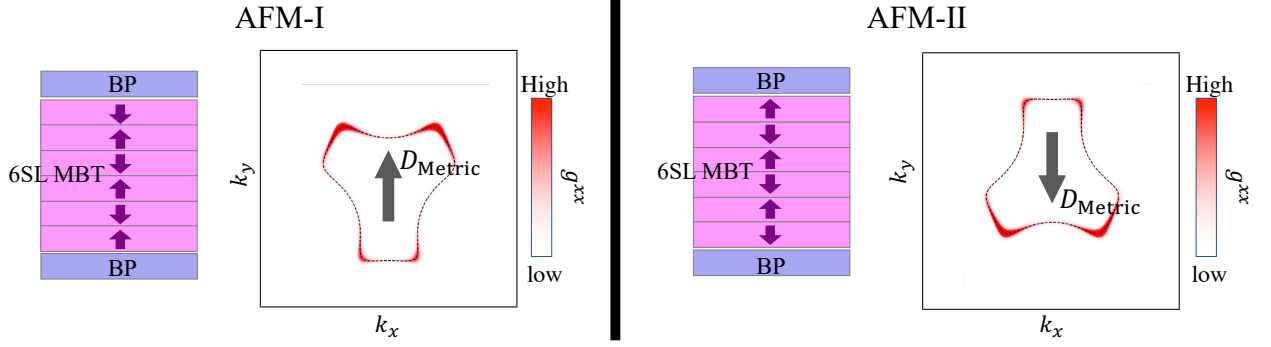


Fig. S38. (A,C) Schematic illustration of the two opposite AFM domains of BP/6L MnBi_2Te_4 /BP. (B,D) Distribution of the quantum metric g_{xx} around the Fermi contour for the two AFM domains. Note that g_{xx} is always positive because it is the distance between nearby wavefunctions (distance is always positive). The quantum metric dipole D_{metric} is a vector that connects from the side with smaller g_{xx} to that with larger g_{xx} . Therefore, it is clear that the opposite AFM domains have opposite quantum metric dipole D_{metric} .

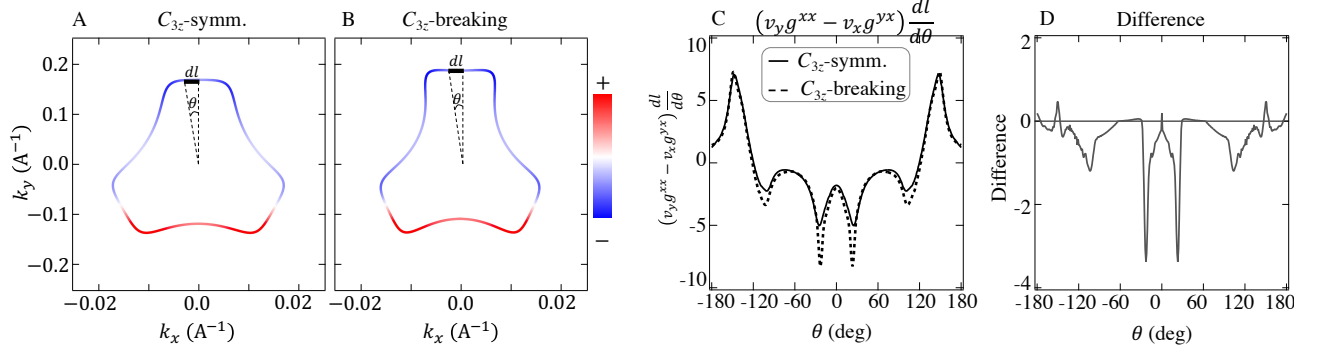
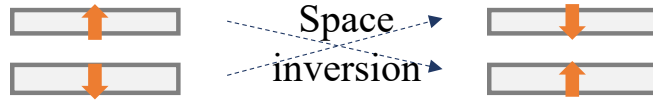


Fig. S39. (A,B) the same as Fig. 4D of main text. The color represents the kernel $(v_y g^{xx} - v_x g^{yx})$. The quantum metric dipole is obtained by integrating the kernel around the Fermi contour $\int (v_y g^{xx} - v_x g^{yx}) dl$. dl is an infinitesimal length along the contour, as shown in the Figure. This integral can be rewritten as an integral of the polar angle, $\int (v_y g^{xx} - v_x g^{yx}) (\frac{dl}{d\theta}) d\theta$. (C) We plot $(v_y g^{xx} - v_x g^{yx}) (\frac{dl}{d\theta})$ as a function of the polar angle θ for both the C_{3z} -symmetric and C_{3z} -breaking cases. (D) The difference between the two curves in panel C, which clearly shows that the integral becomes nonzero when coupling is turned on.

A Even-layered AFM breaks inversion



B Odd-layered AFM preserves inversion

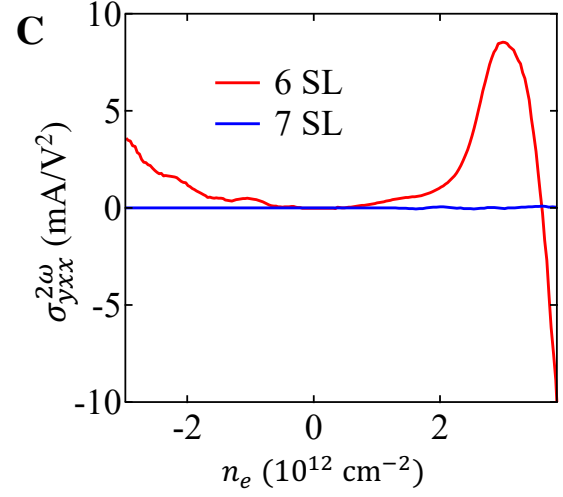
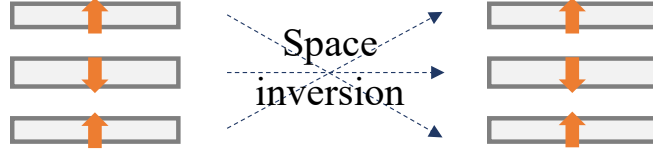


Fig. S40. (A,B) Symmetry of even(odd)-layered AFM state. (C) BP/7SL MnBi₂Te₄ show negligible nonlinear Hall signal.

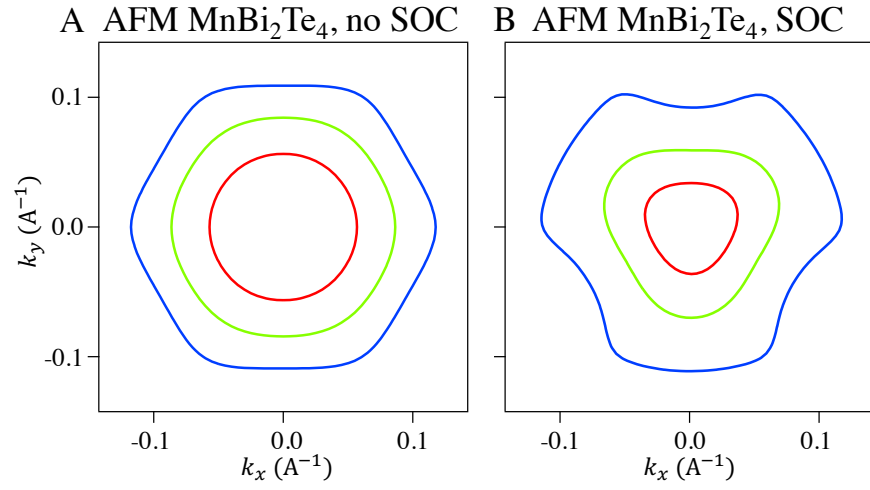


Fig. S41. DFT constant energy contours of 6SL AFM MnBi_2Te_4 without and with SOC.

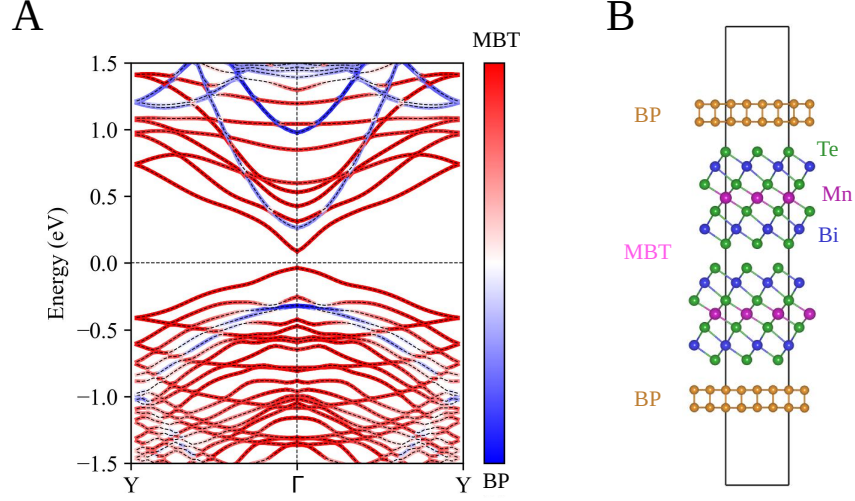


Fig. S42. The DFT computed band structure of the BP/MnBi₂Te₄/BP heterostructure. (A) The MnBi₂Te₄ bands are marked in red, and the BP bands are shown in blue. As evident from the plot, MnBi₂Te₄ and BP share similar work functions, and as a result, the low-energy BP bands hybridize with MnBi₂Te₄ bands around the Fermi level of MnBi₂Te₄, forming a straddling gap (type-I) arrangement. Due to the large lattice mismatch between the two materials, we had to artificially distort the BP unit cell by a lot ($\sim 13\%$ elongation along b) to build the 1×2 supercell. Even then, we could only do BP/2SL MnBi₂Te₄/BP, because BP/6SL MnBi₂Te₄/BP with the super cell was too heavy for DFT. Given the large distortion and the fact that it is only for BP/2SL MnBi₂Te₄/BP, we do not use it to fit our model parameters of BP/6SL MnBi₂Te₄/BP. We only showed this DFT result here for completeness. (B) The heterostructure considered here is created by a rectangular supercell of 2SL MnBi₂Te₄ and a 1×2 supercell of BP. The armchair direction of the MnBi₂Te₄ is aligned along the zigzag direction of the BP.

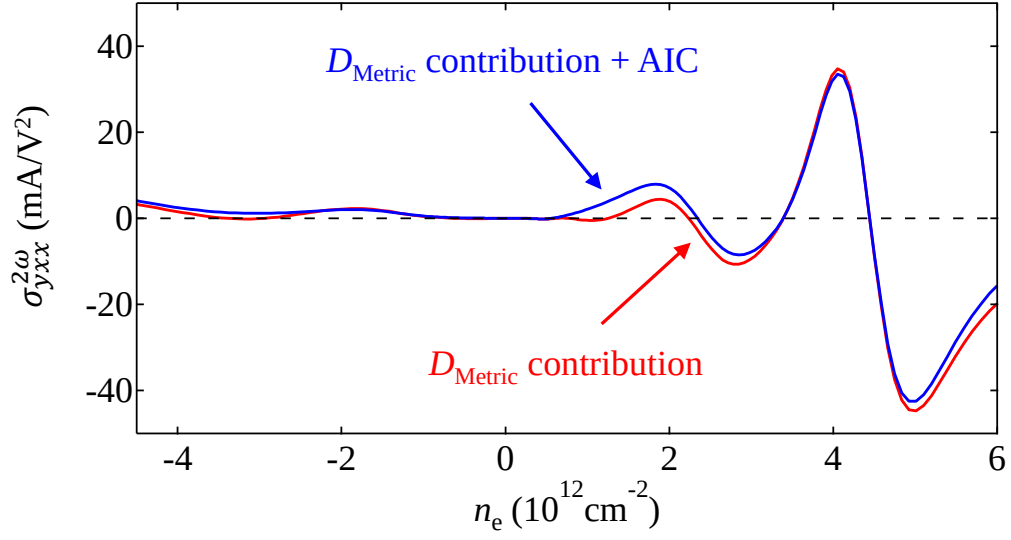


Fig. S43. Quantum metric dipole dominated nonlinear Hall conductivity. Two-band model (quantum metric dipole D_{Metric} contribution) and multiband model (D_{Metric} + additional inter band contributions AIC) calculated nonlinear Hall conductivity as a function of carrier density. The nonlinear Hall signals are dominated by the quantum metric dipole contribution.

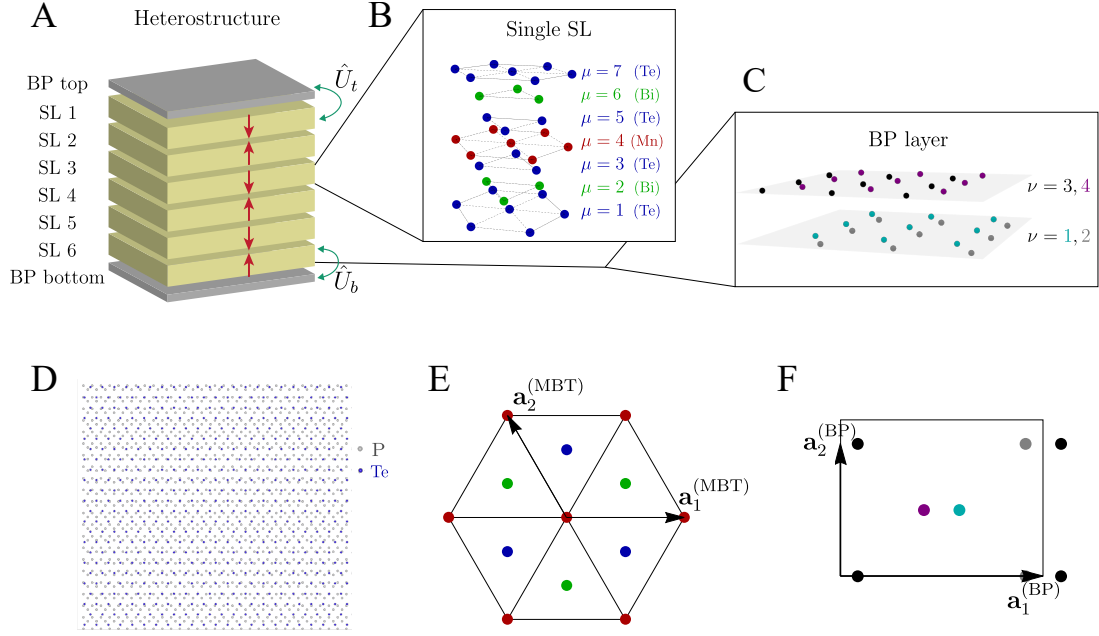


Fig. S44. Real-space lattice structure of the MBT-BP heterostructure. (A) Schematic drawing of the $\mathcal{PT}$ symmetric MBT-BP heterostructure consisting of 6SL MBT encapsulated by BP monolayer. Red arrows indicate AFM order of the MBT SLs. The direct electronic coupling between MBT and BP is dominated by hopping processes between BP top and SL 1 ($\hat{U}_t$) and between BP bottom and SL 6 ($\hat{U}_b$). (B) Lattice structure of a single MBT SL with seven atomic layers labeled by index μ . (C) Lattice structure of a BP monolayer with four basis atoms labeled by index ν . (D) Schematic top view on an incommensurate bilayer structure consisting of bottom layer of MBT (blue) and top P layer of BP (grey) with different lattice constants. (E) Two-dimensional projection of MBT SL with primitive vectors $\mathbf{a}_1^{(\text{MBT})}$ and $\mathbf{a}_2^{(\text{MBT})}$. Color coding corresponds to that used in panel (B). (F) Two-dimensional projection of BP monolayer with primitive vectors $\mathbf{a}_1^{(\text{BP})}$ and $\mathbf{a}_2^{(\text{BP})}$. Color coding corresponds to that used in panel (C).

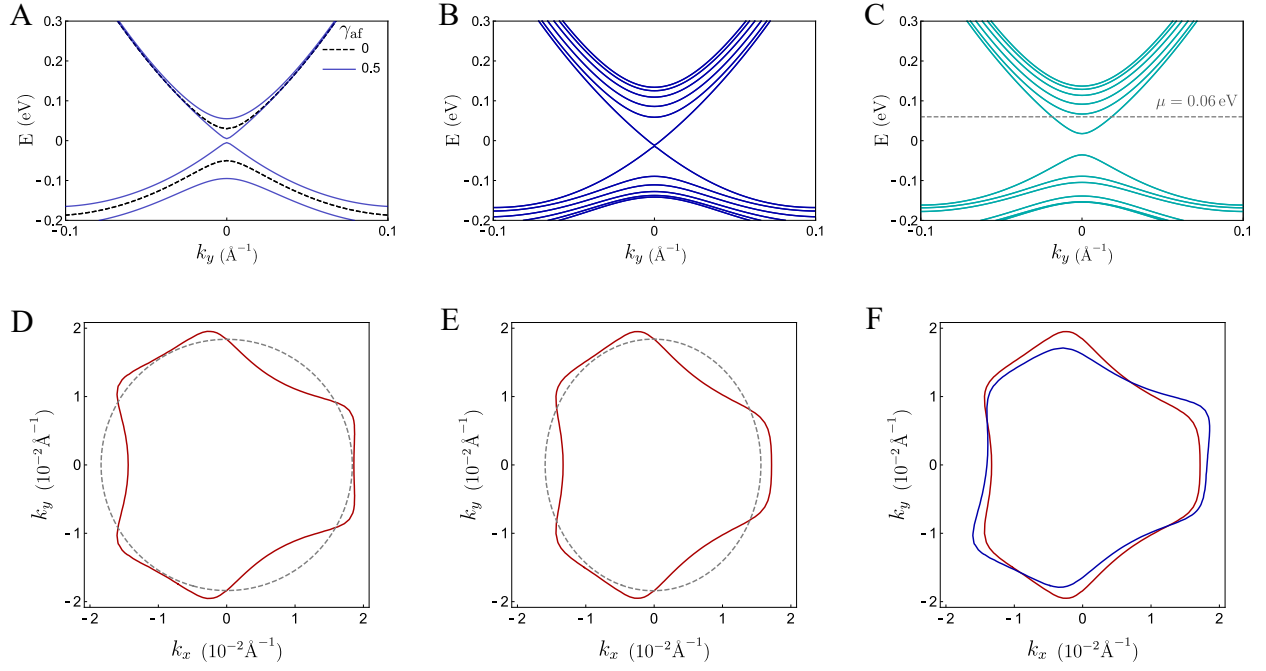


Fig. S45. Energy band Fermi surface of MBT modeled by Eqs.(S14), (S29) and (S30). (A) A cut along k_y of the bands of a single MBT SL in the paramagnetic state (dashed) and after the onset of the AFM order (solid line). The band gap at Γ ($\mathbf{k} = 0$) in the paramagnetic state originates from the hybridization of the surface states on the top and bottom of the SL and is reduced after the onset of magnetism. The hybridization of the surface states is suppressed by stacking multiple SLs along $\hat{\mathbf{z}}$, which increases the system's thickness. (B) The energy bands of a set of six SLs in the paramagnetic state ($\gamma_{af} = 0$), where the surface Dirac cone at Γ can be readily seen. (C) The onset of the AFM order open a gap in the surface Dirac cone. (D to F) The Fermi surface of 6SL MBT when the chemical potential is set to $\mu = 0.06$ meV. The dashed (solid) line in (D) and (E) refer to the Fermi surface without (with) the warping term. The red curve in panel (F) is the same as in (E), while the blue curve is obtained by rotation the red by $2\pi/3$ around $\hat{\mathbf{z}}$ to highlight the break of C_{3z} symmetry induced by strain. We set the parameters of the model to the values listed in Table S3. Note that the choice of k_x and k_y axis is rotated by $\pi/2$ with respect to the main text.

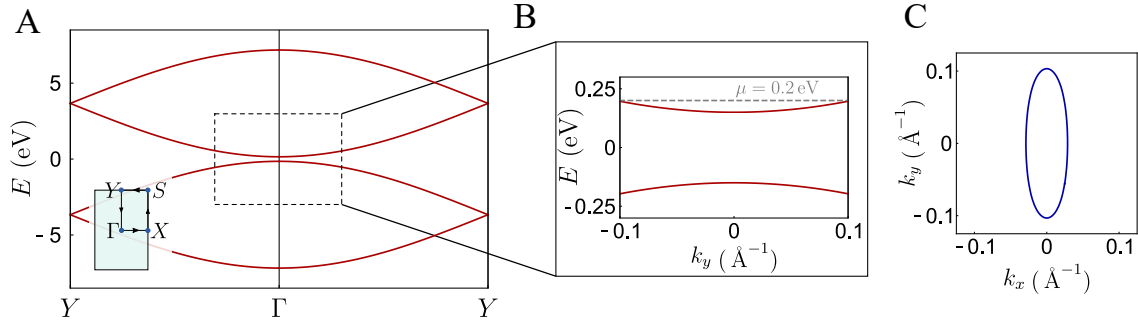


Fig. S46. Energy band and Fermi surface of monolayer BP. (A) Energy bands of BP monolayer modeled by Eq.(S35) along a high-symmetry direction of the Brillouin zone shown in blue as a inset. (B) A zoom-in view of the bands around the Γ point (e.g. $\mathbf{k} = 0$). (C) The Fermi surface of BP where we set the chemical potential to be $\mu = 0.2$ eV (dashed gray line in (B)). BP does not preserve 3-fold rotational symmetry around $\hat{z}$, which is reflected in the elliptical shape of its Fermi surface. Therefore, similar to the role of uniaxial strain, encapsulating the MBT SLs with BP monolayer leads to a breaking of C_{3z} symmetry of the heterostructure. We set the parameters of the model to the values listed in Table S3.

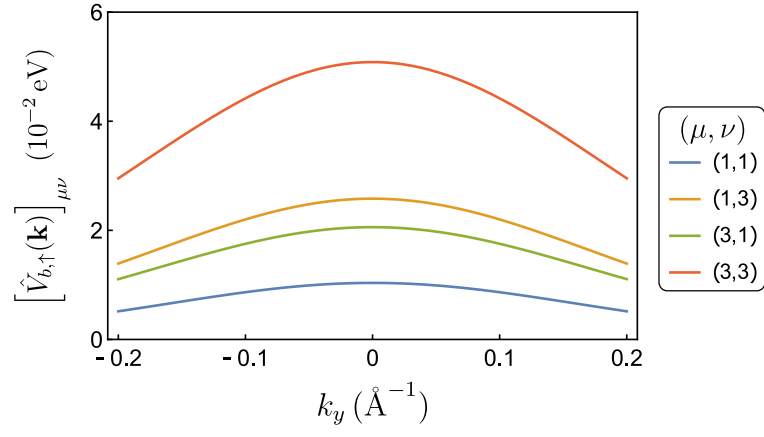


Fig. S47. Matrix elements of $\hat{V}_{b,\uparrow}(\mathbf{k})$ along the k_y axis. We set the parameters of the model to the values listed in Table S3.

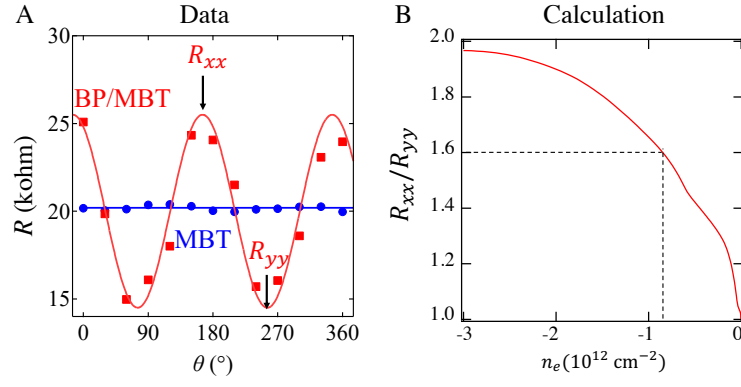


Fig. S48. (A) Measured R_{xx}/R_{yy} at $n_e \simeq -0.8 \times 10^{12} \text{ cm}^{-2}$. (B) Calculated R_{xx}/R_{yy} .

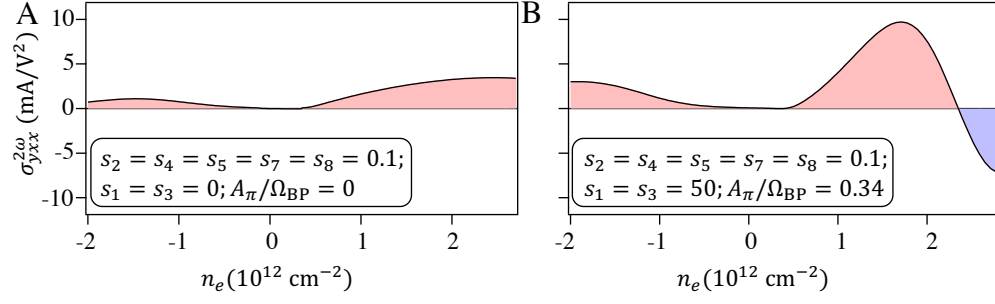


Fig. S49. Calculated $\sigma_{yxx}^{2\omega}$. (A) The linear strain term is set to 0.1 by fitting the R_{xx}/R_{yy} data. The quadratic strain and band hybridization terms are zero. (B) The quadratic strain and band hybridization are adjusted for a detailed agreement with data.

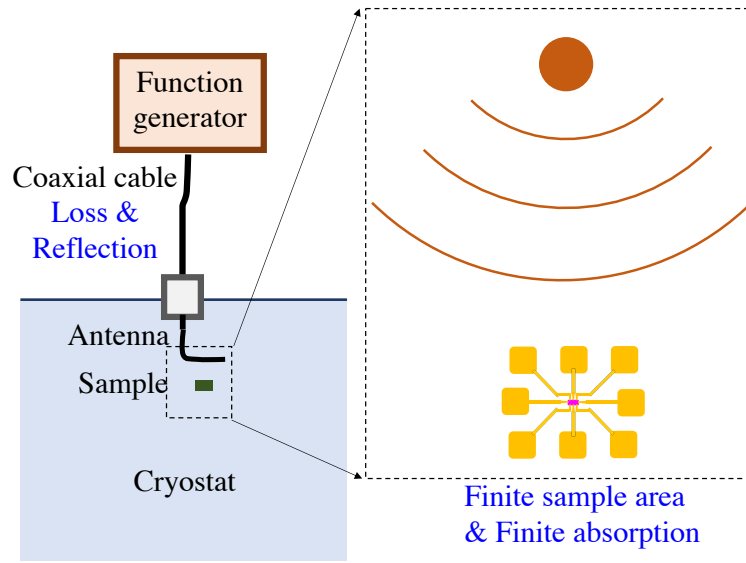


Fig. S50. Schematic illustration of our GHz rectification setup.

Device type	Device number	Contact geometry	Angle between top/bottom BP and MBT	Angle between MBT and contacts	top/bottom BP thickness	σ_{xx}^{ω} (μ S)	$\sigma_{yxx}^{2\omega}$ (mA/V ²)
BP/MBT/BP	BMB1	Hall bar	0/0	0	2L/2L	227	8.93
	BMB2	Hall bar	0/-1	0	2L/1L	671	6.62
	BMB3	Hall bar	-11/-6	1	2L/ $\sim$ 10nm	249	10.25
	BMB4	Hall bar	53/15	27	1L/ $\sim$ 15nm	585	10.5
	BMB5	Hall bar	-16/-17	16	2L/ $\sim$ 10nm	1466	15.68
BP/MBT	BM1	Circular	-5	0	$\sim$ 12nm	154	13.97
	BM2	Hall bar	-	-	-	500	6.24
	BM3	Hall bar	-30	25	-	141	9.70
	BM4	Hall bar	-	-	-	119	19.91
	BM5	Hall bar	-	-	-	403	10.42
	BM6	Hall bar	-	-	-	206	9.06
	BM7	Hall bar	-5	-2	-	498	8.82
	BM8	Hall bar	-1	1	-	608	10.26
	BM9	Hall bar	-	-	4L	1070	8.37
	BM10	Hall bar	-	-	-	2890	12.22
	BM11	Hall bar	-	-	-	307	6.24
	BM12	Hall bar	-	-	5-7L	398	7.82
	BM13	Hall bar	-	-	-	450	5.95
	BM14	Hall bar	-	-	-	943	8.75
	BM15	Hall bar	-	-	-	483	7.97
	BM16	Hall bar	-26	-2	$\sim$ 8nm	238	5.10
	BM17	Hall bar	-	-	-	357	4.60
	BM18	Hall bar	-35	54	-	500	7.00
	BM19	Hall bar	16	26	-	357	8.70
	BM20	Hall bar	-	-	-	700	5.86
	BM21	Hall bar	-	-	-	930	5.69
	BM22	Hall bar	-	-	-	2698	12.05
	BM23	Hall bar	-	-	-	3600	17.15
	BM24	Hall bar	-	-	-	133	11.90
	BM25	Circular	-	-	-	1698	7.82

Table S1. Device summary of 30 heterostructure devices. Among the 30 devices, 5 are encapsulated BP/MnBi₂Te₄/BP devices (BMB) and 25 are BP on one side BP/MnBi₂Te₄ devices (BM).

P basis atom	Components (Å)		
	$\hat{x}$	$\hat{y}$	$\hat{z}$
ρ_1	2.71	1.65	15.05
ρ_2	4.2	0	15.05
ρ_3	0.41	0	17.15
ρ_4	1.9	1.65	17.15

Table S2. Cartesian components of the position relative to the origin of the unit cell of the four nonequivalent phosphorus atoms in the BP monolayer.

Bare MnBi_2Te_4 parameters						
γ_0 (eV)	γ_1 (eVA ²)	m_0 (eV)	β_0 (eVA ²)	β_1 (eVA ³)	α (eVA)	t_1 (eV)
-0.01	17	-0.04	9.4	8000	3.2	-0.0533
t_2 (eV)	λ (eV)	α_3 (eVA ²)	γ_{AF}	m_1 (eV)	m_2 (eV)	
0.0463	0.0577	0.0	0.5	0.05	0.09	
Bare BP parameters						
t_1^{BP} (eV)	t_2^{BP} (eV)					
-1.7575	3.665					
MnBi_2Te_4 -BP coupling parameters						
Strain on MnBi_2Te_4	s_1 (eVA ²)	s_2 (eVA)	s_3 (eVA ²)	s_4 (eVA)	s_5 (eVA)	
	0.1	0.1	50	0.1	0.1	
MnBi ₂ Te ₄ -BP band hybridization	s_6 (eV)	s_7 (eVA)	s_8 (eVA)	s_9 (eV)		
	0	0.1	0.1	0.0		
	$A_\sigma/\Omega_{\text{BP}}$ (eVA ²)	A_π/Ω_{BP} (eVA ²)	a_σ (A)	a_π (A)		
	0.34	0.0	2.0	2.0		

Tab. S3. A table of the model parameters.

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